# *The Frequency-Response Design Method*

Professor: Juan David Bastidas Rodríguez

Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

# Content

1. Frequency Response
2. Neutral Stability
3. The Nyquist Stability Criterion
4. Stability Margins
5. Bode's Gain–Phase Relationship
6. Closed-Loop Frequency Response
7. Compensation

# 1. Frequency Response

- A linear system's response to sinusoidal inputs
- It results from the analysis of the steady-state response
- It can be obtained from knowledge the poles and zeros

$$\frac{Y(s)}{U(s)} = G(s)$$

$$Y(s) = \frac{\alpha_1}{s-p_1} + \frac{\alpha_2}{s-p_2} + \cdots + \frac{\alpha_n}{s-p_n} + \frac{\alpha_o}{s+j\omega_o} + \frac{\alpha_o^*}{s-j\omega_o}$$

$$u(t) = A\sin(\omega_o t)1(t) \qquad U(s) = \frac{A\omega_o}{s^2+\omega_o^2} \qquad y(t) = \alpha_1 e^{p_1 t} + \alpha_2 e^{p_2 t} + \cdots + \alpha_n e^{p_n t} + 2|\alpha_o|\cos(\omega_o t + \phi), \quad t \geq 0$$

$$Y(s) = G(s)\frac{A\omega_o}{s^2+\omega_o^2}$$

$$\phi = \tan^{-1}\left[\frac{\mathrm{Im}(\alpha_o)}{\mathrm{Re}(\alpha_o)}\right]$$

# 1. Frequency Response

Example: frequency response of a lead compensator

- Analytical evaluation

$$D_c(s) = K\frac{Ts+1}{\alpha Ts+1}, \quad \alpha < 1$$

Substituting s = jω

$$D_c(j\omega) = K\frac{Tj\omega+1}{\alpha Tj\omega+1}$$

$$M = |D_c| = |K|\frac{\sqrt{1+(\omega T)^2}}{\sqrt{1+(\alpha\omega T)^2}}$$

$$\begin{aligned}\phi &= \angle(1+j\omega T) - \angle(1+j\alpha\omega T)\\ &= \tan^{-1}(\omega T) - \tan^{-1}(\alpha\omega T).\end{aligned}$$

M = |K| low frequencies
M = |K/α| high frequencies

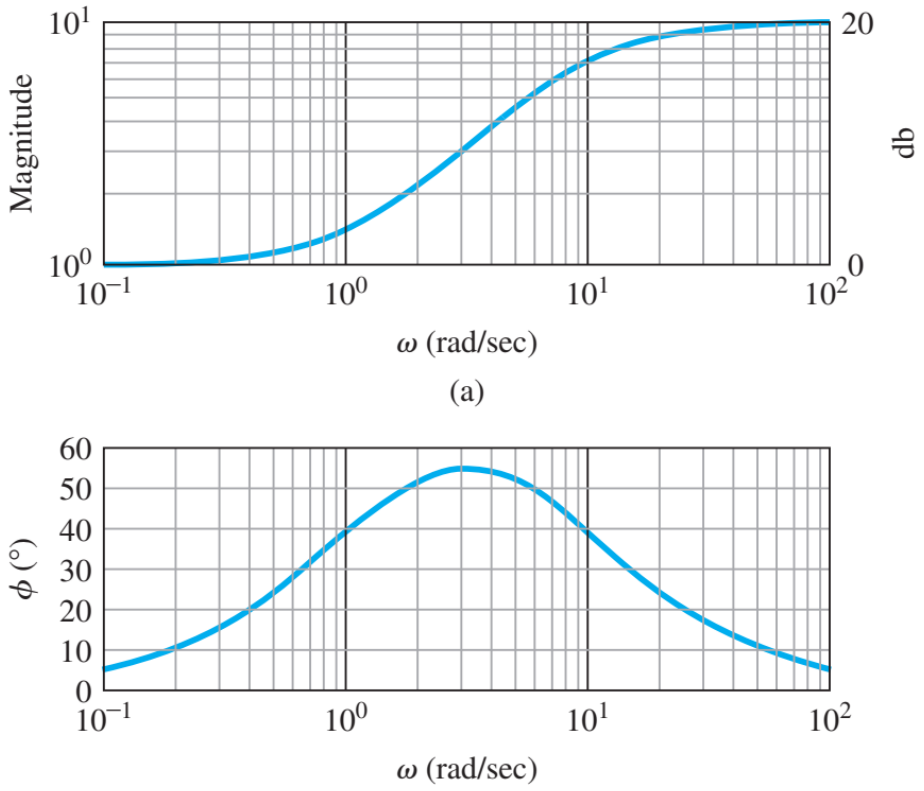
$\phi$ = 0 low frequencies
$\phi$ = 0 high frequencies

# 1. Frequency Response

Example: frequency response of a lead compensator

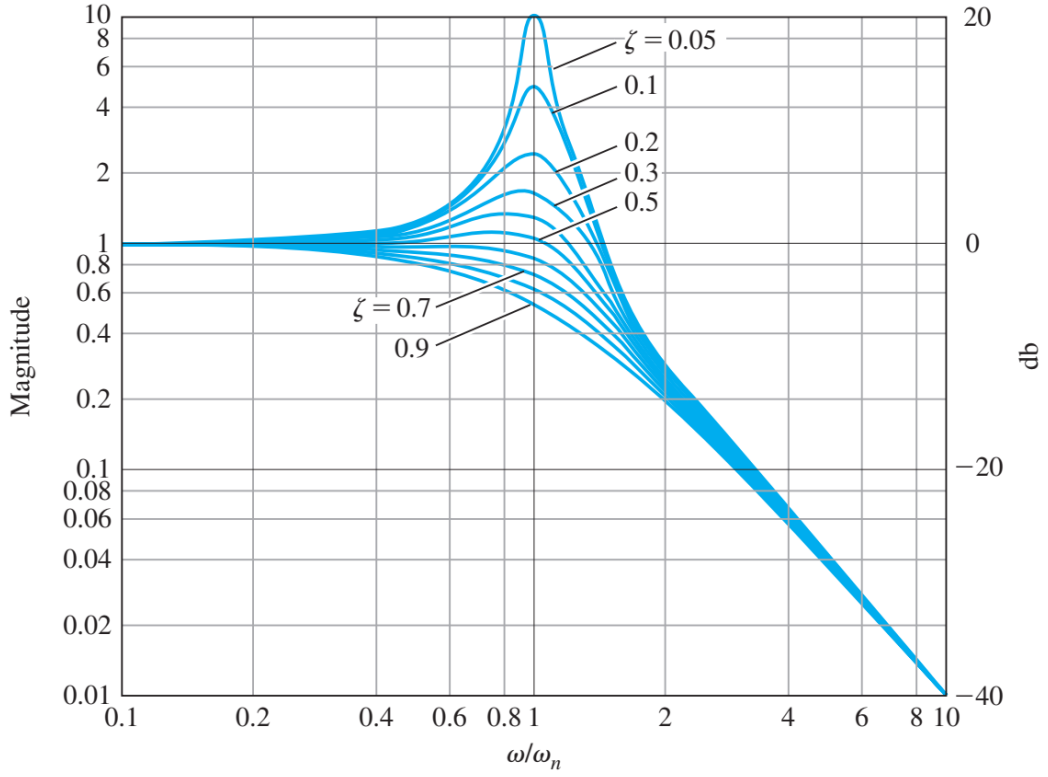
- Numerical evaluation(Matlab)

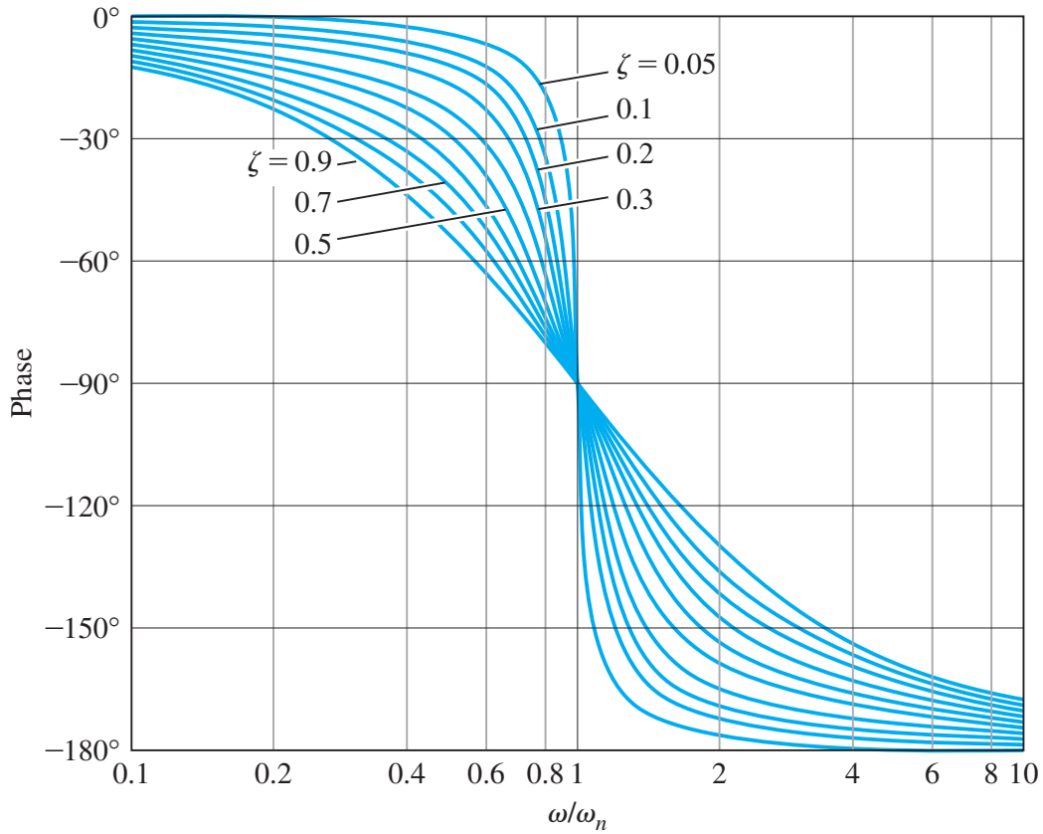
```
s = tf('s');
sysD = (s + 1)/ (s/10 + 1);
w=logspace(-1,2); % determines frequencies over range of interest
[mag,phase] = bode(sysD,w); % computes magnitude and phase over frequency range of interest
figure(1);
loglog(w,squeeze(mag)),grid;
axis([0.1 100 1 10])
figure(2);
semilogx(w,squeeze(phase)),grid;
axis([0.1 100 0 60])
```



# 1. Frequency Response

Frequency response for a 2nd order system: $G(s) = \dfrac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$

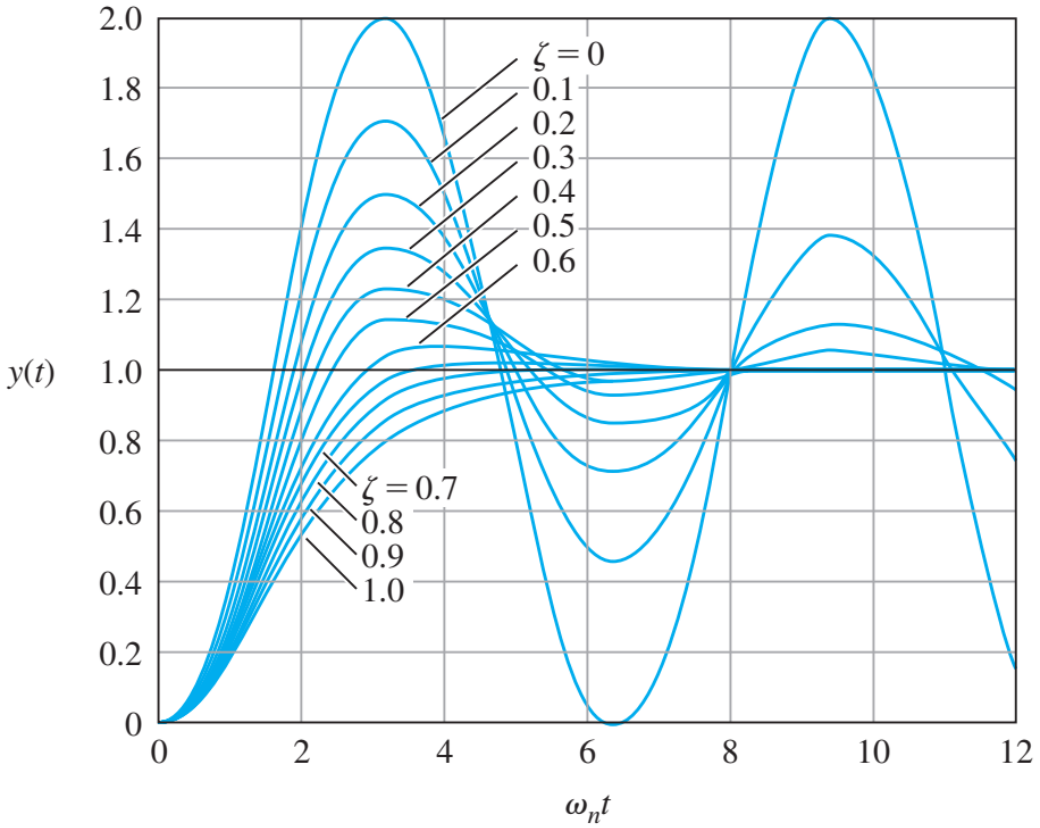


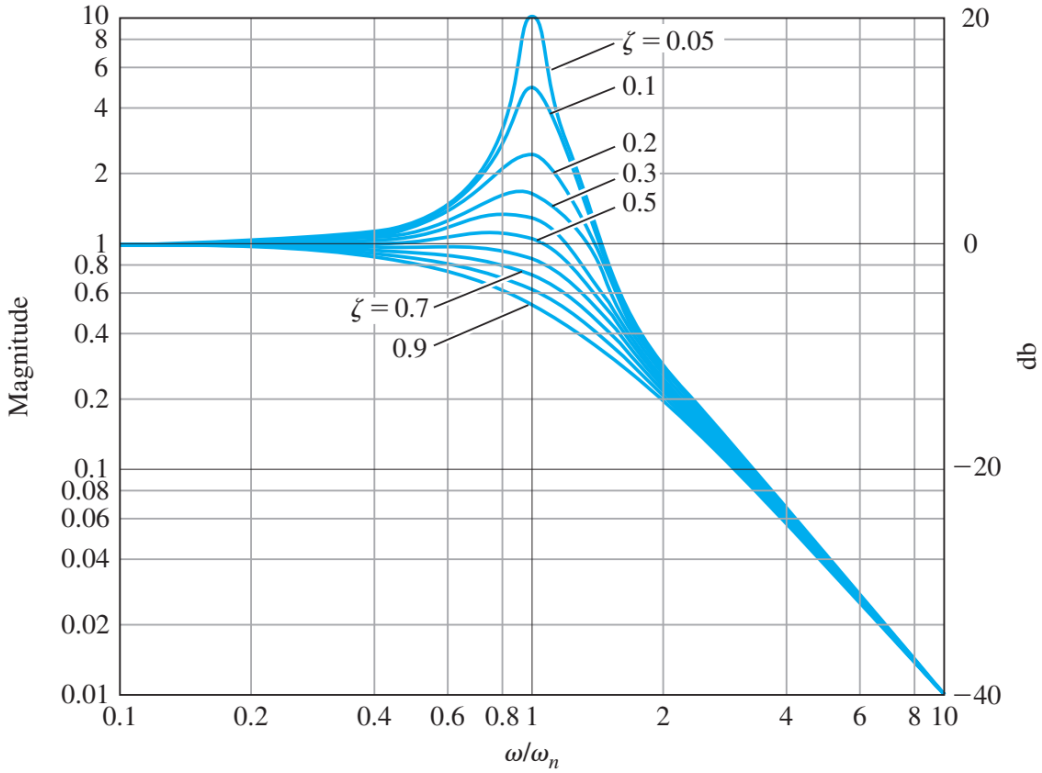



# 1. Frequency Response

Step response for a 2nd order system:

$$G(s) = \frac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$



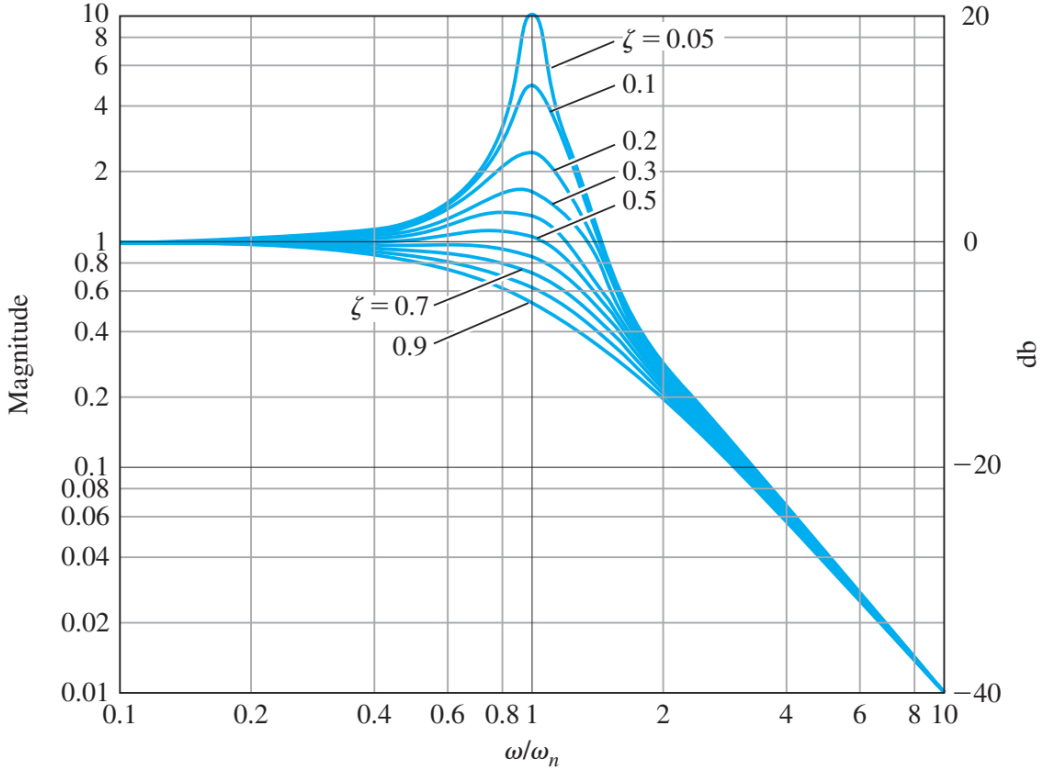



# 1. Frequency Response

Step response for a 2nd order system:

$$G(s) = \frac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

- The damping of the system can be determined from the peak in the magnitude of the frequency response
- $\omega_n$ is approximately equal to the bandwidth
- The rise time can be estimated from the bandwidth
- The peak overshoot in the step response can be estimated from the peak overshoot in the frequency response



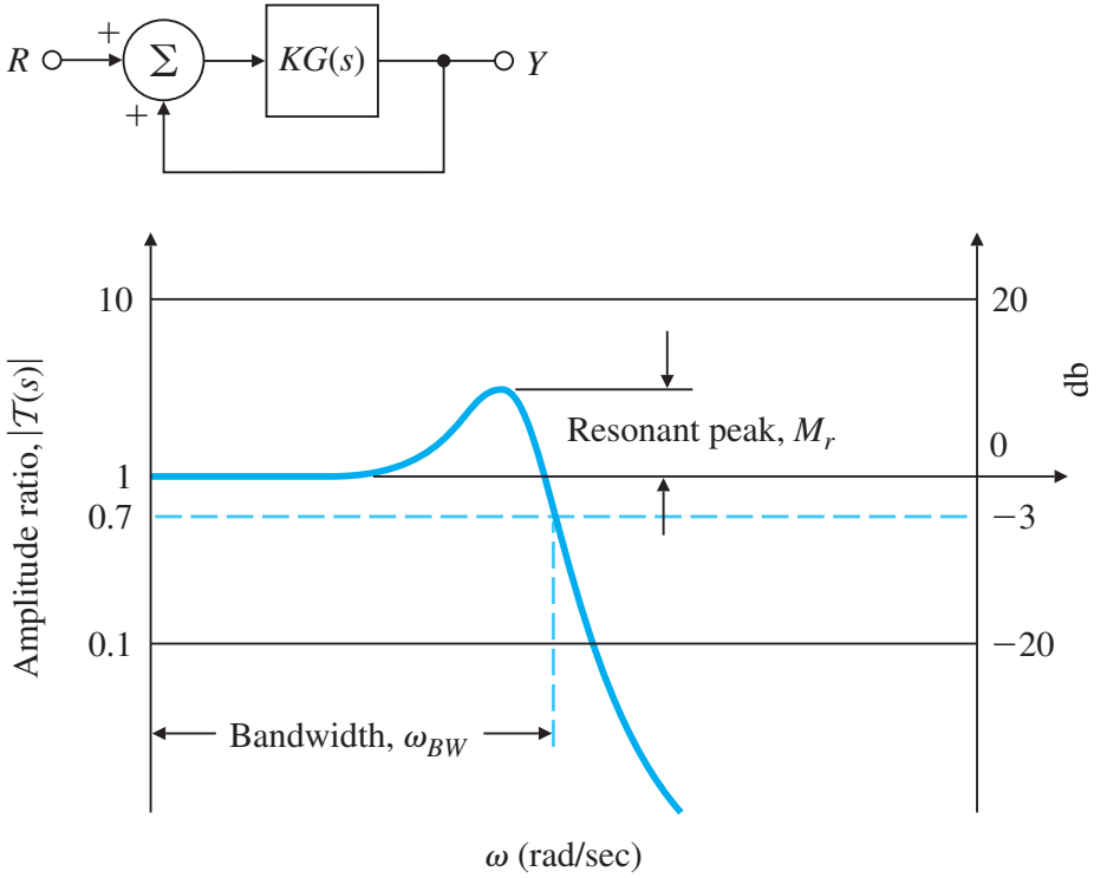
# 1. Frequency Response

**Bandwidth**:

- The maximum frequency at which the output of a system will track an input sinusoid in a satisfactory manner.
- For a closed-loop system, the bandwidth is the frequency of **r** at which the output **y** is **attenuated** to a factor of **0.707 (-3db)** times the input

**Resonant peak**:

- The maximum value of the magnitude



# 1. Frequency Response

**Bode plot techniques:**

- Plot magnitude curves using a **logarithmic** scale
- Plot the phase curves using a **linear** scale
- FdT for s = jw can be written in polar form (phasors)

$$G(j\omega) = \frac{\vec{s}_1\vec{s}_2}{\vec{s}_3\vec{s}_4\vec{s}_5} = \frac{r_1 e^{j\theta_1} r_2 e^{j\theta_2}}{r_3 e^{j\theta_3} r_4 e^{j\theta_4} r_5 e^{j\theta_5}} = \left(\frac{r_1 r_2}{r_3 r_4 r_5}\right) e^{j(\theta_1+\theta_2-\theta_3-\theta_4-\theta_5)}$$

$$|G(j\omega)| = \frac{r_1 r_2}{r_3 r_4 r_5} \qquad \log_{10}|G(j\omega)| = \log_{10} r_1 + \log_{10} r_2 - \log_{10} r_3 - \log_{10} r_4 - \log_{10} r_5$$

# 1. Frequency Response

## Bode plot techniques:

Before we used this FdT representation

$$KG(s) = K\frac{(s-z_1)(s-z_2)\cdots}{(s-p_1)(s-p_2)\cdots}$$

Now, Bode form of the FdT

$$KG(j\omega) = K_o(j\omega)^n\frac{(j\omega\tau_1+1)(j\omega\tau_2+1)\cdots}{(j\omega\tau_a+1)(j\omega\tau_b+1)\cdots}$$

All transfer functions are composed by 3 types of terms:

1. $K_o(j\omega)^n$.
2. $(j\omega\tau+1)^{\pm1}$.
3. $\left[\left(\frac{j\omega}{\omega_n}\right)^2+2\zeta\frac{j\omega}{\omega_n}+1\right]^{\pm1}$

# 1. Frequency Response

## **Bode plot techniques:** example

1. Plot each term
2. Generate the composite curve

$$KG(j\omega) = K_o \frac{j\omega\tau_1 + 1}{(j\omega)^2(j\omega\tau_a + 1)}$$

$$\angle KG(j\omega) = \angle K_o + \angle(j\omega\tau_1 + 1) - \angle(j\omega)^2 - \angle(j\omega\tau_a + 1)$$

$$\log|KG(j\omega)| = \log|K_o| + \log|j\omega\tau_1 + 1| - \log|(j\omega)^2| - \log|j\omega\tau_a + 1|.$$

$$|KG(j\omega)|_{\text{db}} = 20\log|K_o| + 20\log|j\omega\tau_1 + 1| - 20\log|(j\omega)^2| - 20\log|j\omega\tau_a + 1|.$$

# 1. Frequency Response

## Bode plot techniques:

1. $K_o(j\omega)^n$: Class 1: The magnitude plot of this term is a **straight line** with a **slope** n*(20 db per decade)

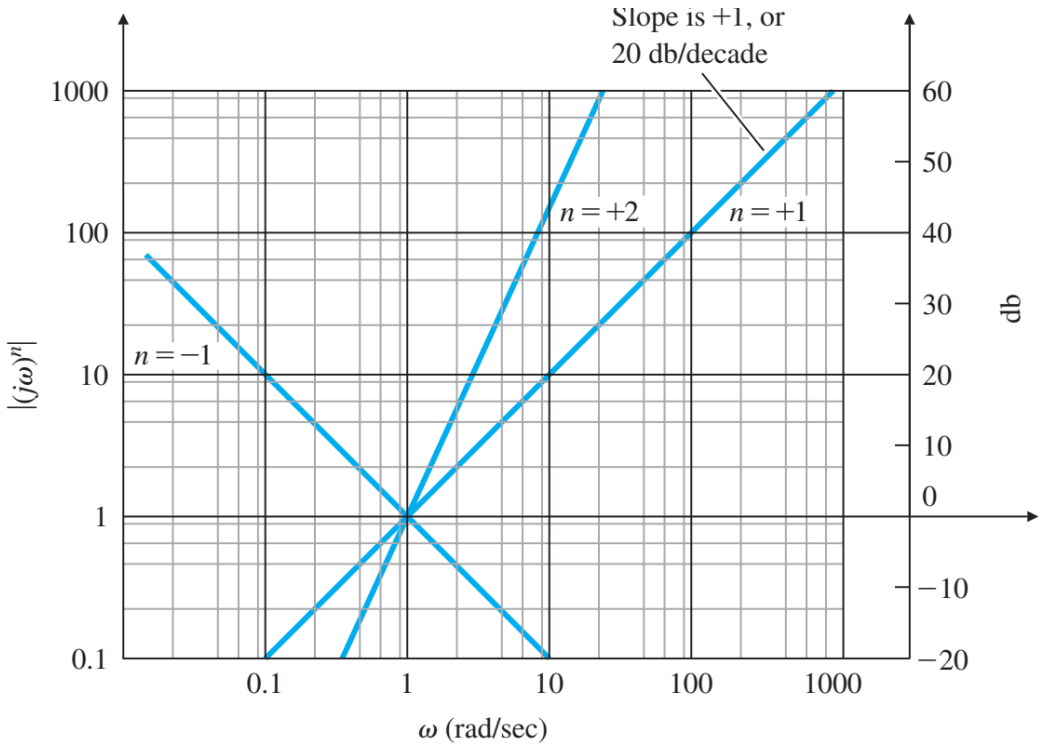
$$\log K_o|(j\omega)^n| = \log K_o + n \log |j\omega|$$

To draw the line (magnitude plot):

1. Locate ω = 1 and draw a point at $20*\log_{10}(Ko)$ [db]
2. Then draw the line with slope n*20 db/decade through that point

To draw the line (phase plot):

- Horizontal line at ϕ = n × 90∘



# 1. Frequency Response

## Bode plot techniques:

2. $(j\omega\tau + 1)$

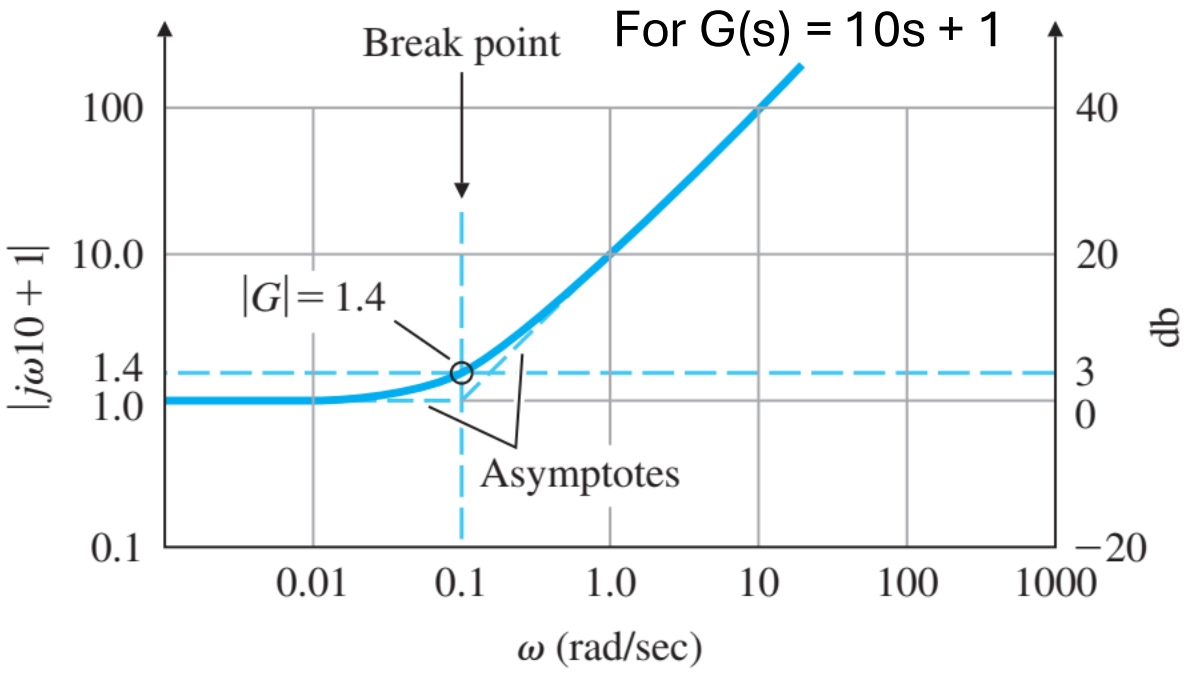
The magnitude approaches:

- one asymptote at very low frequencies
- another asymptote at very high frequencies

$$\text{For } \omega\tau \ll 1, j\omega\tau + 1 \cong 1.$$
$$\text{For } \omega\tau \gg 1, j\omega\tau + 1 \cong j\omega\tau$$

If we call ω = 1/τ the break point

- Magnitude = 1 before w = 1 /τ
- Magnitude = as class 1 (**-20 db/decade or +20 db/decade**)



# 1. Frequency Response

## Bode plot techniques:

2. $(j\omega\tau + 1)$

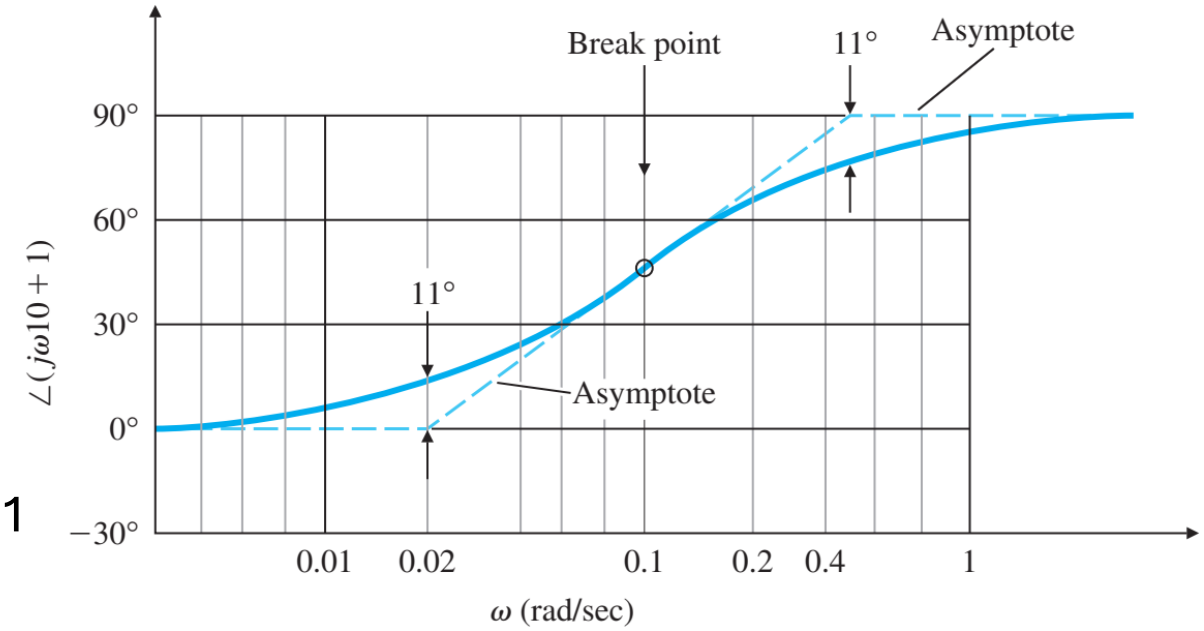
The phase asymptotes

(a) For $\omega\tau \ll 1, \angle 1 = 0°$.
(b) For $\omega\tau \gg 1, \angle j\omega\tau = 90°$.
(c) For $\omega\tau \cong 1, \angle(j\omega\tau + 1) \cong 45°$

If we call ω = 1/τ the break point

- Phase = **45° at break point**
- Asymptote going from 0° at ωτ = 0.2 to 90° at ωτ = 5

For G(s) = 10s + 1



# 1. Frequency Response

## Bode plot techniques:

3. $[(j\omega/\omega_n)^2 + 2\zeta(j\omega/\omega_n) + 1]^{\pm 1}$

The magnitude approaches:

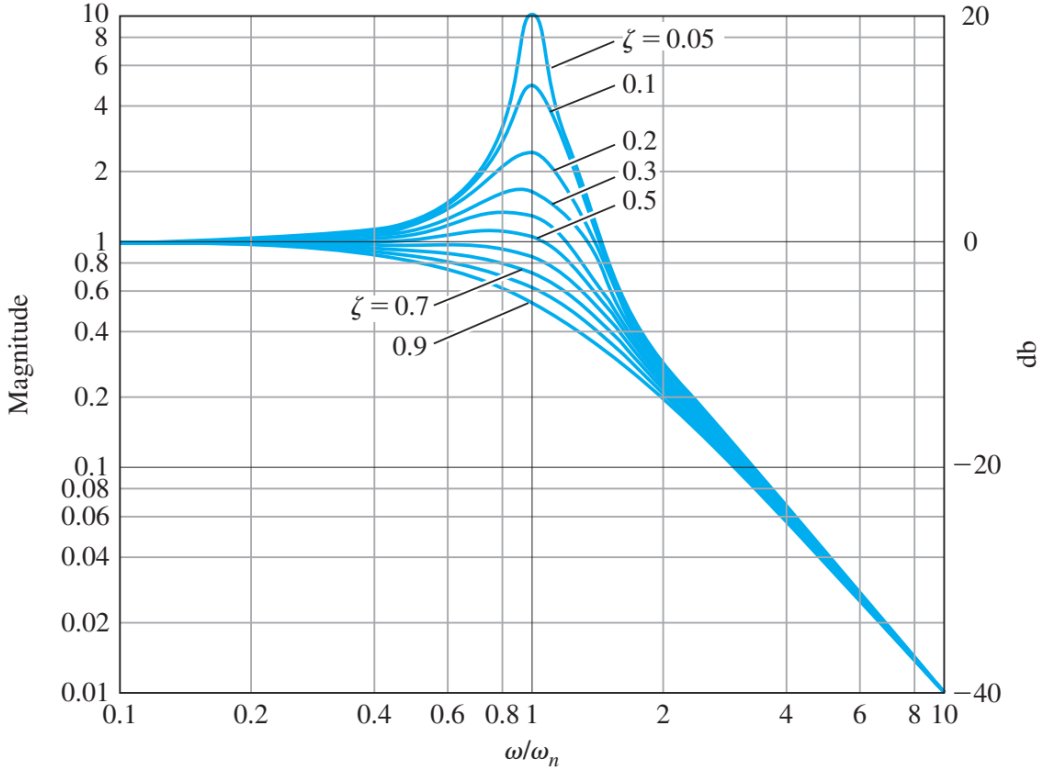
- one asymptote at very low frequencies
- another asymptote at very high frequencies
- Approximation at w = $w_n$

$$|G(j\omega)| = \frac{1}{2\zeta} \quad \text{at} \quad \omega = \omega_n$$ For poles

$$|G(j\omega)| = 2\zeta$$ For zeros

Considering $w_n$ as a key point

- Magnitude = 1 (**0 db**) before w = $w_n$
- Magnitude = **-40 db/decade or +40 db/decade**



# 1. Frequency Response

## Bode plot techniques:

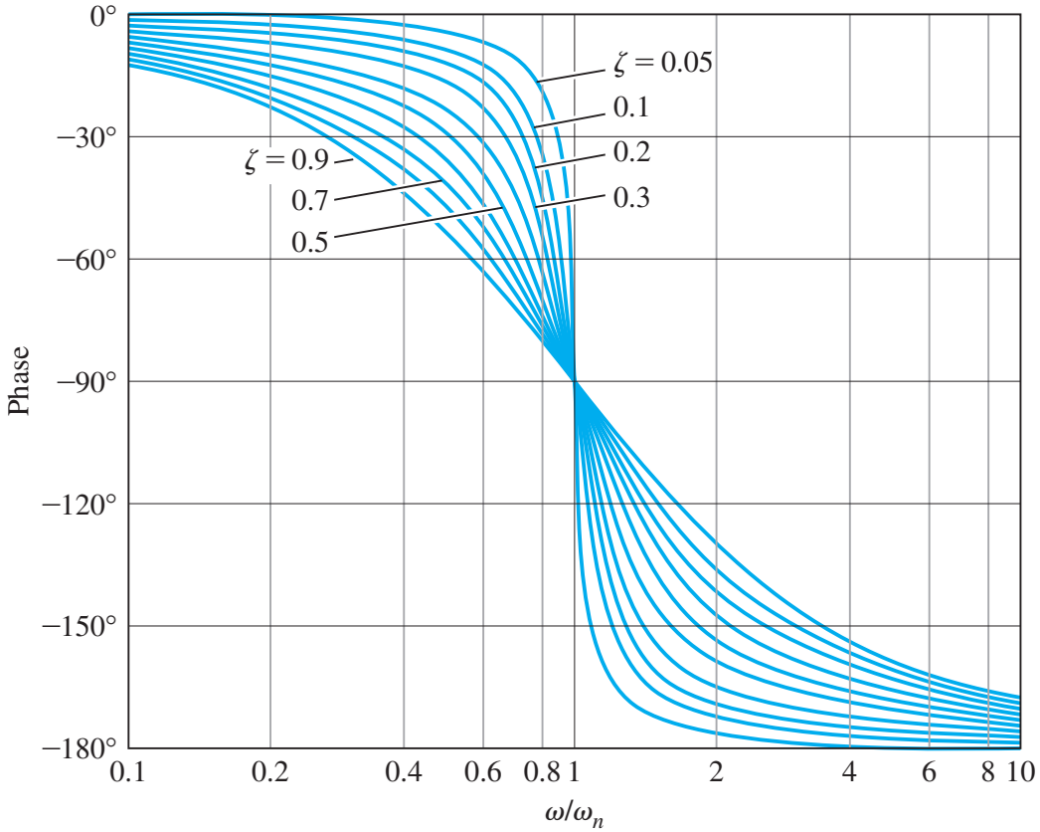
3. $[(j\omega/\omega_n)^2 + 2\zeta(j\omega/\omega_n) + 1]^{\pm 1}$

The phase asymptotes:

- 0° for w << $w_n$
- - 180° (+ 180°) for w >> $w_n$
- - 90° (+ 90°) at w = $w_n$

For sketching the transition:

- For ζ = 0: step function
- For ζ = 1: 0° at ω = 0.2*$w_n$ to -180∘ (+180°) at ω = 5*$w_n$



# 1. Frequency Response

## Bode plot techniques: summary

1. Put G(s) in the Bode form
2. Determine n for the $Ko(j\omega)^n$ and plot Ko at $\omega = 1$ with a slope of n*20 db/decade
3. Extend the asymptote until the first frequency break point. Adjust the slope:
   - Real zero/pole: +20/-20 db/decade
   - Complex zeros/poles: +40/-40 db/decade
4. Adjust the transitions at the break points:
   - Real zero/pole: +3/-3 db
   - Complex zeros (resonance): $|G(j\omega)| = 2\zeta$
   - Complex poles (resonance): $|G(j\omega)| = 1/2\zeta$

**Phase**:

1. Plot the low-frequency asymptote $\phi = n \times 90°$.
2. Extend the asymptote until the first frequency break point. Adjust the angle:
   1. Real zero/pole: +90°/-90°
   2. Complex zeros/poles: +180°/-180°
3. Approximate the symptotes for each phase for the transitions:
   - At break points real zeros/poles (from low w asymptotes): +45°/-45°
   - At $w_n$ complex zeros/poles (from low w asymptotes): +90°/-90°

# 1. Frequency Response

## Bode plot techniques: example – real poles and zeros

$$KG(s) = \frac{2000(s+0.5)}{s(s+10)(s+50)}$$

1. Bode form of G(S)

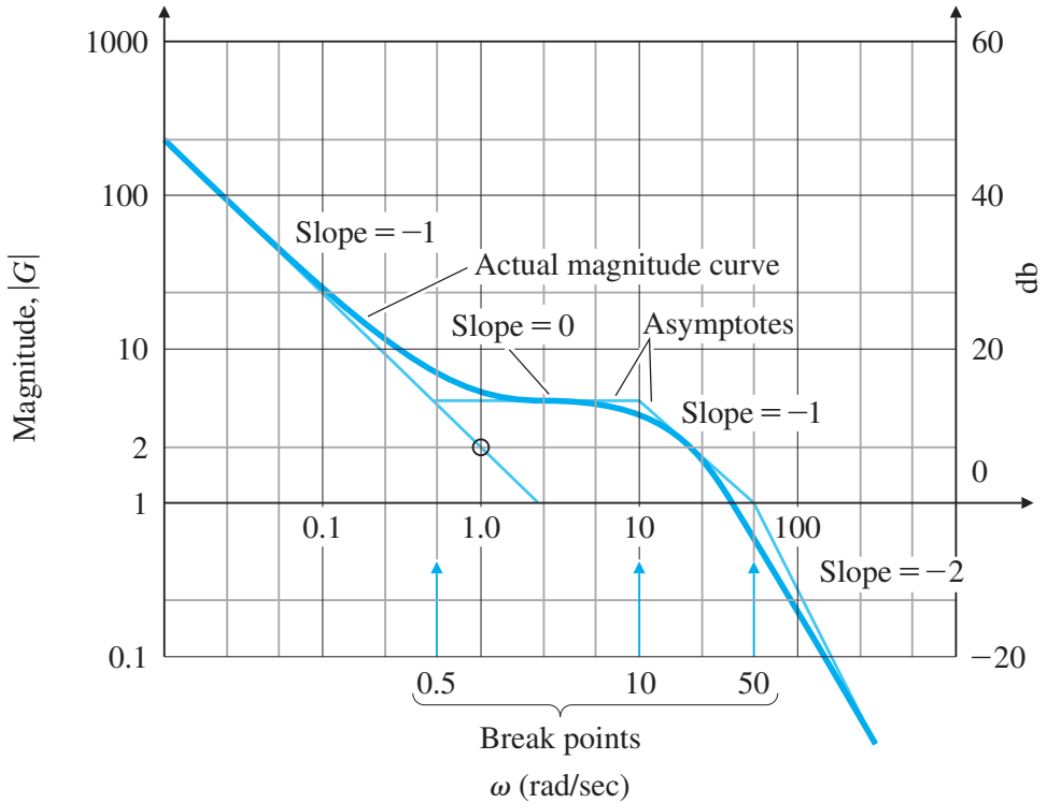
$$KG(j\omega) = \frac{2[(j\omega/0.5)+1]}{j\omega[(j\omega/10)+1][(j\omega/50)+1]}$$

2. n = -1, plot the point at w = 1 => 2 (6.0206 db), slope -20 db/decade

$$KG(j\omega) = \frac{2}{j\omega}$$

3. Break points at 0.5, 10 and 50 rad/sec

4. Adjust the transitions at the break points



# 1. Frequency Response

## Bode plot techniques: example – real poles and zeros

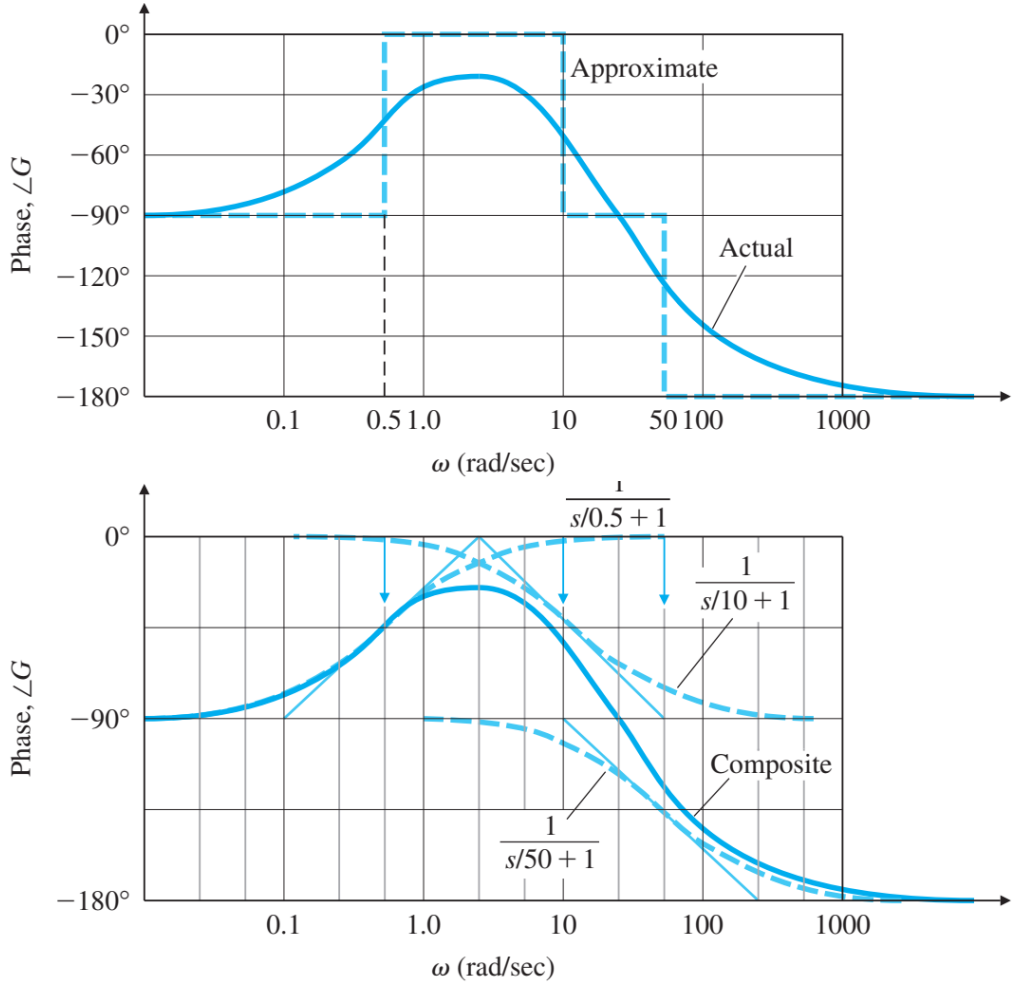
$$KG(j\omega) = \frac{2[(j\omega/0.5)+1]}{j\omega[(j\omega/10)+1][(j\omega/50)+1]}$$

5. Phase starts at -90° (n = 1)

6. Steps in the asymptotes for each pole and zero

7. Individual phase curves for each pole and zero

8. Add the curves



# 1. Frequency Response

## Bode plot techniques: example – complex poles

$$KG(s) = \frac{10}{s[s^2 + 0.4s + 4]}$$

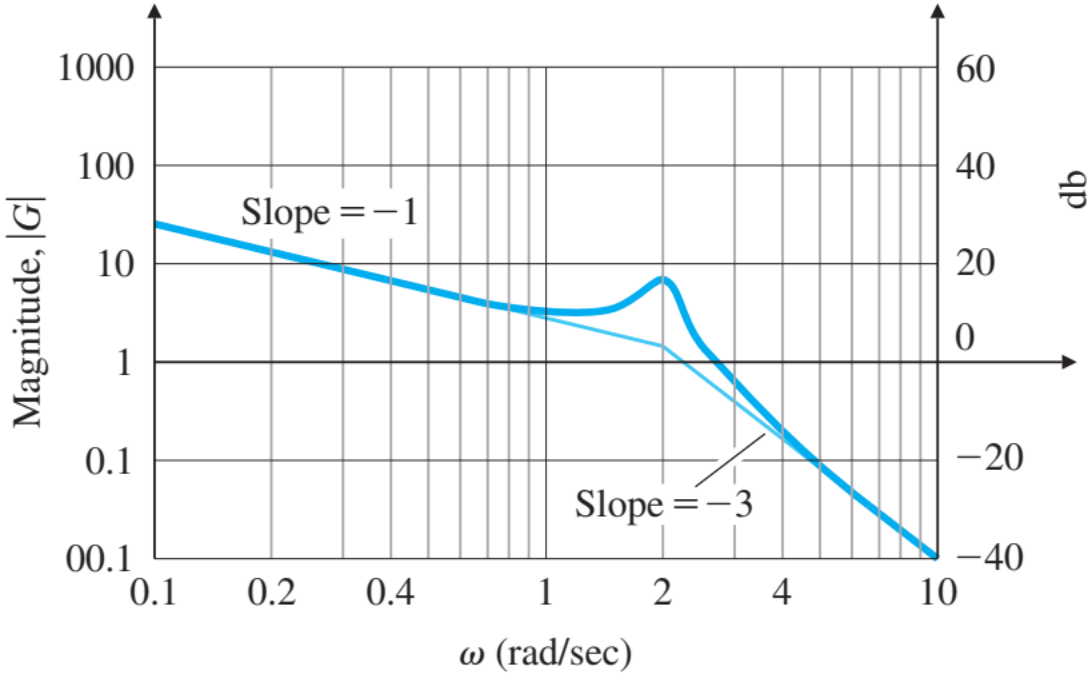
1. Bode form of G(S)

$$KG(s) = \frac{10}{4} \frac{1}{s(s^2/4 + 2(0.1)s/2 + 1)}$$

2. n = -1, plot the point at w = 1 => 2.5 (7.959 db), slope -20 db/decade

3. $w_n$ = 2 rad/sec, ζ = 0.1, -60 db/decade after $w_n$

4. Adjust the transitions at the break points: resonance 1/2ζ = 1/0.2 = 5 = 13.97 db



# 1. Frequency Response

## Bode plot techniques: example – complex poles

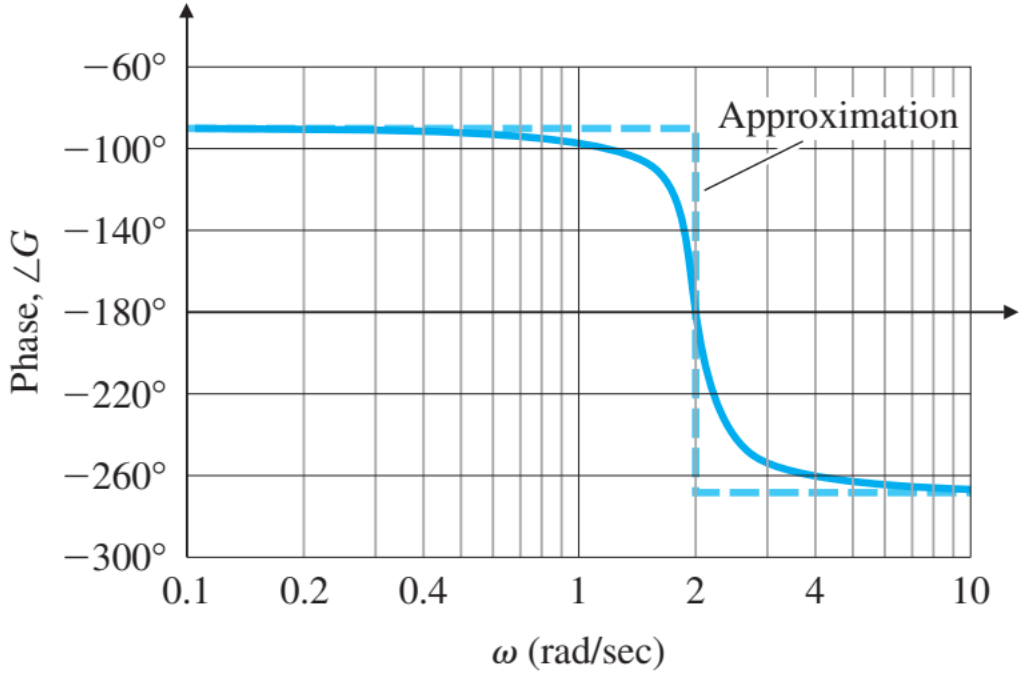
$$KG(s) = \frac{10}{s[s^2 + 0.4s + 4]}$$

5. Phase starts at -90° (n = 1)

6. Step in the asymptotes of -180°, tends to -270°

7. Individual phase curves for each pole and zero

8. Add the curves



# 1. Frequency Response

## Bode plot techniques: example – complex poles and zeros

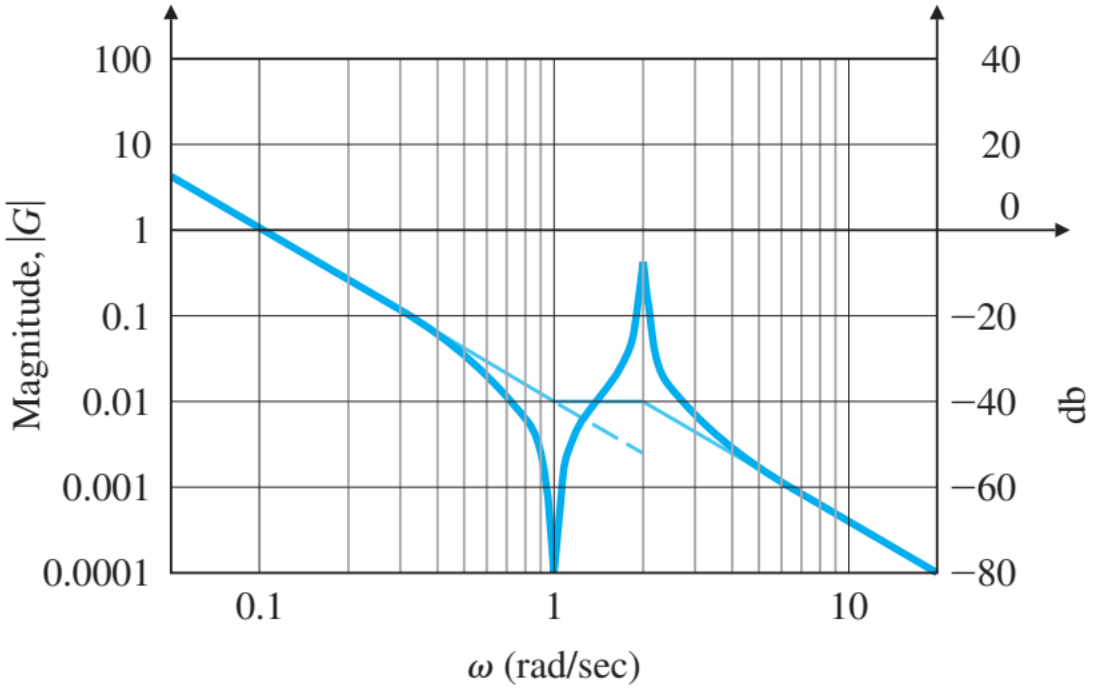
1. Bode form of G(S)

$$KG(s) = \frac{0.01(s^2 + 0.01s + 1)}{s^2[(s^2/4) + 0.02(s/2) + 1]}$$

2. n = -2, plot the point at w = 1 =>
0.01 (-40 db), slope -40 db/decade

3. Zeros: $w_n$ = 1 rad/sec, ζ = 0.005, 2ζ = 0.01, +40 db/decade after $w_n$
Poles: $w_n$ = 2 rad/sec, ζ = 0.005, 2ζ = 0.02, -40 db/decade after $w_n$

4. Adjust the transitions at the break points:
Zeros' resonance 2ζ = 0.01 = -40 db
poles' resonance 1/2ζ = 1/0.02 = +33.84 db



# 1. Frequency Response

## Bode plot techniques: example – complex poles

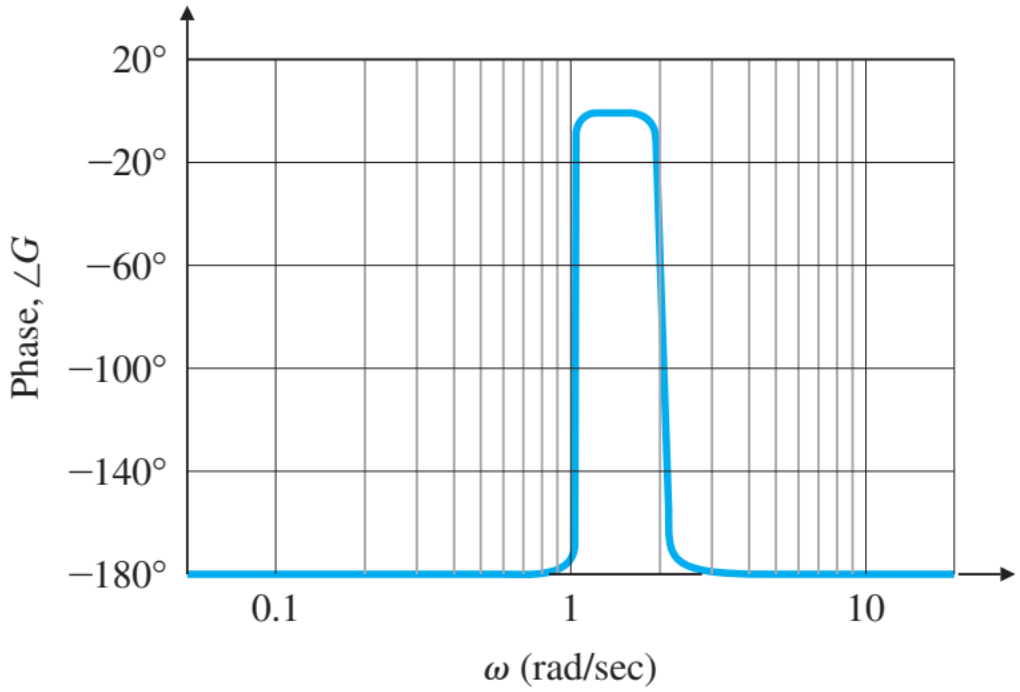
$$KG(s) = \frac{0.01(s^2 + 0.01s + 1)}{s^2[(s^2/4) + 0.02(s/2) + 1]}$$

5. Phase starts at -180° (n = 2)

6. Steps in the asymptotes of +180° at w = 1 and -180° at w = 2

7. Individual phase curves for each pole and zero

8. Add the curves



# 1. Frequency Response

## Steady-State Errors

- The larger the value of the magnitude on the low-frequency asymptote, the lower the steady-state errors will be for the closed-loop system

$$KG(j\omega) = K_o(j\omega)^n \frac{(j\omega\tau_1 + 1)(j\omega\tau_2 + 1)\cdots}{(j\omega\tau_a + 1)(j\omega\tau_b + 1)\cdots}$$

For low frequencies:

$$KG(j\omega) \cong K_o(j\omega)^n$$

Type 0 system (n = 0)
Low frequency magnitude $\mathbf{K_0}$
**$K_0 = K_p$ (position error constant)**
**Eeasily obtained from Bode plot**

$$K_p = \lim_{s\to 0} GD_{cl}(s)$$

$$e_{ss} = \frac{1}{1 + K_p}$$

# 1. Frequency Response

## Steady-State Errors

$$KG(j\omega) = K_o(j\omega)^n \frac{(j\omega\tau_1 + 1)(j\omega\tau_2 + 1)\cdots}{(j\omega\tau_a + 1)(j\omega\tau_b + 1)\cdots}$$

For low frequencies:

$$KG(j\omega) \cong K_o(j\omega)^n$$

Type 1 system (n = -1)
Low frequency magnitude **$K_0$/w**
**$K_0 = K_v$ (velocity error constant)**
**Eeasily obtained from Bode plot**
**|G(s)| at w = 1 or**

$$K_v = \lim_{s\to 0} sGD_{cl}(s), \qquad e_{ss} = \frac{1}{K_v}$$

$K_v$ can also be obtained by extending the low freq. asymptote

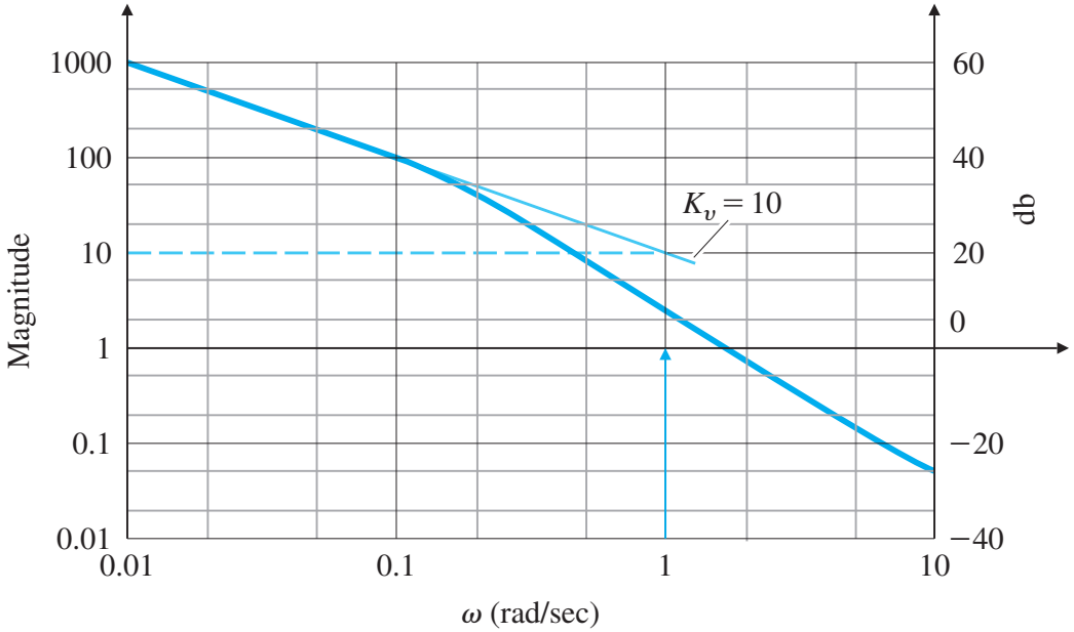
$K_v$ can also be obtained from the low freq. asymptote (A(w)) because:
$A(w) = K_v/w = K_0/w$

# 1. Frequency Response

## Steady-State Errors: example

$$KG(s) = \frac{10}{s(s+1)}$$

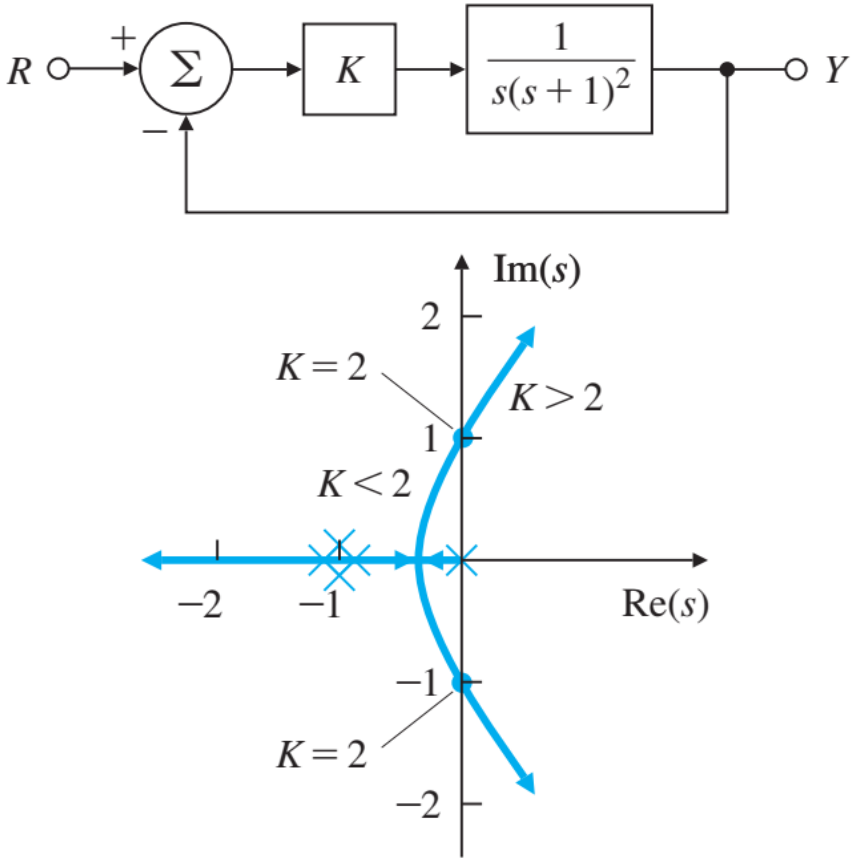
- The system is type 1
- -20 db/decade at low freq.
- Low frequency asymptote does not passes through w = 1
- 2 options to calculate Kv
  1. Extendign the low frequency asymptote to w = 1: $K_v$ = 20 db = 10
  2. Evaluating the low frequency asymptote A(w) at w = 0.1



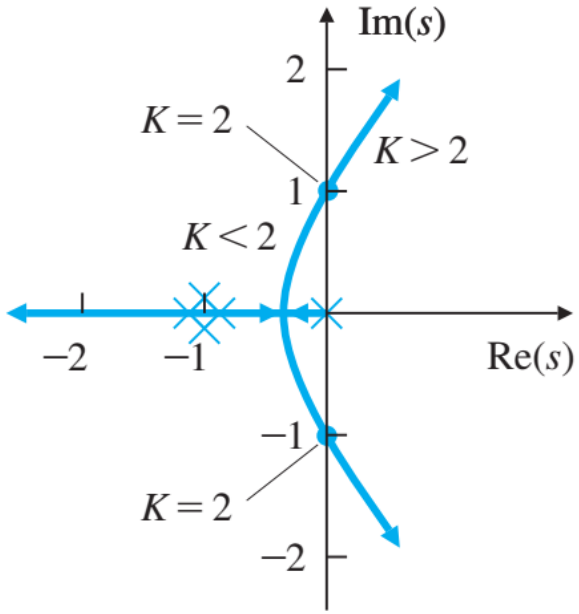

$$K_o = K_v \cong \omega|A(\omega)| = 0.01(1000) = 10$$

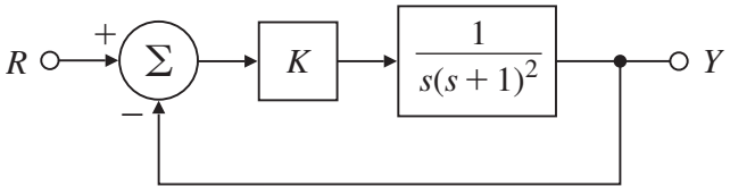
# 2. Neutral Stability

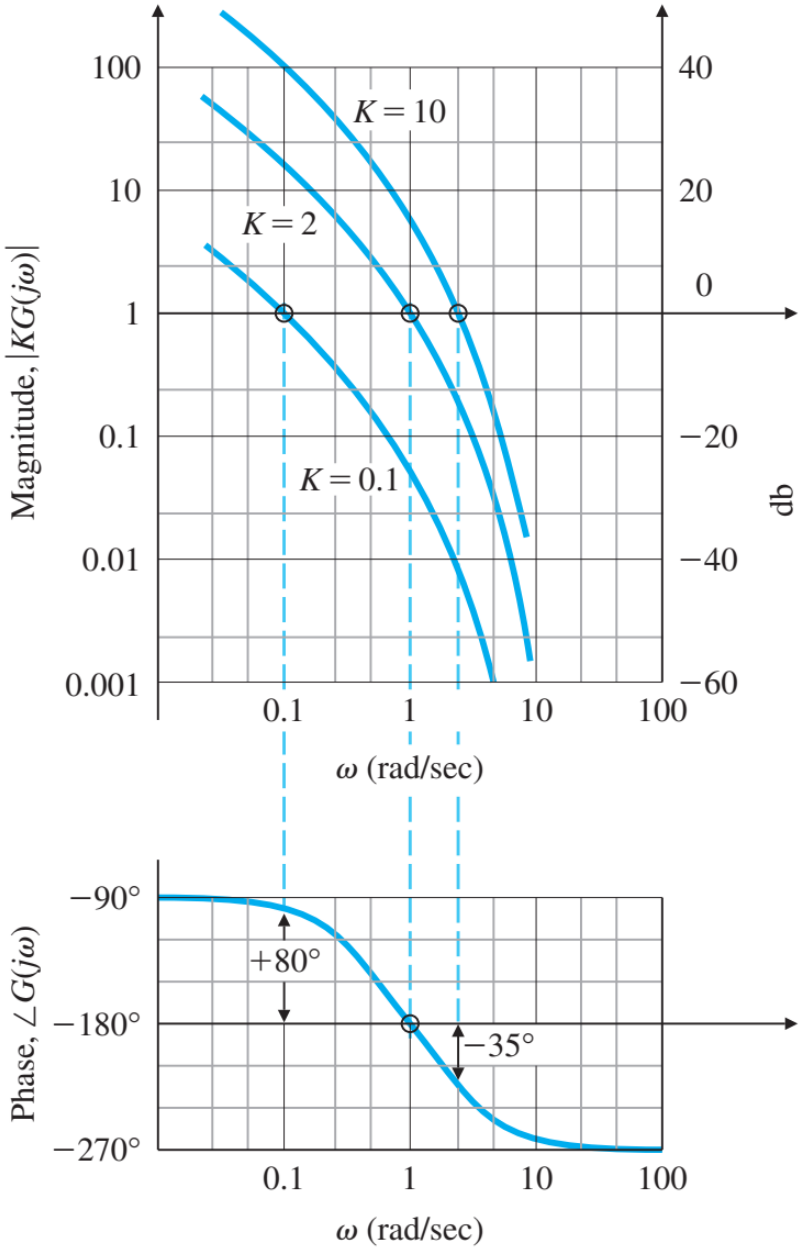
- It is possible to determine **closed-loop stability** by evaluating the **frequency response** of the **open-loop transfer function** KG(jω)
- It is usefull because the **tranfer function** is **not always known**
- Example



# 2. Neutral Stability




Stability condition when:

$|KG(j\omega)| <$ 0 db at $\angle G(j\omega) = -180°$ Increasing K leads to **instability**

$|KG(j\omega)| >$ 0 db at $\angle G(j\omega) = -180°$ Increasing K leads to **stability**

# 3. The Nyquist Stability Criterion

- It is useful when the Neutral Stability analysis is not enough
- A more general criterion to determine stability
- Relates the **open-loop frequency response** to the number of **closed-loop poles** of the system in the **RHP**
- Also applies when there are multiple crosees of
  - $|KG(s)| = 0$ db
  - $\angle G(s) = 180°$

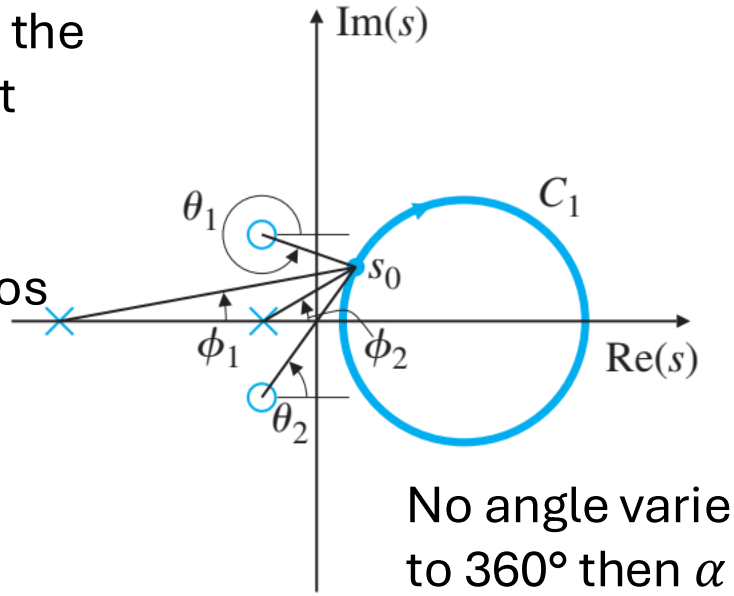
# 3. The Nyquist Stability Criterion

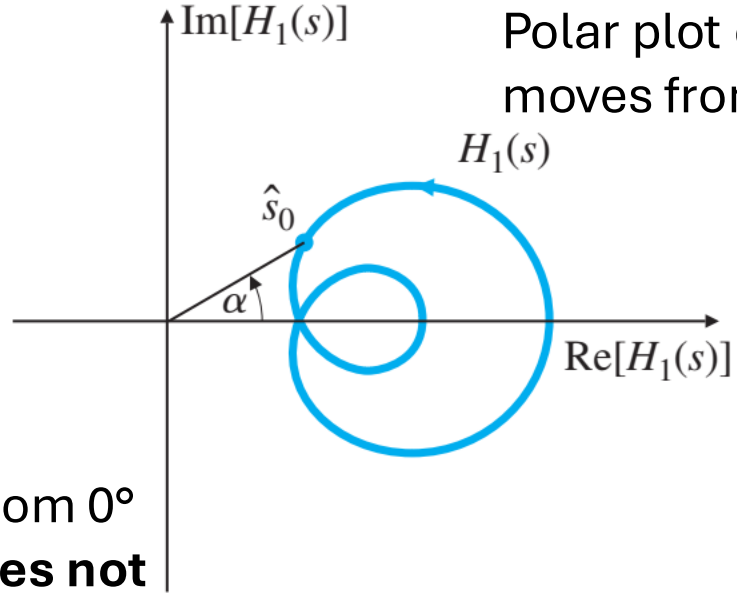
## The Argument Principle

$$H_1(s_o) = \vec{v} = |\vec{v}|e^{j\alpha} \qquad \alpha = \theta_1 + \theta_2 - (\phi_1 + \phi_2)$$

Contour **$C_1$** in the imaginary plot

Open-loop poles and zeros




Polar plot of $H_1(s)$ as **s** moves from **$s_0$** trough **$C_1$**

No angle varies from 0° to 360° then $\alpha$ **does not** encircles the origin

# 3. The Nyquist Stability Criterion

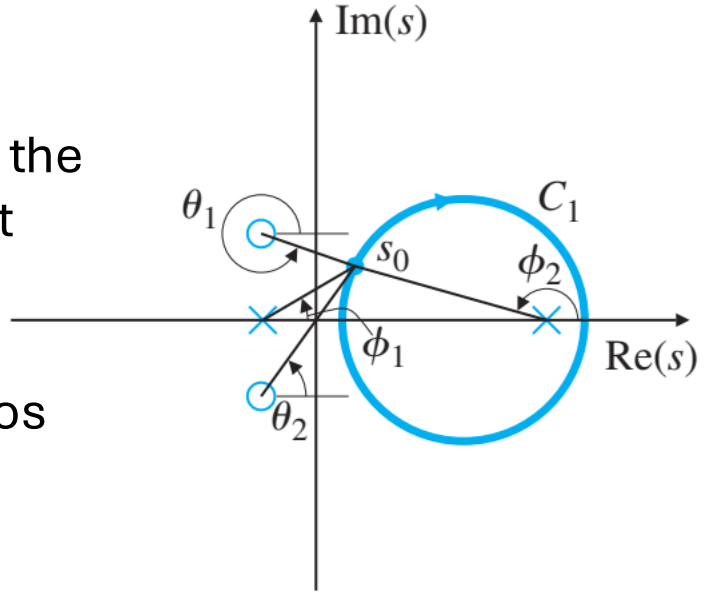
## The Argument Principle

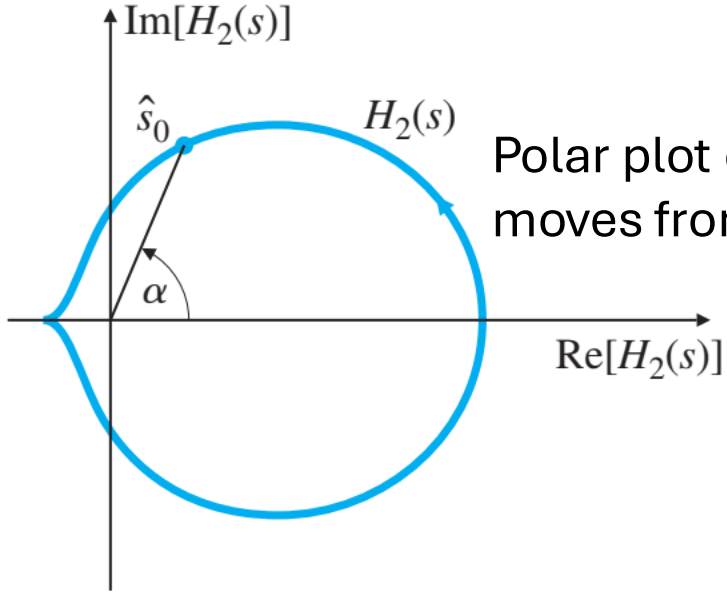
$H_2(s)$

$$\alpha = \theta_1 + \theta_2 - (\phi_1 + \phi_2)$$

Contour **$C_1$** in the imaginary plot

Open-loop poles and zeros




Polar plot of $H_2(s)$ as **s** moves from **$s_0$** trough **$C_1$**

$\phi_2$ varies from 0° to 360° then $\alpha$ encircles the origin

# 3. The Nyquist Stability Criterion

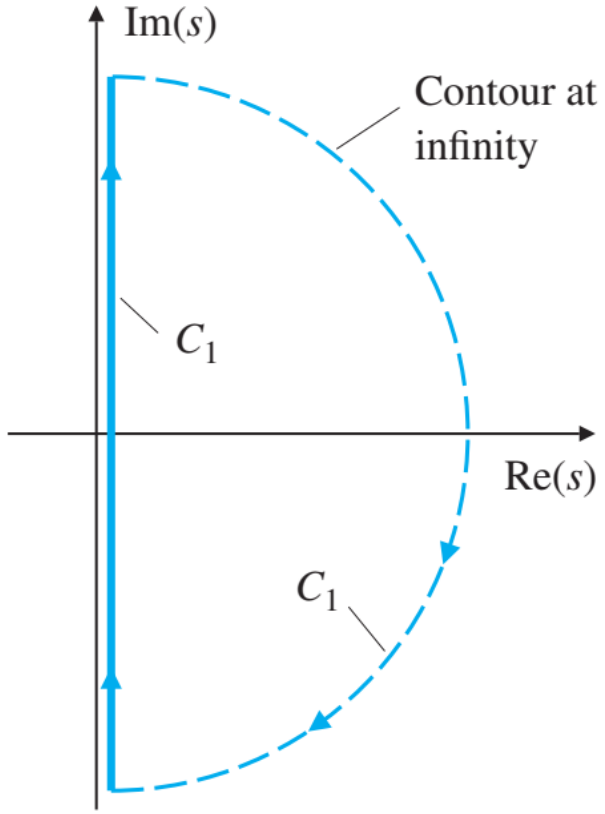
## The Argument Principle

A contour map of a complex function will encircle the origin $Z-P$ times, where $Z$ is the number of zeros and $P$ is the number of poles of the function inside the contour.

# 3. The Nyquist Stability Criterion

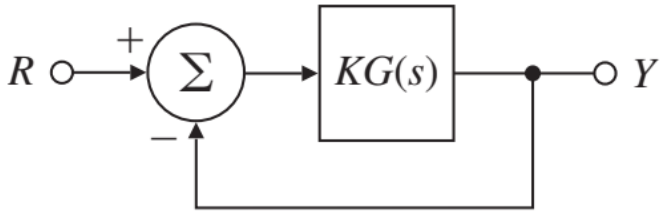
## Application of the Argument Principle to Control Design

H(s) (polar plot) will encircle the origin only if H(s) has an RHP pole or zero



# 3. The Nyquist Stability Criterion

## Application of the Argument Principle to Control Design



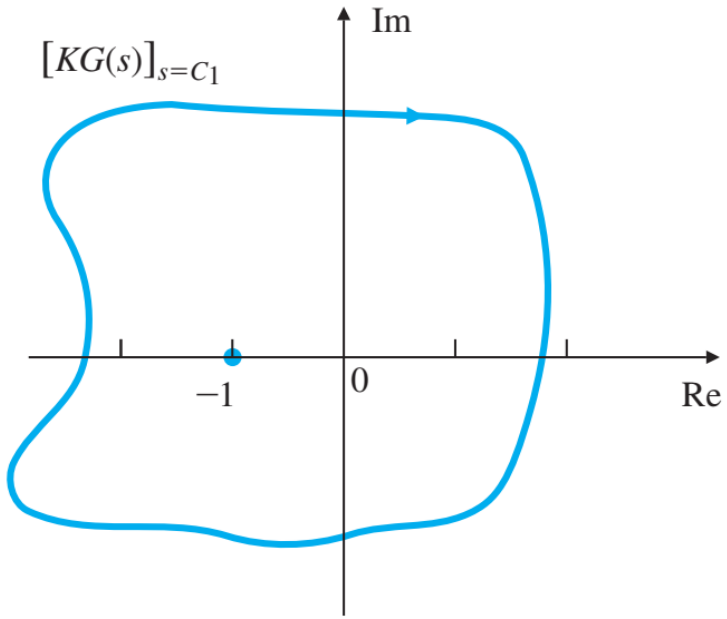

$$\frac{Y(s)}{R(s)} = \mathcal{T}(s) = \frac{KG(s)}{1 + KG(s)}$$

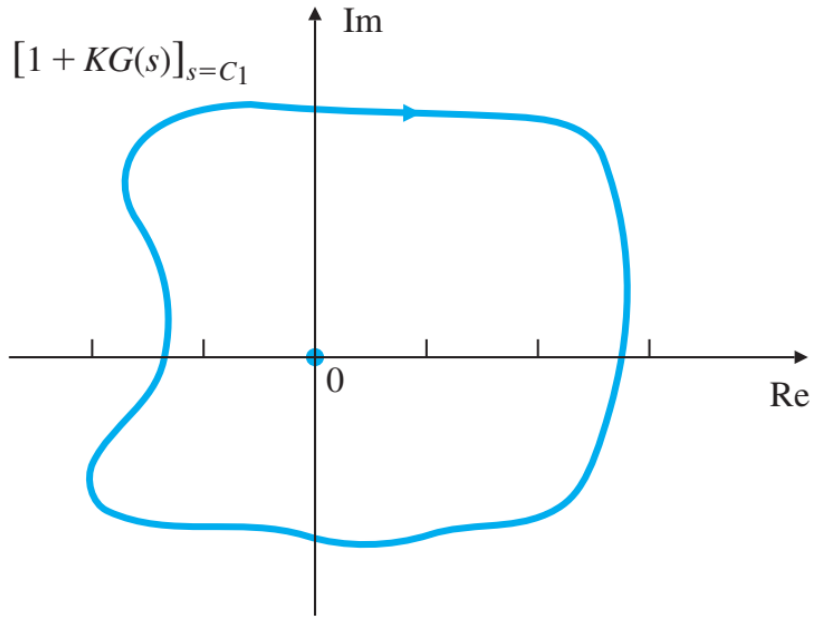
$$1 + KG(s) = 0.$$

- Applying the principle of the argument to **1 + KG(s)**, the contour encircles the **point (0,0)** if it has RHP poles or zeros
- The contour of 1 + KG(s) is the contour of KG(s) shifted 1 unit to the right
- Applying the principle of the argument to **KG(s)**, the contour encircles the **point (-1,0)** if it has RHP poles or zeros
- The contour of KG(s) is called **Nyquist plot**, or **polar plot**

# 3. The Nyquist Stability Criterion

## Application of the Argument Principle to Control Design



# 3. The Nyquist Stability Criterion

**Application of the Argument Principle to Control Design**

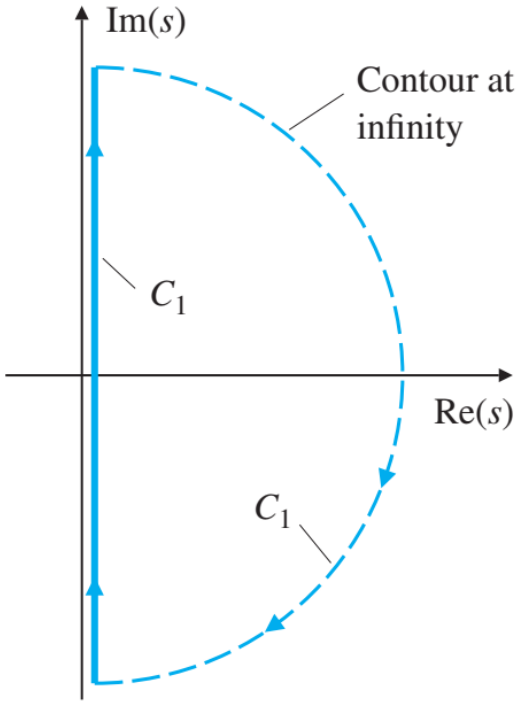
- If C1 encloses a RHP zero of 1 + KG(s) => KG(s) encircles the −1 clockwise
- If C1 encloses a RHP pole of 1 + KG(s) => KG(s) encircles the −1 counterclockwise
- The net number of clockwise encirclements (N), equals the number of RHP zeros (Z) minus the number of RHP poles (P)

$$N = Z - P$$

# 3. The Nyquist Stability Criterion

## Application of the Argument Principle to Control Design

- The contour can be generated from the frequency response
- For physical systems usually there are more poles than zeros => output tend to 0 when s -> ∞
  - **When num. Zeros = num. Poles => the arc at infinity has to be evaluated (dashed line)**
- The contour can be calculated from the response for s varying from −j∞ to +j∞ (i.e. the frequency response 0< s < +j∞)



# 3. The Nyquist Stability Criterion

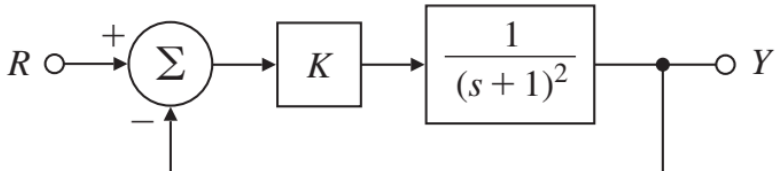
**Procedure for determining Nyquist Stability**

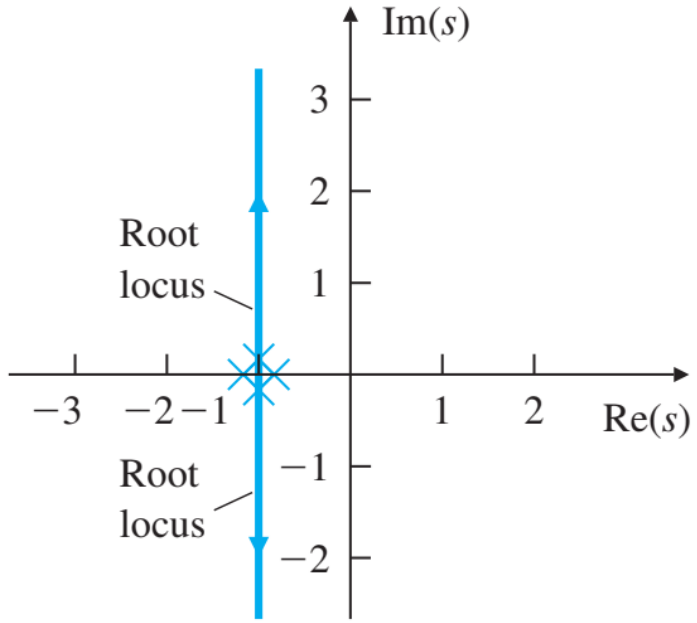
1. Generate the Nyquist plot KG(s): evaluate KG(jw) for $0 < w < w_h$ up to the values tend to 0 (regarding low w values). Reflect the plot about the real axis.
2. Evaluate the number of **clockwise encirclements** of -1 (i.e. **N**): draw a **stright line in any direction from -1 to** ∞; count the net number of **left-to-right crossings** of the straight line. Encirclements are closewise => N > 0
3. Determine the number of **unstable (RHP) poles of KG(s)**, and call that number **P**
4. Calculate the number of **unstable closed-loop roots Z**

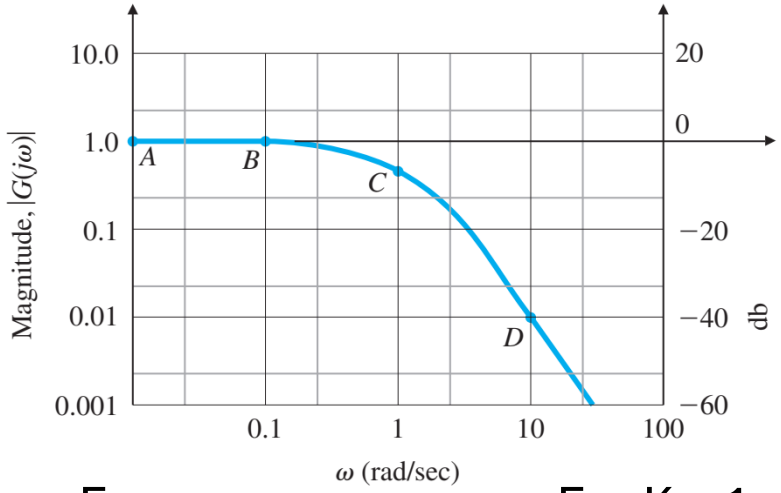
$$Z = N + P.$$

# 3. The Nyquist Stability Criterion

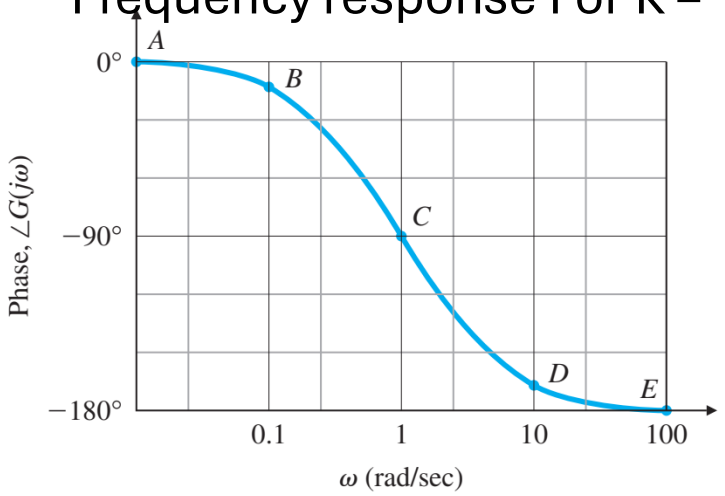
Example: 2nd order system



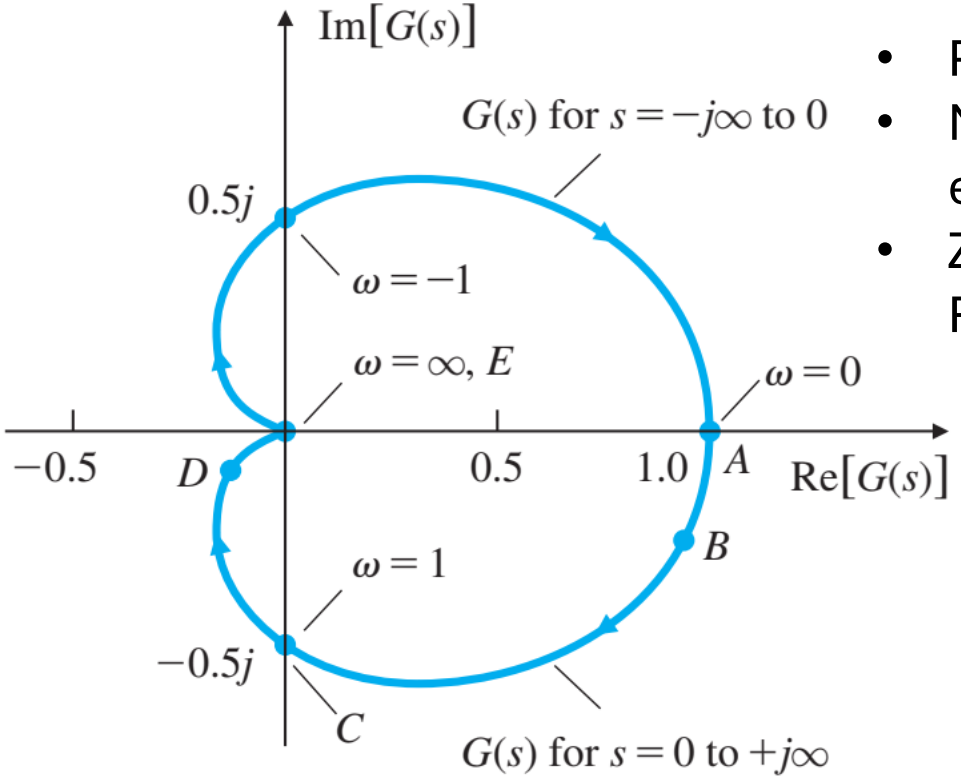




Frequency response For K = 1

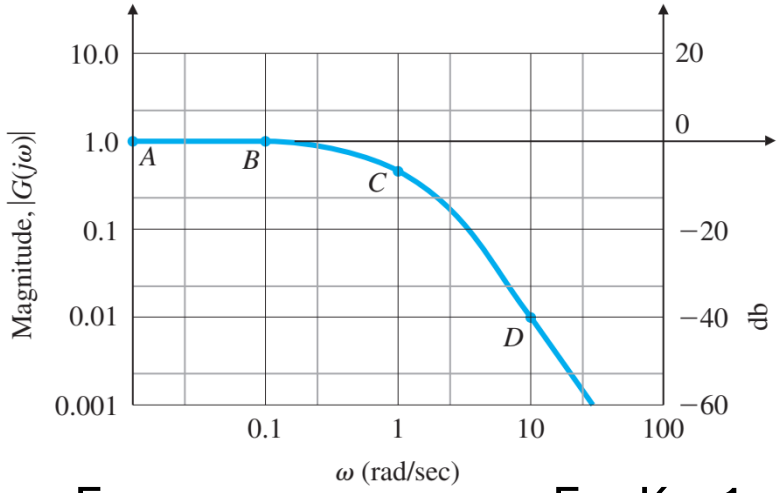


# 3. The Nyquist Stability Criterion

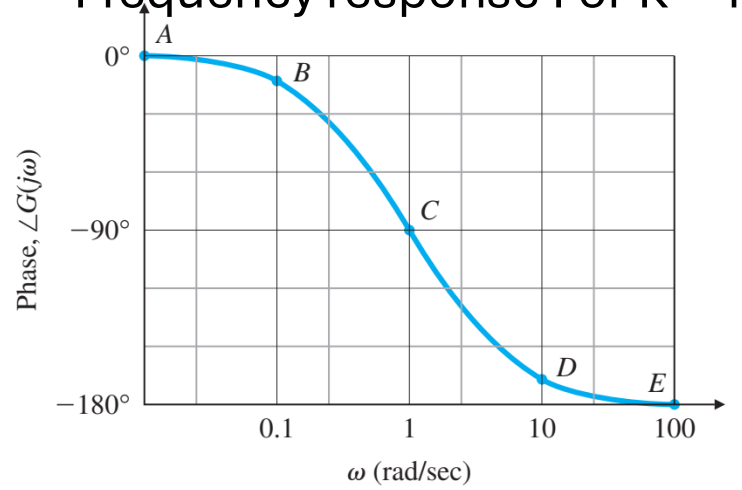
Example: 2nd order system




- P = 0 (no RHP poles)
- N = 0 (no encirclements of -1)
- Z = N + P = 0 (no RHP zeros)



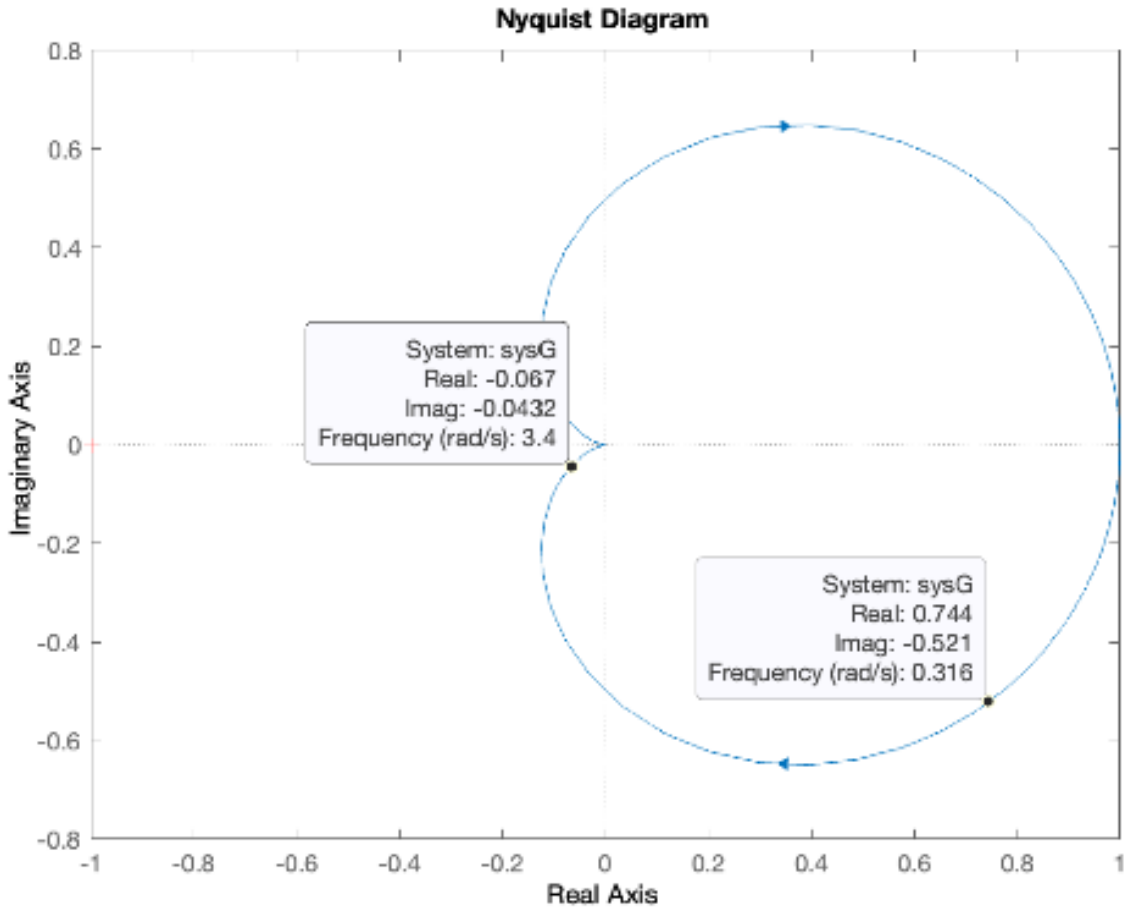

Frequency response For K = 1



# 3. The Nyquist Stability Criterion

Example: 2nd order system

```
s = tf('s');
sysG = 1/(s+1)^2;
nyquist(sysG);
```



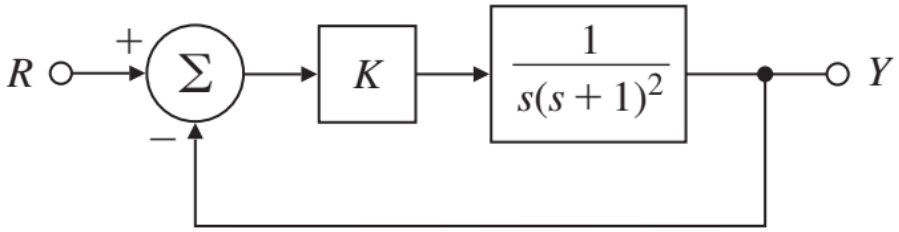
# 3. The Nyquist Stability Criterion

Nyquist stability for a range of gains K

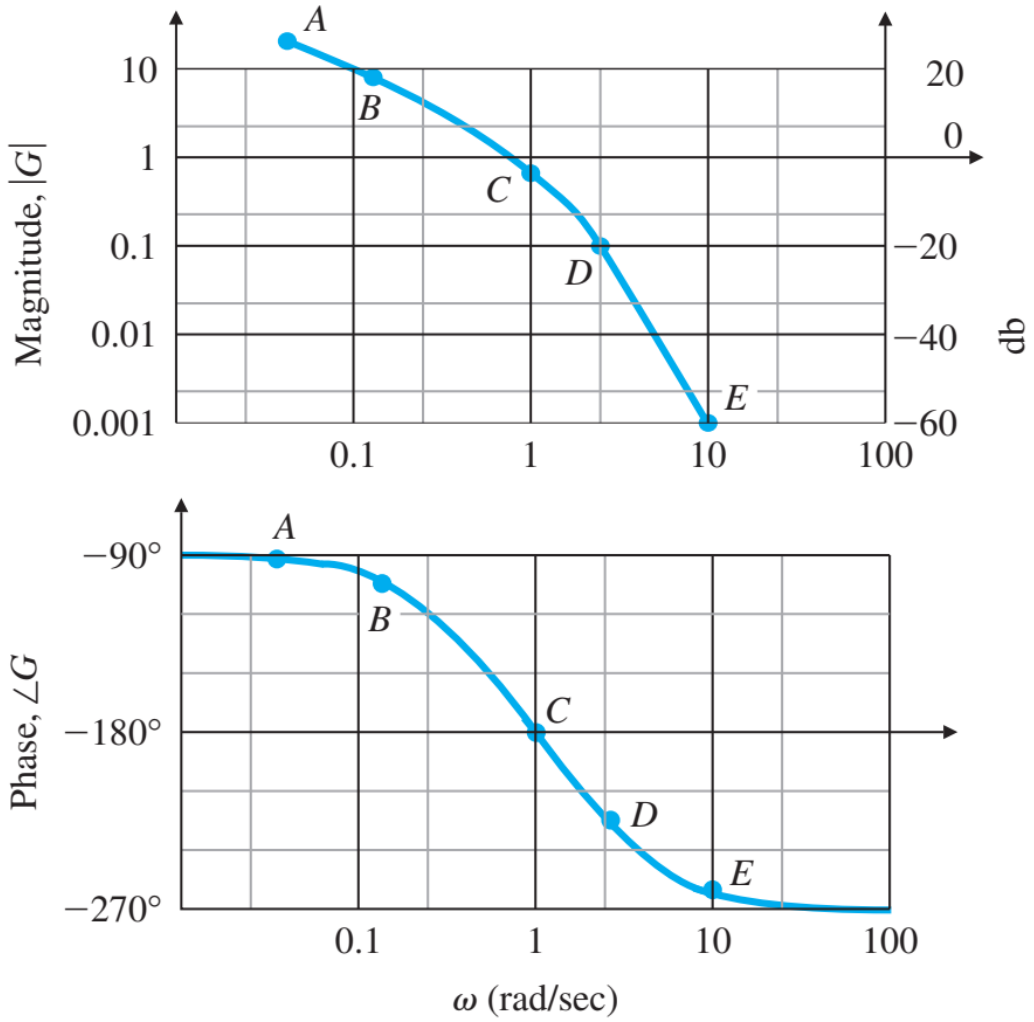
- An encirclement of −1 by KG(s) is equivalent to an encirclement of −1/K by G(s)
- It is necessary consider only:
  - The Nyquist plot of G(s)
  - The number of the encirclements of the −1/K point
- We are evaluating the stability for a range of K values

# 3. The Nyquist Stability Criterion

Example: 3rd order system



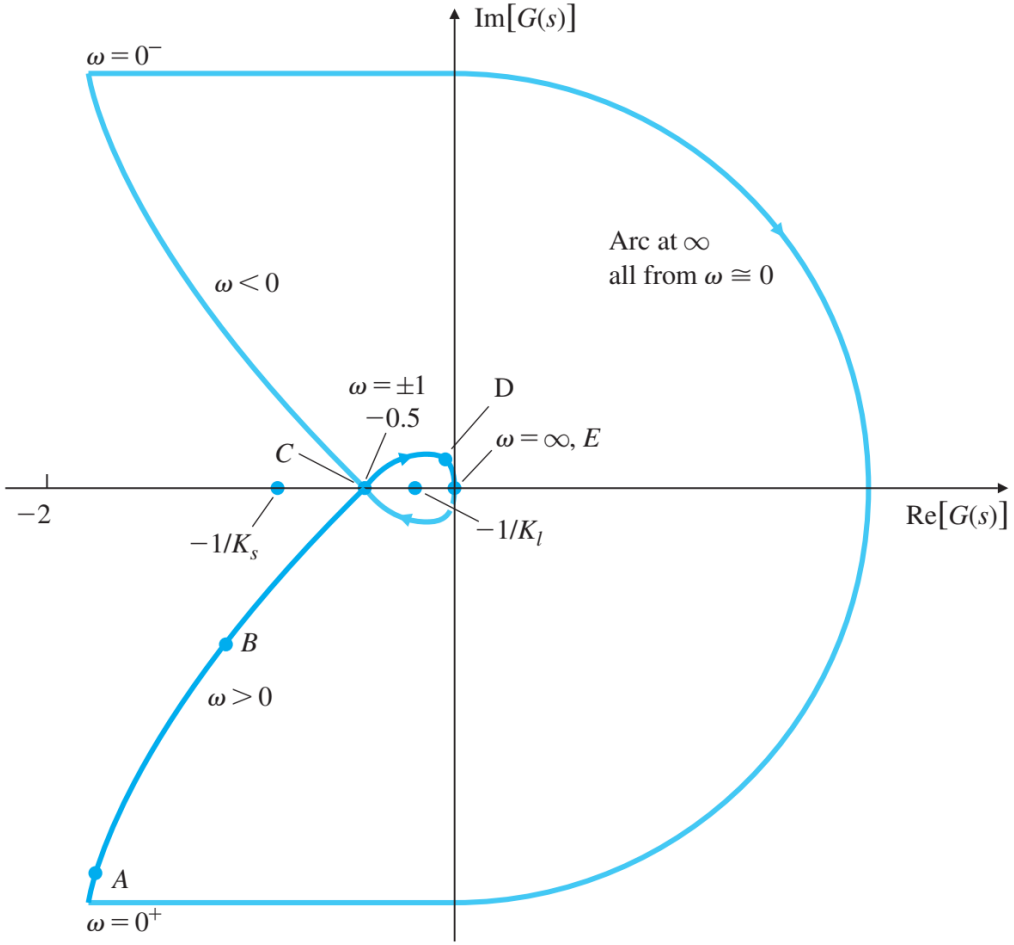

Determine its stability properties using the Nyquist criterion



# 3. The Nyquist Stability Criterion

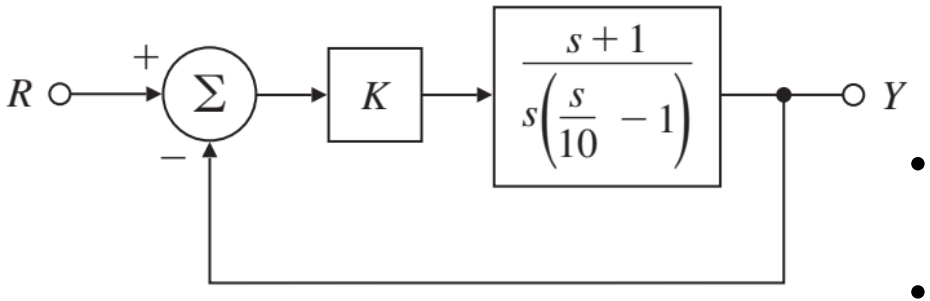
Example: 3rd order system

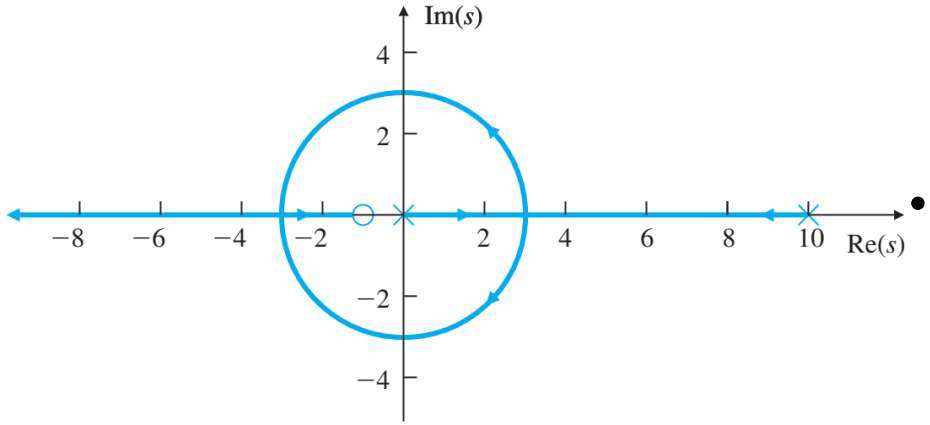
- Note the large arc at infinity that arises from the pole at s = 0
- Any pole on the imaginary axis produe an arc at infinity
- The Nyquist plot crosses the real axis at ω = 1 with |G| = 0.5(Bode plot)
- For large values of K: −0.5 < −1/Kl < 0 will lie inside the two loops (N=2, P=0 => Z = 2)
- For small values of K: 1/Ks lies outside the two loops (N=0, P=0 => Z = 0)



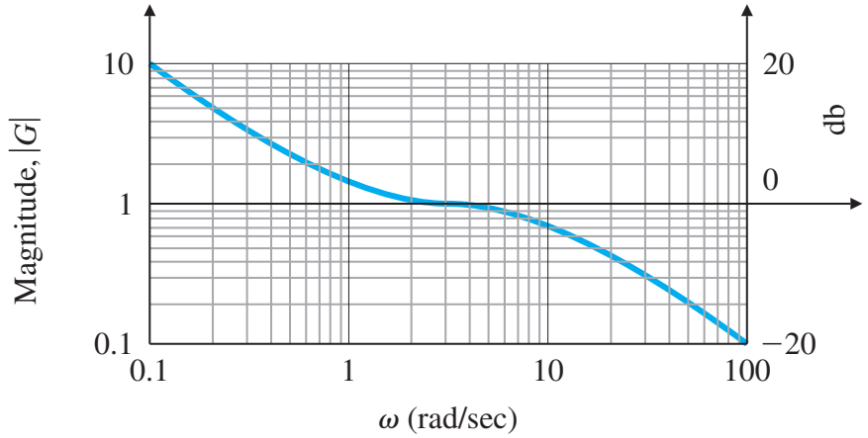
# 3. The Nyquist Stability Criterion

Example: Unstable system

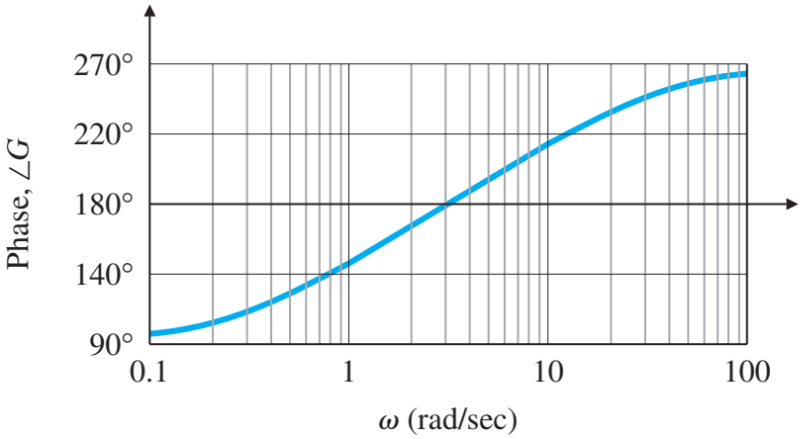



- Magnitude: as a pole in 10 rad/s
- **Phase: increment of phase instead of reducction**
- Not possible to obtain the freq. Response experimentally

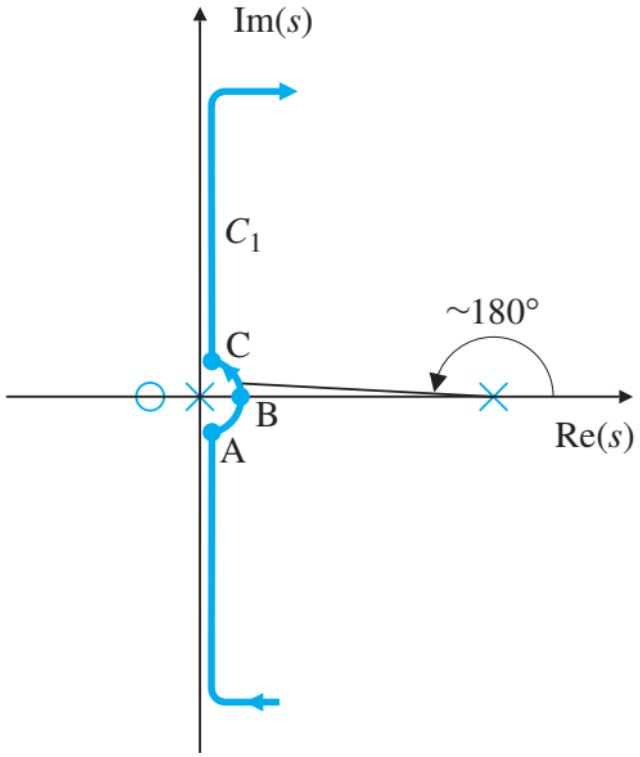



(a)

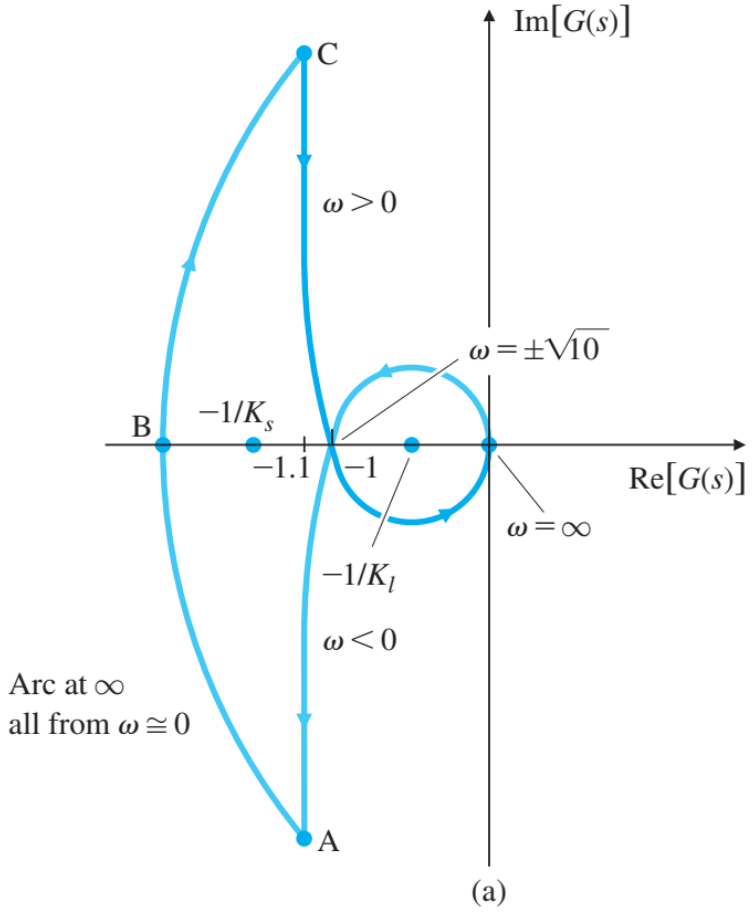


# 3. The Nyquist Stability Criterion

## Example: Unstable system




- For K > 0 and high:
  - N = -1 (counterclockwise encirclement of -1/Kl)
  - P = 1 (RHP pole)
  - Z = -1 + 1 = 0 unstable roots
- For K > 0 and small:
  - N = 1 (clockwise encirclement of -1/Ks)
  - P = 1 (RHP pole)
  - Z = 1 + 1 = 2 unstable roots

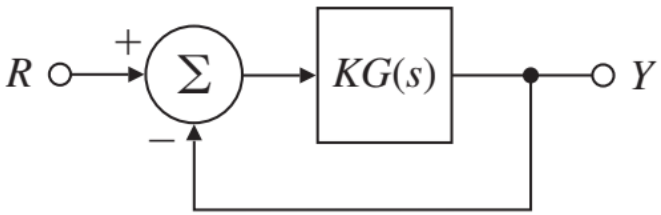


# 3. The Nyquist Stability Criterion

Example: Nyquist Plot Characteristics

- Determine whether the system is stable for all positive values of K.

$$G(s) = \frac{s^2 + 3}{(s+1)^2}$$



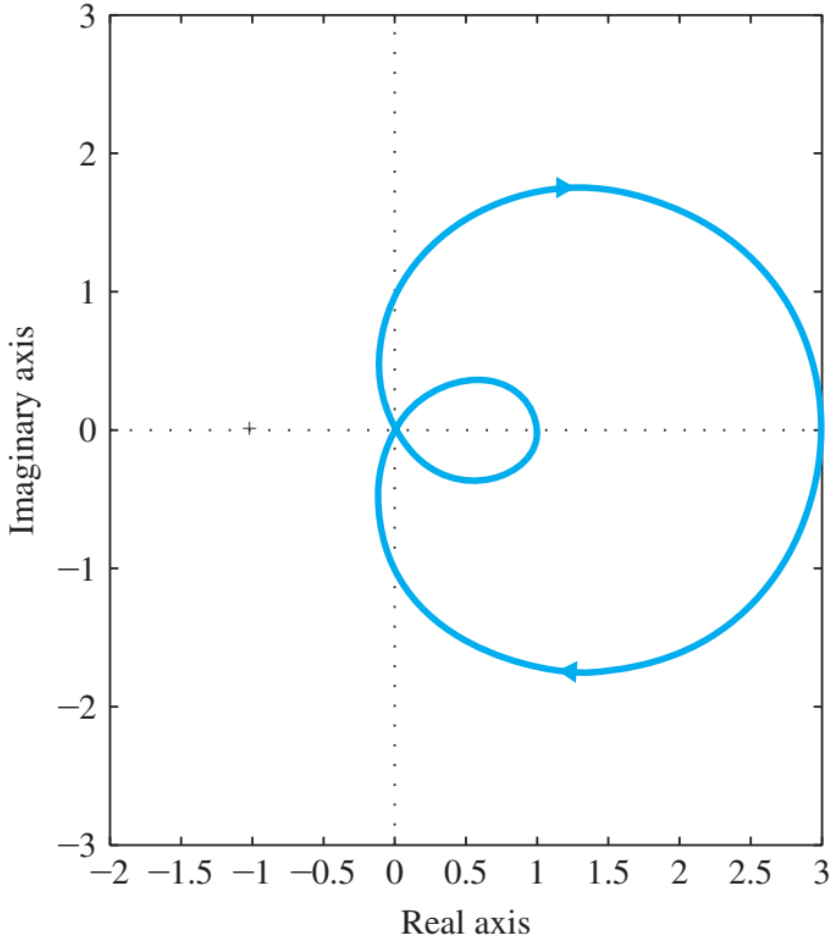

```
s = tf('s');
sysG = (s^2 + 3)/(s + 1)^2;
nyquist(sysG);
axis([-2 3 -3 3]);
```

# 3. The Nyquist Stability Criterion

## Example: Nyquist Plot Characteristics

$$G(s) = \frac{s^2 + 3}{(s+1)^2}$$

- Start at (3,0) and end at (1,0)
- Phase initially decreases due to the efect of the poles
- No poles/zeros at the origin: begin and end with 0°
- Angle jump close to zero (-120° to 60°)
- Changing K will increase or decrease the magnitude of the Nyquist plot but it can never cross the negative-real axis



# 4. Stability Margins

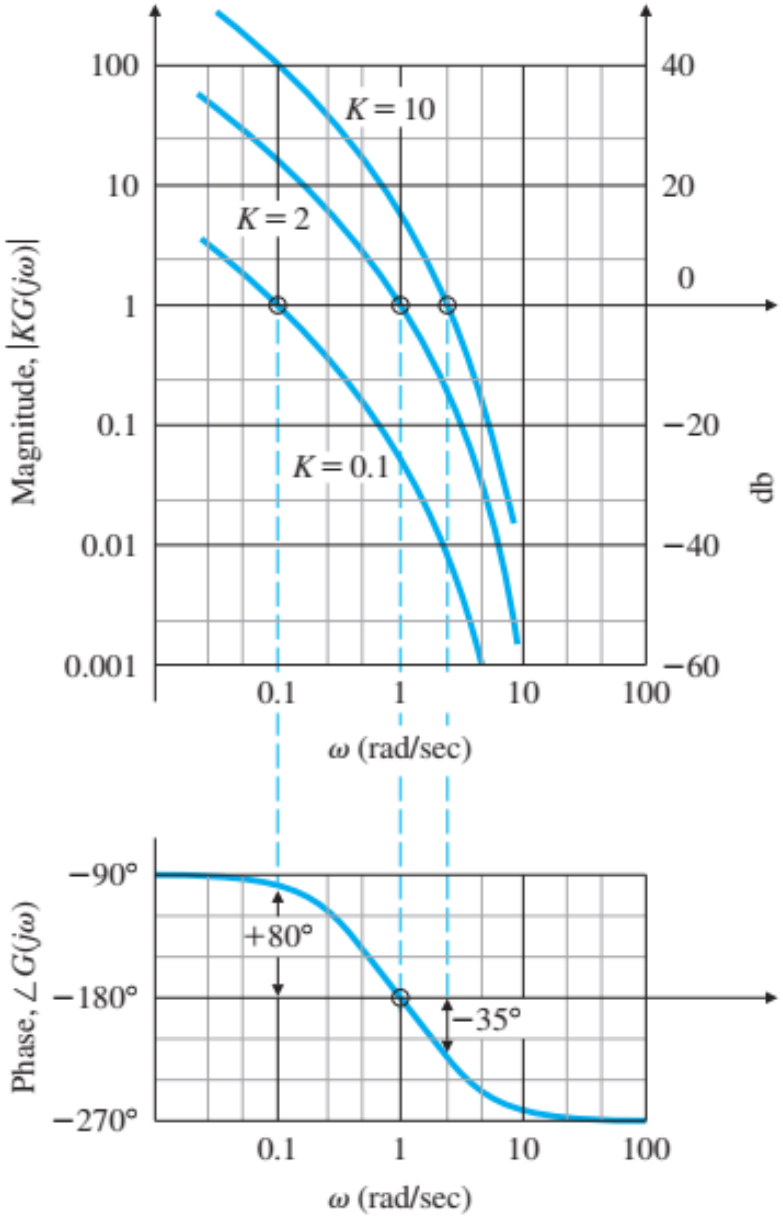
- The **gain margin** (GM): GM = - $|KG(j\omega_y)|$ where $\omega_y$ is the frequency for $\angle KG(j\omega_y) = -180°$
- **$|GM| < 0$ db (or $< 1$) => unstable system**

- The **phase margin** (PM): PM = $\angle KG(j\omega_x) + 180°$ where $\omega_x$ is the frequency for $|G(j\omega)| = 0$ db (or $= 1$)
- **PM < 0° => unstable system**

# 4. Stability Margins

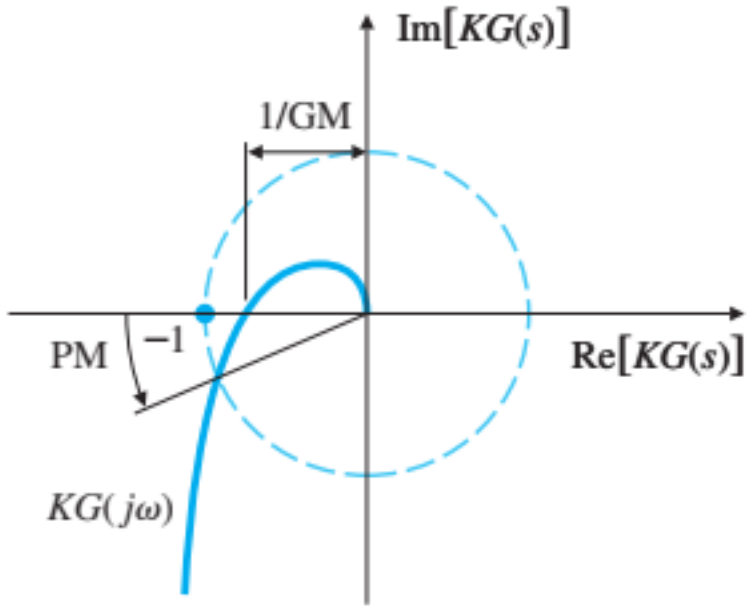
- The **gain margin** (GM): GM = - $|KG(j\omega_y)|$ where $\omega_y$ is the frequency for $\angle KG(j\omega_y) = -180°$
- **$|GM| < 0$ db (or $< 1$) => unstable system**

- The **phase margin** (PM): PM = $\angle KG(j\omega_x) + 180°$ where $\omega_x$ is the frequency for $|G(j\omega)| = 0$ db (or $= 1$)
- **PM < 0° => unstable system**



# 4. Stability Margins

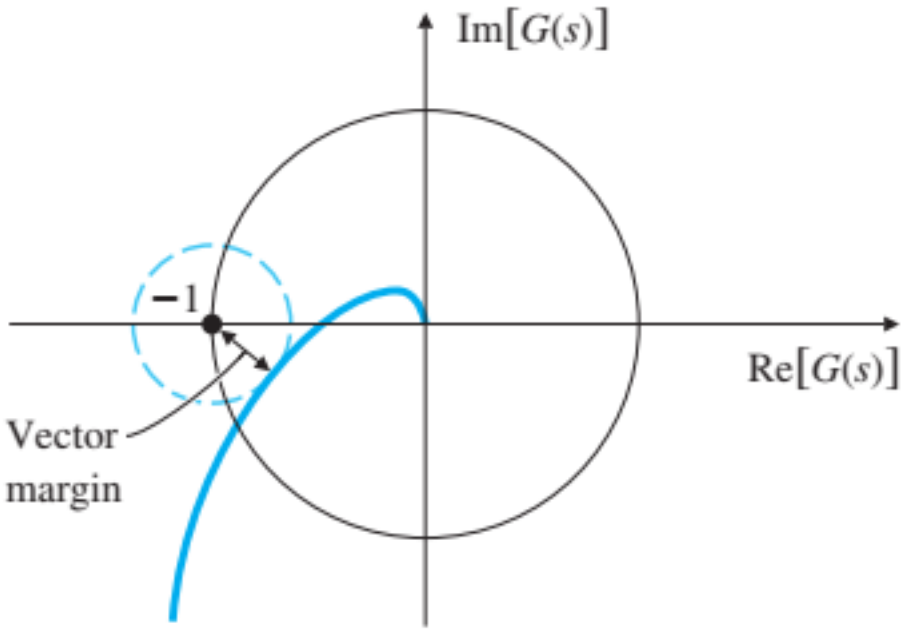
- **GM** and **PM** can be calculated from the Nyquist plot
- **1/GM**: the distance between the cross of the negative real axis and the origin
- **PM**: the angle between the negative real axis and the cross of the nyquist plot and the circle of radious 1



# 4. Stability Margins

**Vector Margin (or Complex Margin)**

- It is the distance to the −1 point from the closest approach of the Nyquist plot
- Calculated with the computational tools
- Usefull when there are multiple crossings through |KG(jw)| = 0 db (= 1)



# 4. Stability Margins

## Relationship between PM and time response

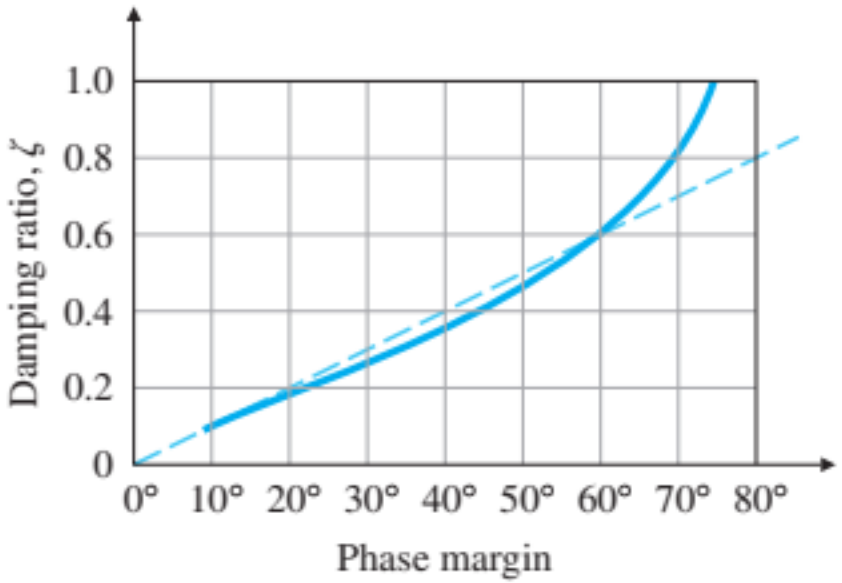
- Taking a 2nd order system with unitary feedback as a reference

$$G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$ Open-loop TF

$$\mathcal{T}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$ Closed-loop TF

$$\text{PM} = \tan^{-1}\left[\frac{2\zeta}{\sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}}\right]$$ Relationship between PM and ζ

Approximation up to PM = 60° $\zeta \cong \frac{\text{PM}}{100}$



# 4. Stability Margins

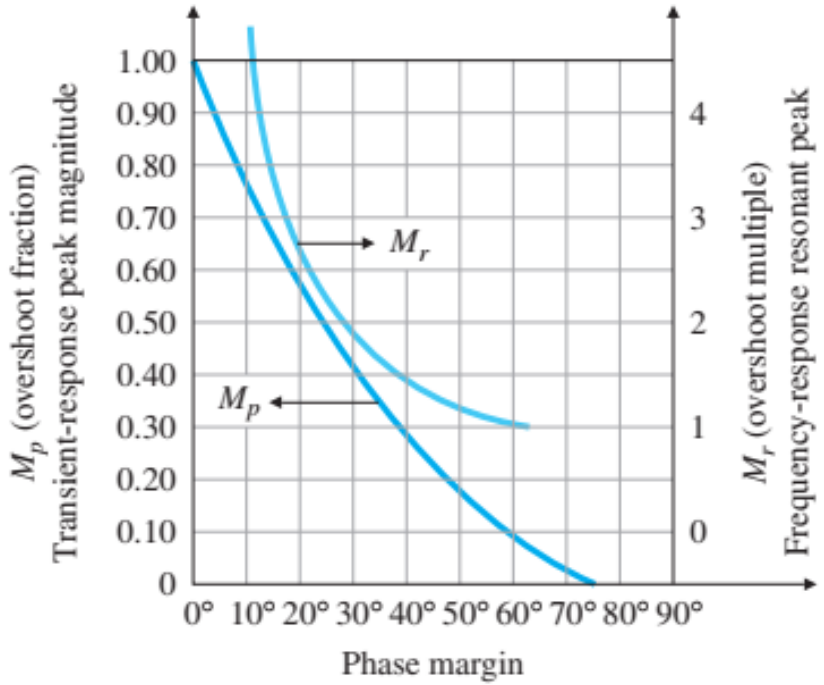
## Relationship between PM and time response

- Taking a 2nd order system with unitary feedback as a reference

$$G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$ Open-loop TF

$$\mathcal{T}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$ Closed-loop TF

$$\text{PM} = \tan^{-1}\left[\frac{2\zeta}{\sqrt{\sqrt{1 + 4\zeta^4} - 2\zeta^2}}\right]$$ Relationship between PM and ζ



# 4. Stability Margins

## Example: Stability Properties for a Conditionally Stable System

Determine the stability properties as a function of the gain K for the system with the open-loop transfer function

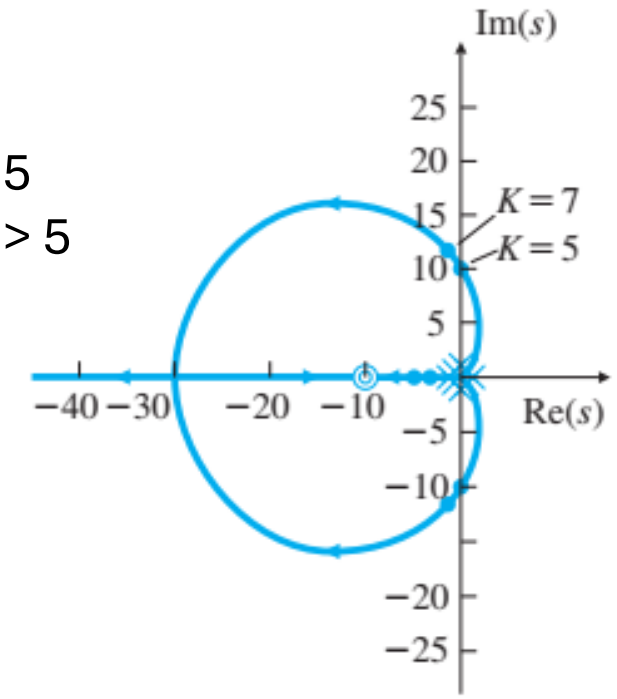
$$KG(s) = \frac{K(s+10)^2}{s^3}$$

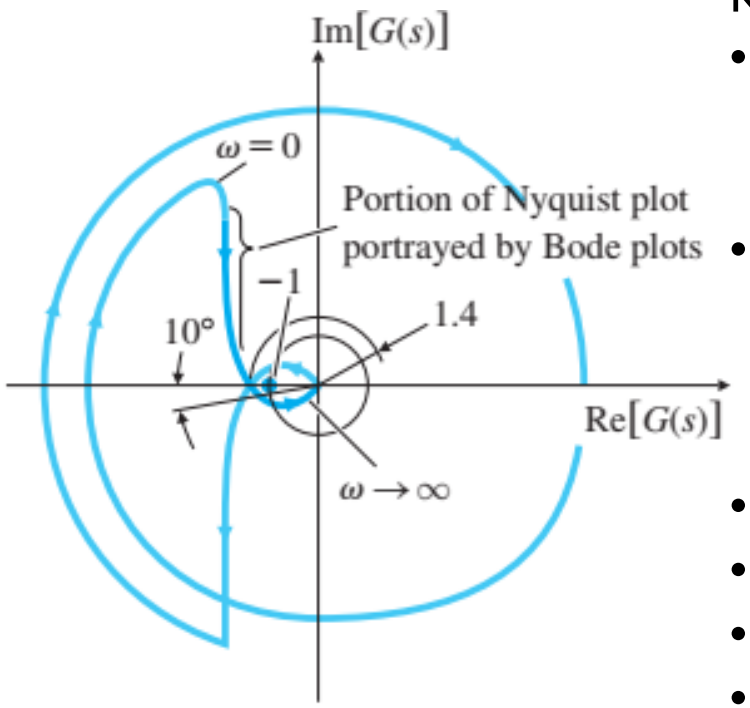
# 4. Stability Margins

## Example: Stability Properties for a Conditionally Stable System

$$KG(s) = \frac{K(s+10)^2}{s^3}$$

The system is unstable for K < 5 and stable for K > 5




Nyquist plot for K=7

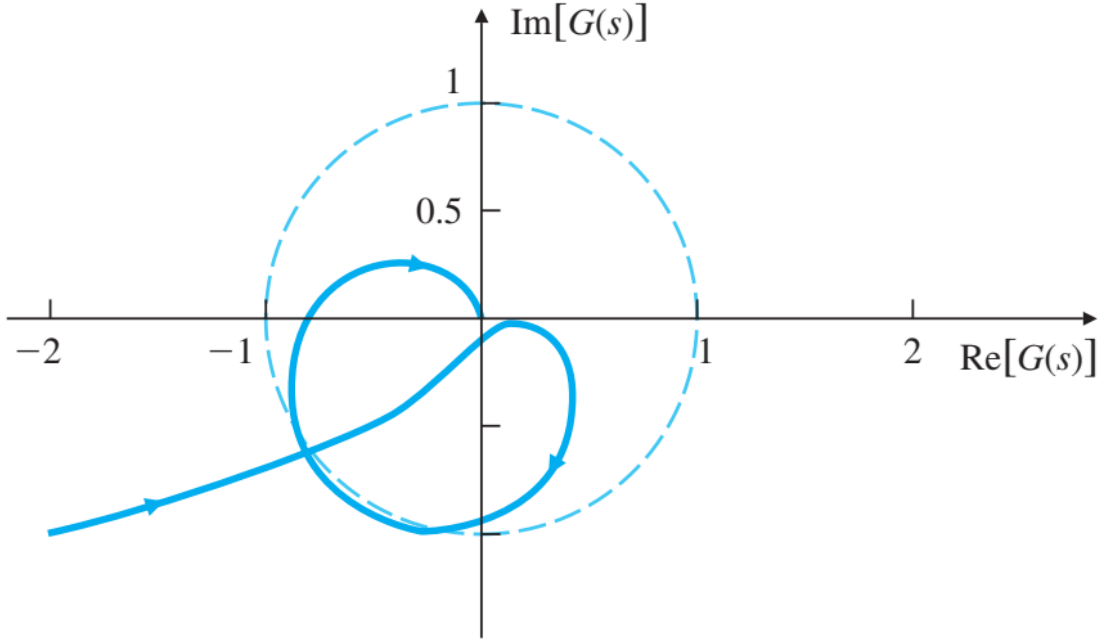
- One encirclement clockwise
- One encirclement counterclock wise
- N=+1-1=0
- P = 0
- Z = 0
- **Stable**

# 4. Stability Margins

## Example: Nyquist Plot for a System with Multiple Crossover Frequencies

Determine the stability margins from the Nyquist plot

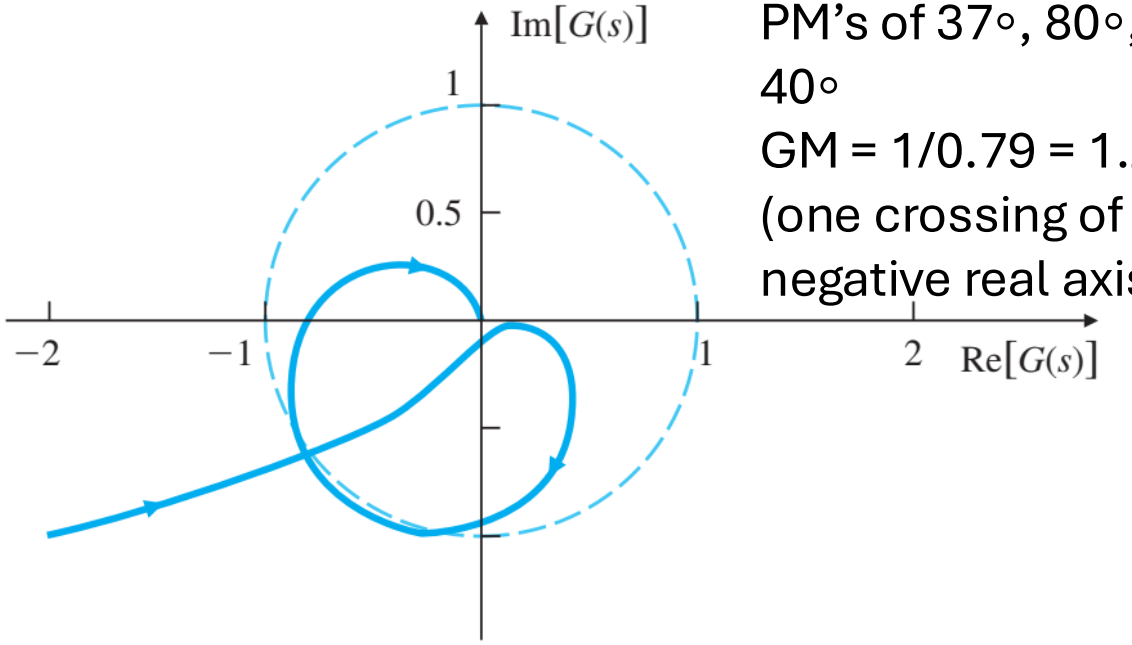
$$G(s) = \frac{85(s+1)(s^2+2s+43.25)}{s^2(s^2+2s+82)(s^2+2s+101)}$$
$$= \frac{85(s+1)(s+1\pm 6.5j)}{s^2(s+1\pm 9j)(s+1\pm 10j)},$$




3 crossover frequencies of the magnitude = 1 circle

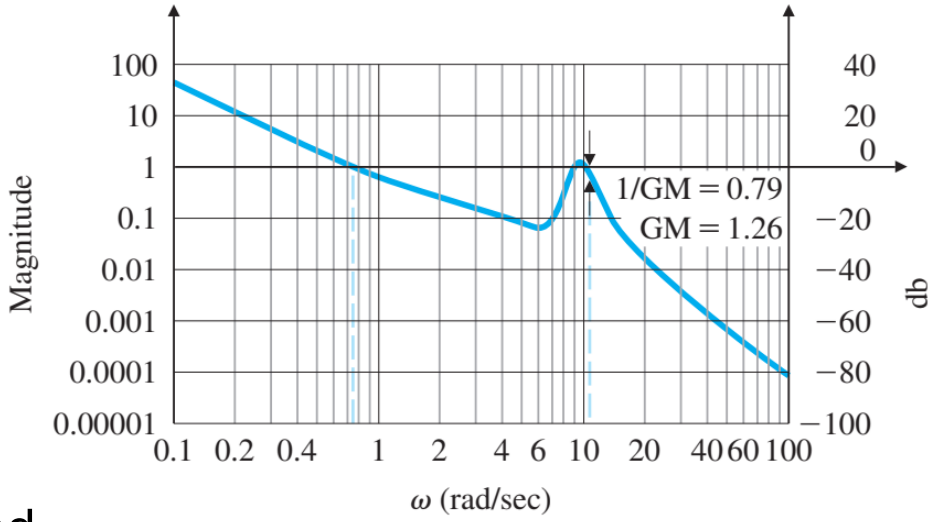
# 4. Stability Margins

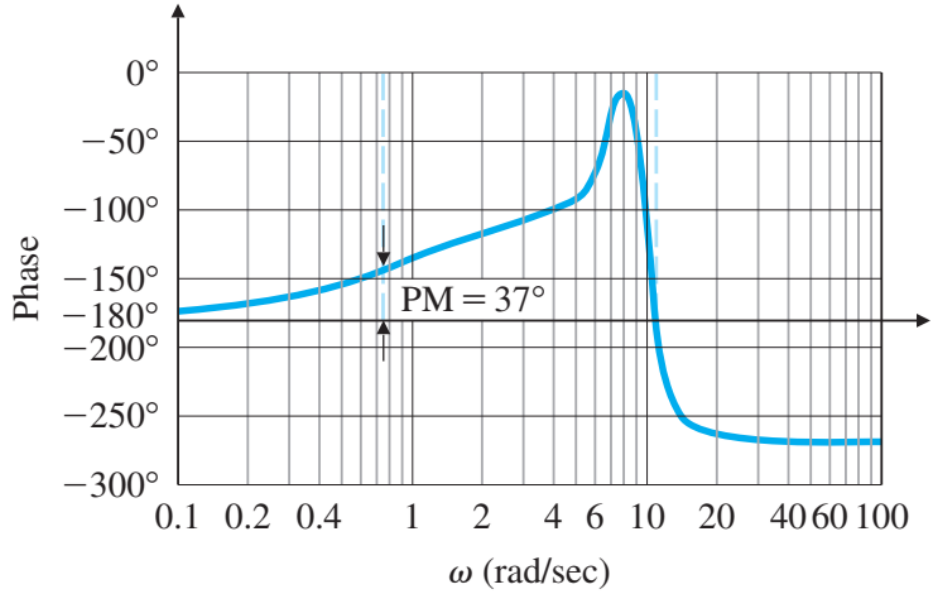
## Example: Nyquist Plot for a System with Multiple Crossover Frequencies




PM's of 37°, 80°, and 40°

GM = 1/0.79 = 1.26 (one crossing of negative real axis)



# 5. Bode's Gain–Phase Relationship

For any stable minimum-phase system (i.e., one with no RHP zeros or poles), the phase of $G(j\omega)$ is uniquely related to the magnitude of $G(j\omega)$.

When the slope of |G(jω)| versus ω on a log–log scale persists at a constant value for approximately a decade of frequency, the relationship is particularly simple and is given by:

$$\angle G(j\omega) \cong n \times 90^\circ$$

# 5. Bode's Gain–Phase Relationship

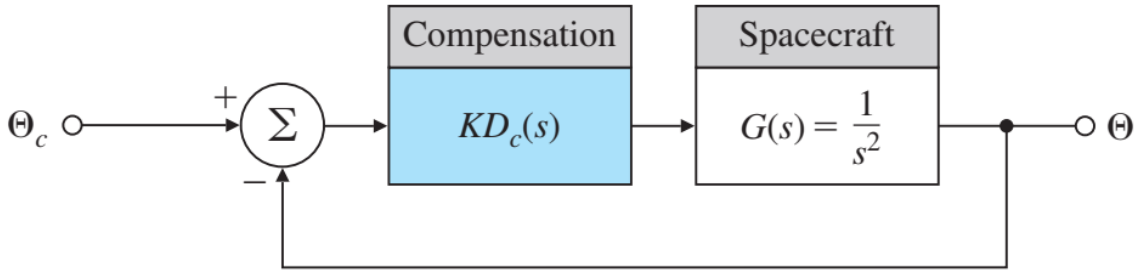
**Simple criterion: to provide acceptable PM**

Adjust the slope of the magnitude curve $|KG(j\omega)|$ so it crosses over magnitude 0 db (1) with a slope of -20 db/dec for a decade around $\omega c$ (crossover frequency)

# 5. Bode's Gain–Phase Relationship

**Example:**

Find a suitable expression for KDc(s) that will provide good damping and a bandwidth of approximately 0.2 rad/sec. Also determine the frequency where the sensitivity function |S| = 0.7 (= −3 db). This frequency is often referred to as the "Disturbance Rejection Bandwidth," or $\omega_{DRB}$.



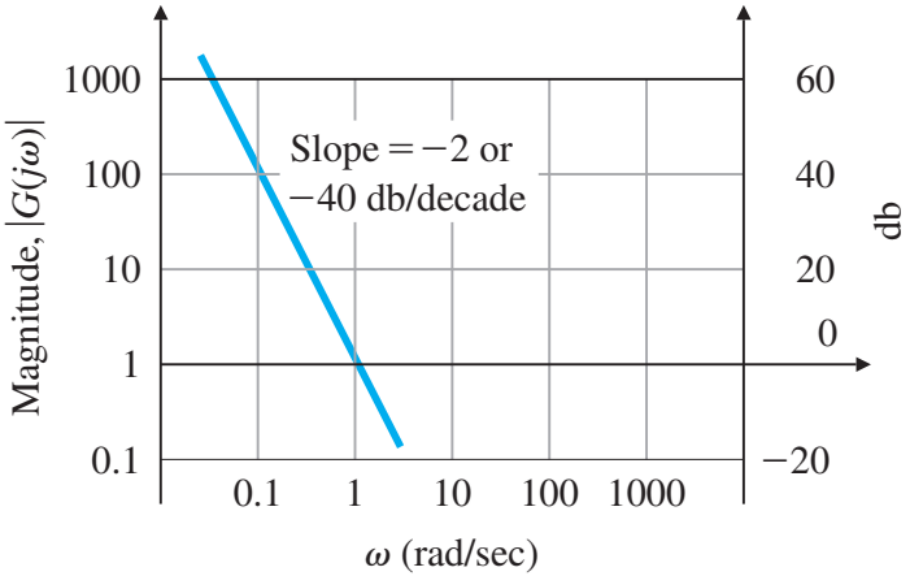
# 5. Bode's Gain–Phase Relationship

**Example:**

- The PD is the simplest compensator

$$KD_c(s) = K(T_D s + 1)$$

- Adjust the gain K to produce the desired bandwidth
- Adjust the break point ω1 = 1/TD to provide the −20 db/dec slope at the crossover frequency



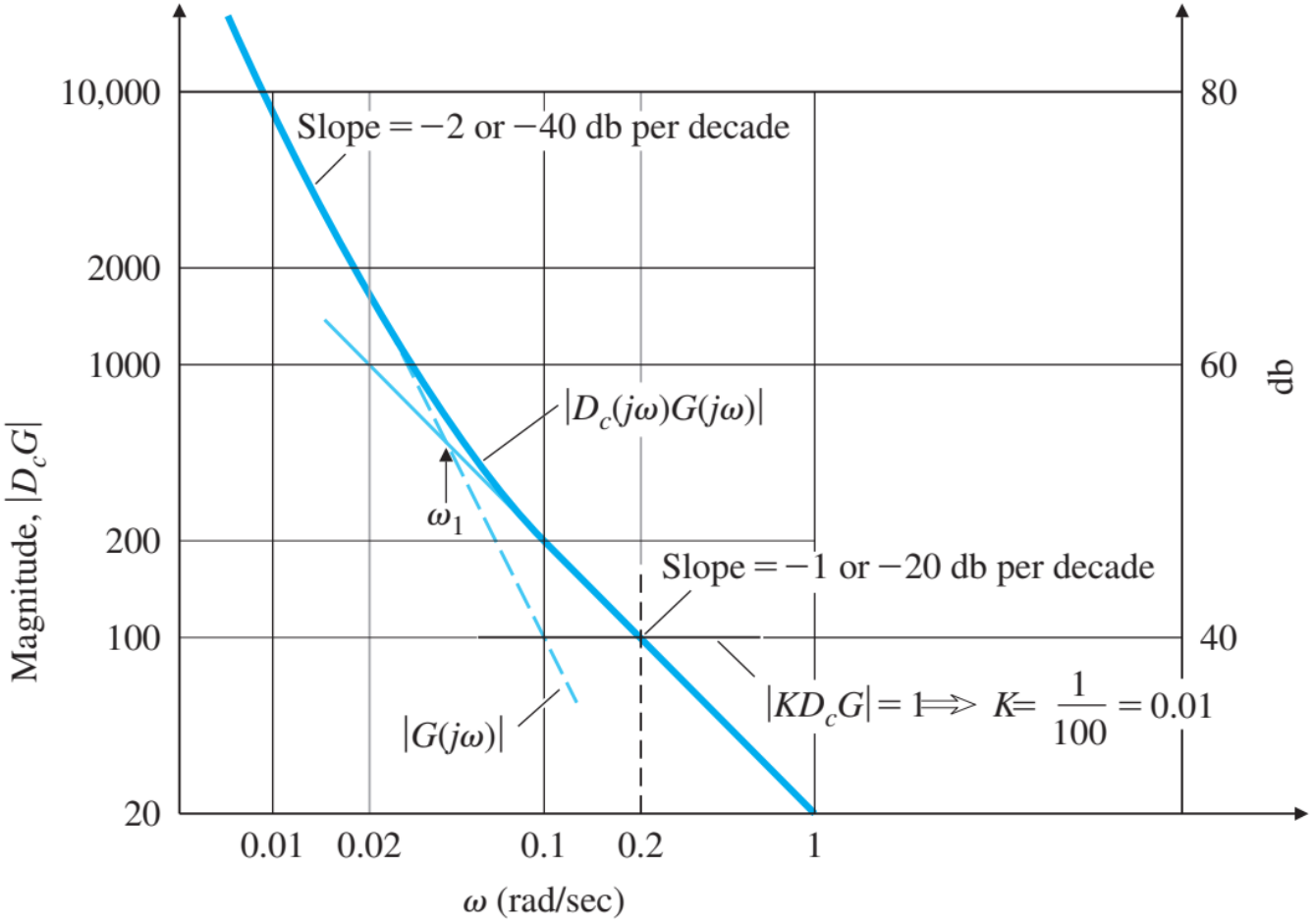
# 5. Bode's Gain–Phase Relationship

**Example:**

1. Plot |G( jω)|
2. Modify the plot to include |Dc(jω)|, with $\omega_1$ = 0.05 rad/sec (TD = 20) => slope −20 db at ω = 0.2 rad/sec.
3. Determine that |DcG| where the |DcG| curve crosses the line ω = 0.2 rad/sec (|DcG| = 40 db = 100)
4. In order for crossover to be at ω = 0.2 rad/sec, compute the required K

$$K = \frac{1}{|D_c G|_{\omega=0.2}} = \frac{1}{100} = 0.01$$

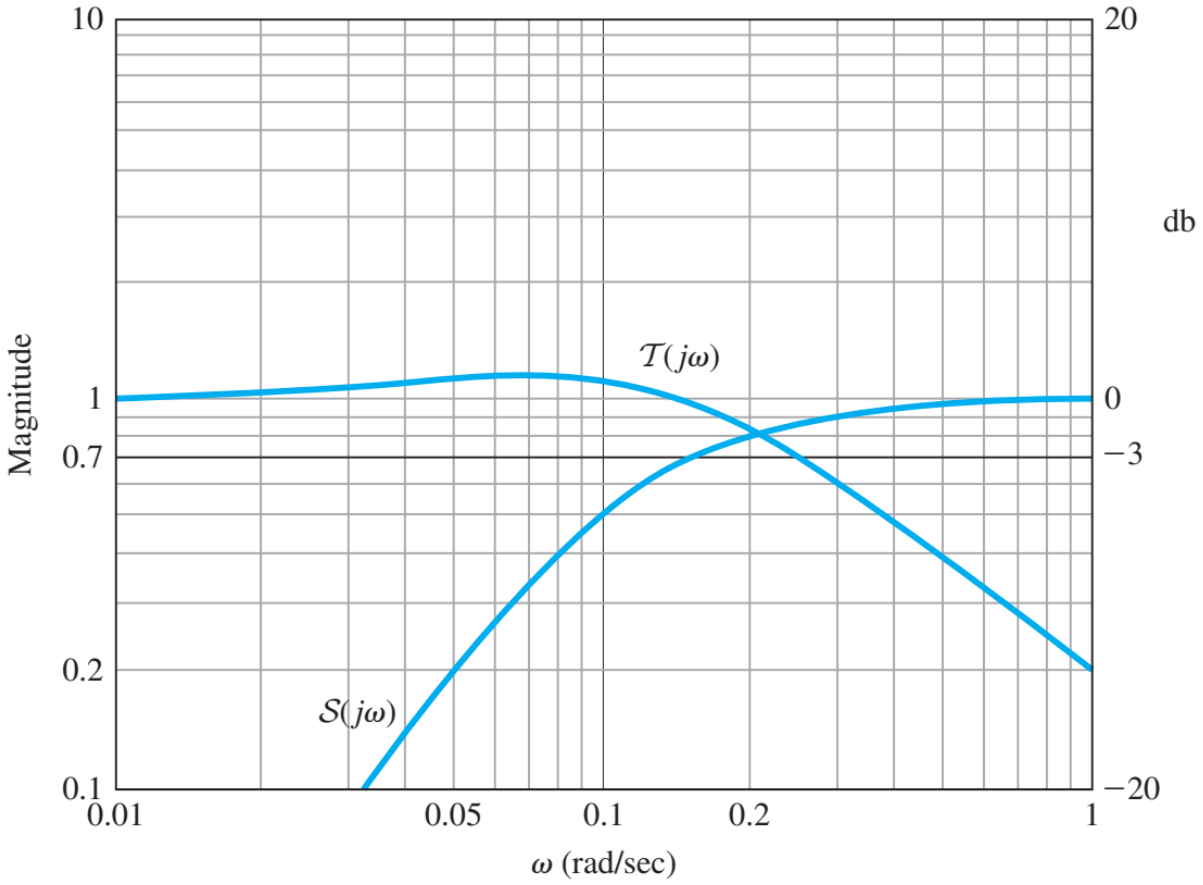
$$KD_c(s) = 0.01(20s + 1)$$



# 5. Bode's Gain–Phase Relationship

**Example:**

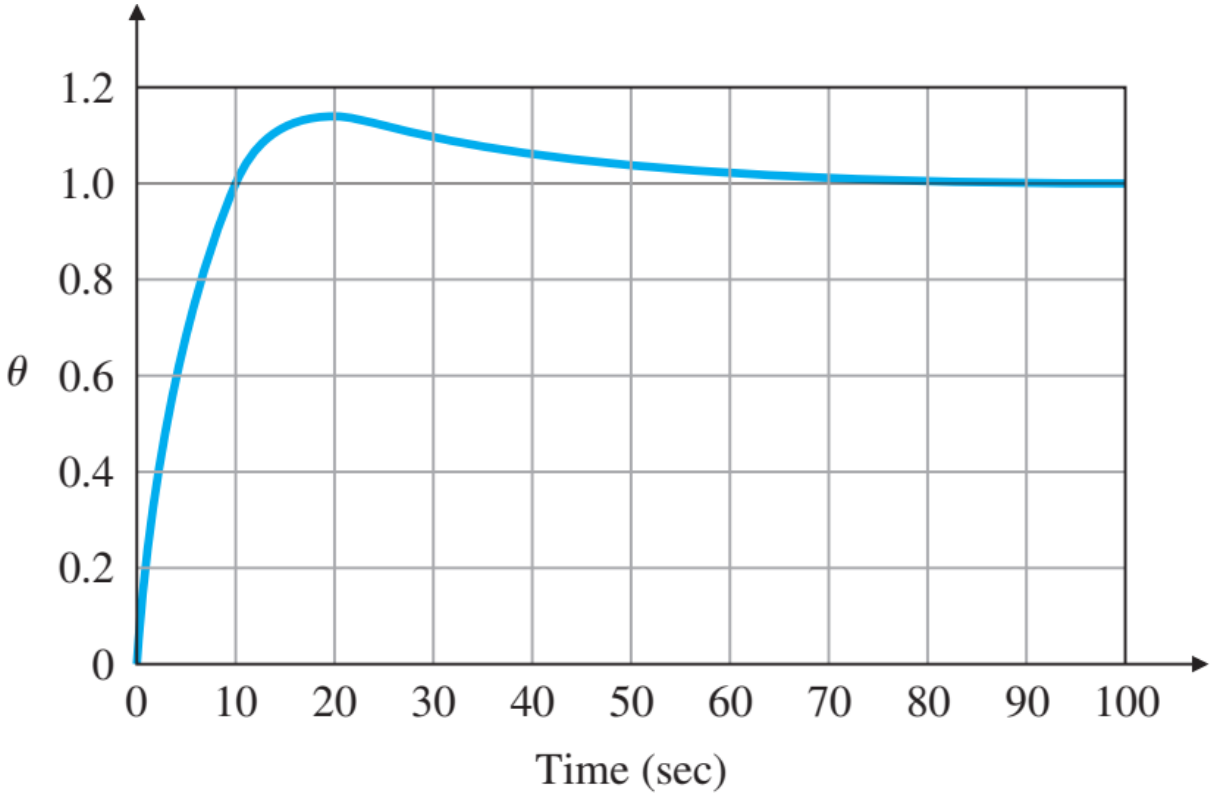
- PM = 75◦ (from Bode plot)
- Slope -20 db/dec for 1 dec around the desired bandwidth
- |S| has a value of 0.7 or −3db at ω = 0.15 rad/sec => $\omega_{DRB}$
- The maximum frequency at which the disturbance rejection



# 5. Bode's Gain–Phase Relationship

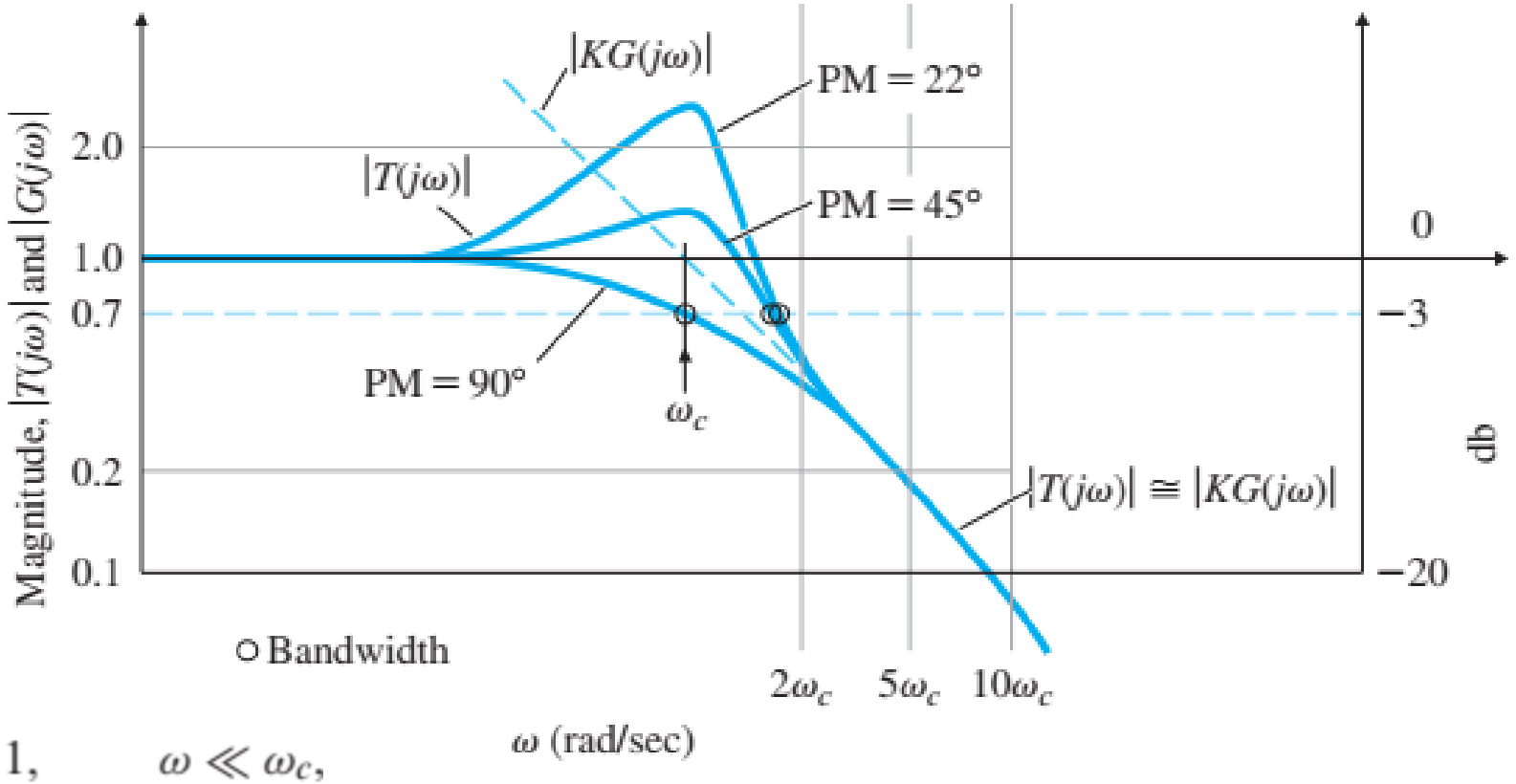
**Example:**

- PM = 75◦ (from Bode plot)
- Step response
- 14% overshoot confirms the adequate damping



# 6. Closed-Loop Frequency Response

- Bandwidth varies with the PM




$$\omega_c \leq \omega_{BW} \leq 2\omega_c.$$

$$|KG(j\omega)| \gg 1 \quad \text{for} \quad \omega \ll \omega_c$$

$$|KG(j\omega)| \ll 1 \quad \text{for} \quad \omega \gg \omega_c$$

$$|\mathcal{T}(j\omega)| = \left| \frac{KG(j\omega)}{1 + KG(j\omega)} \right| \cong \begin{cases} 1, & \omega \ll \omega_c, \\ |KG|, & \omega \gg \omega_c. \end{cases}$$

# 7. Compensation

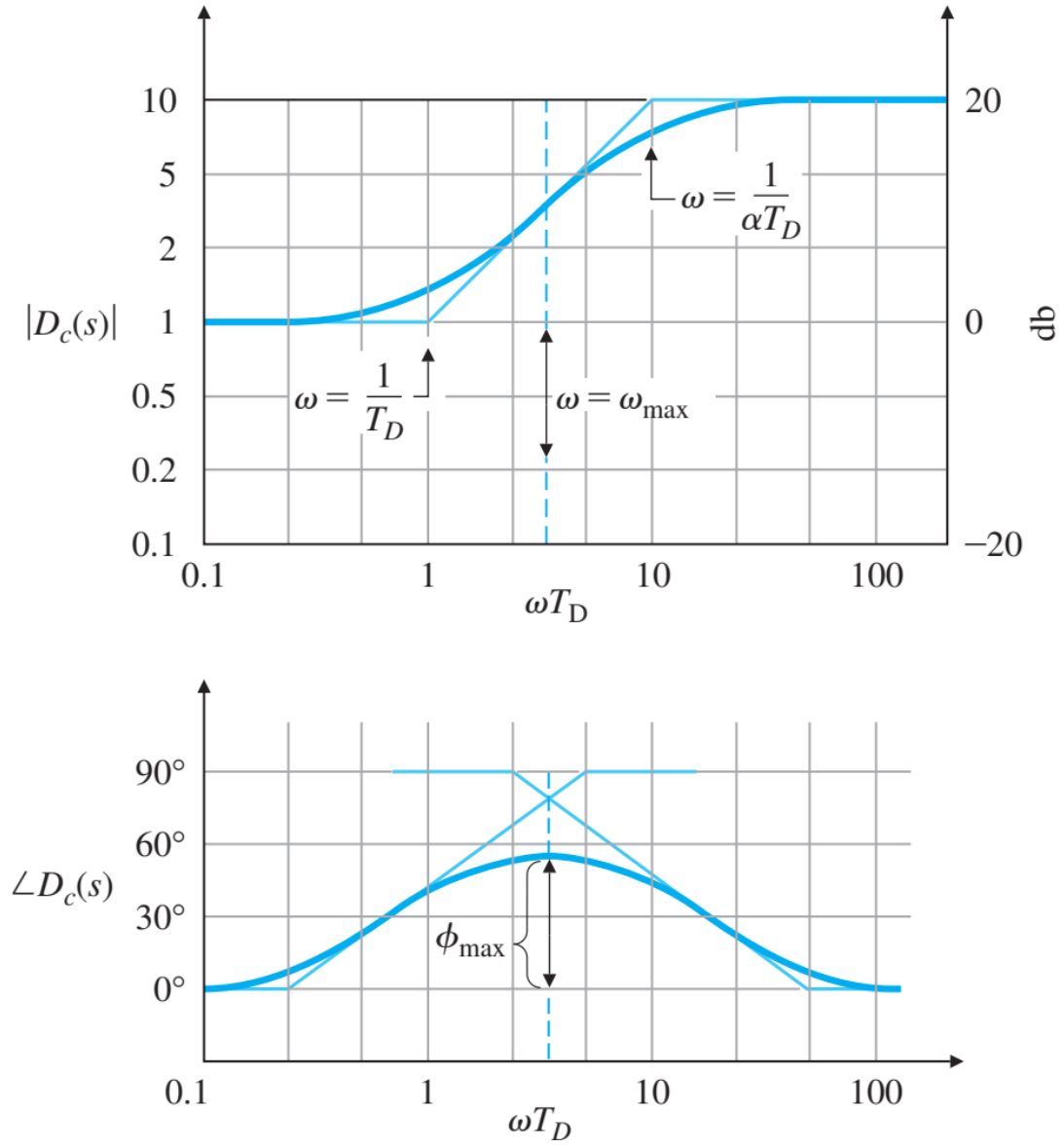
- Lead compensator
- PI compensator
- Lag compensator
- PID compensator

# 7. Compensation

**Lead compensator**:

- Increase in the phase => improve the PM
- Limited effect in the amplitude for high frequencies
- 1/α is the ratio between the pole/zero break-point frequencies

$$D_c(s) = \frac{T_D s + 1}{\alpha T_D s + 1}, \qquad \alpha < 1$$



# 7. Compensation

## Lead compensator:

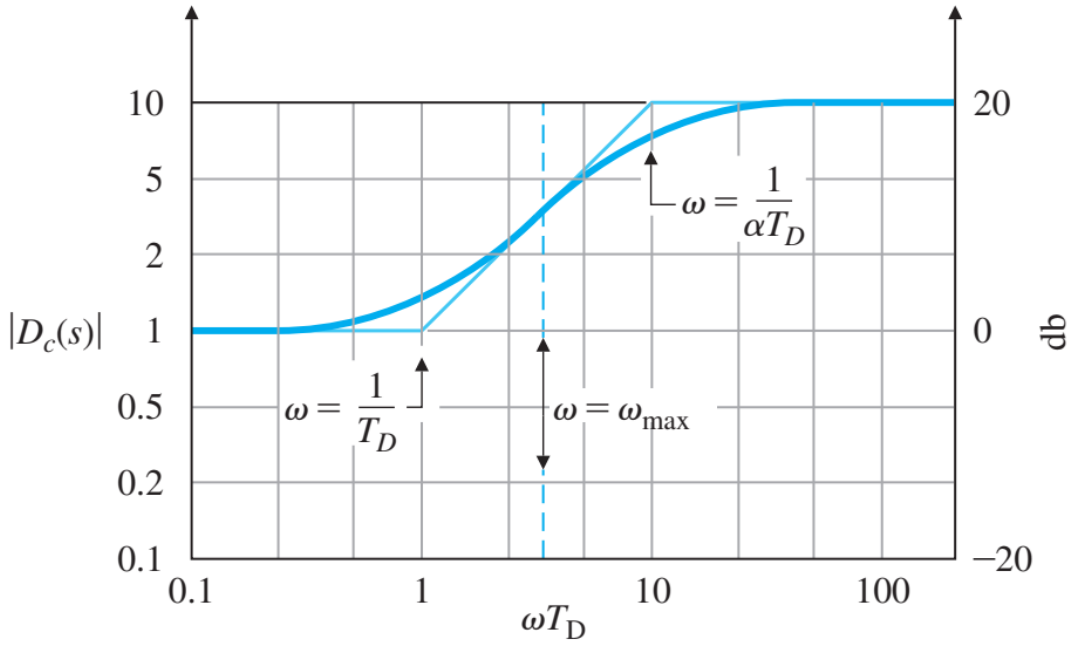
$$D_c(s) = \frac{T_D s + 1}{\alpha T_D s + 1}, \qquad \alpha < 1$$

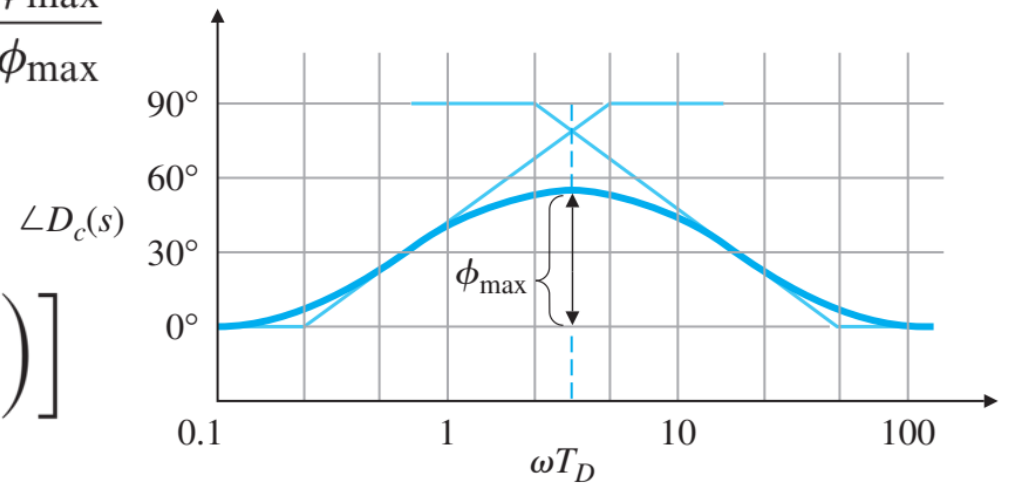
$$\phi = \tan^{-1}(T_D\,\omega) - \tan^{-1}(\alpha T_D\,\omega)$$

$$\omega_{\max} = \frac{1}{T_D\sqrt{\alpha}} \qquad \sin\phi_{\max} = \frac{1-\alpha}{1+\alpha} \qquad \alpha = \frac{1-\sin\phi_{\max}}{1+\sin\phi_{\max}}$$

$\phi_{max}$: midway between $1/T_D$ and $1/\alpha T_D$

$$\log\omega_{\max} = \frac{1}{2}\left[\log\left(\frac{1}{T_D}\right) + \log\left(\frac{1}{\alpha T_D}\right)\right]$$



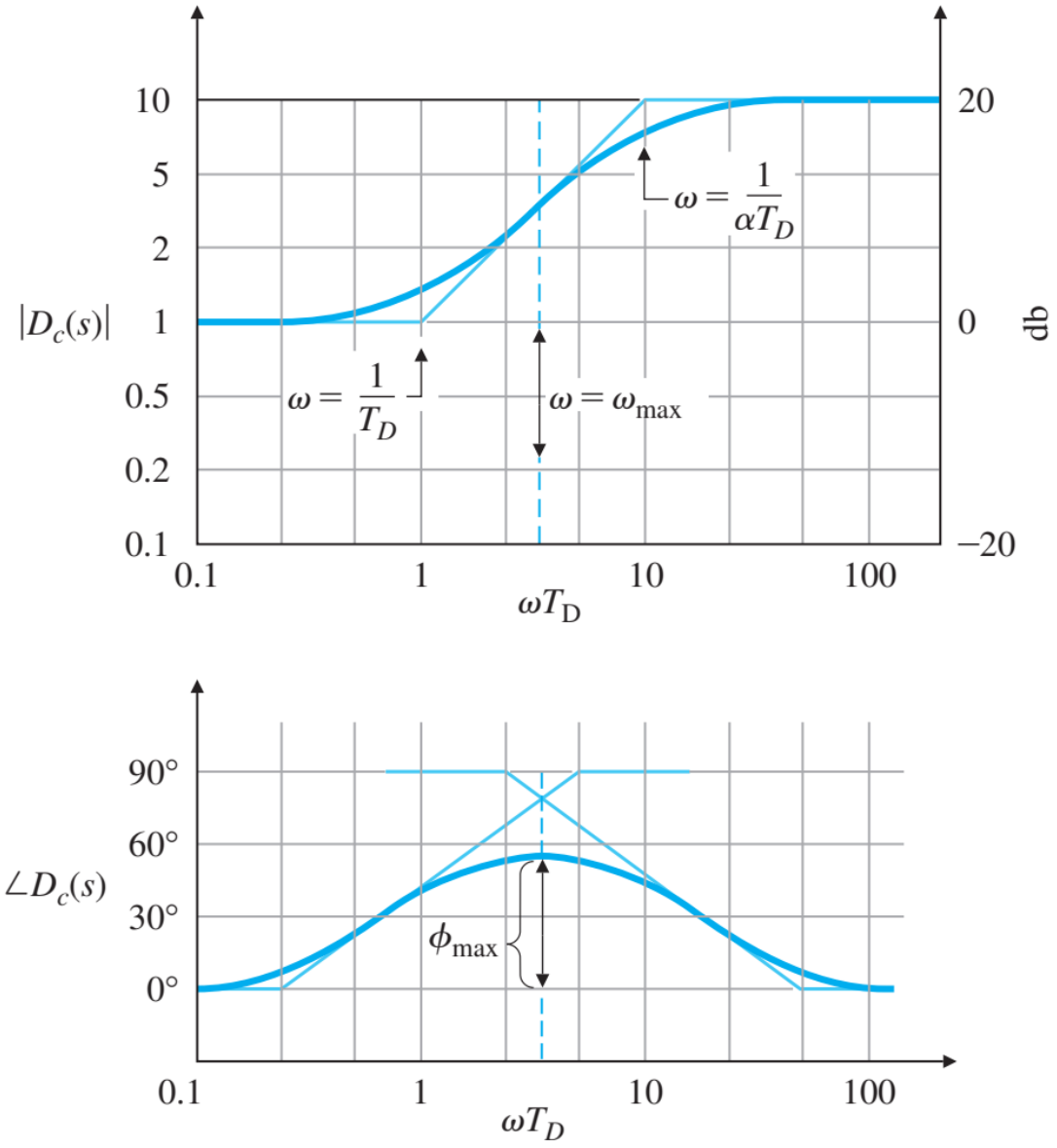
# 7. Compensation

**Lead compensator**: (zpk)

$$D_c(s) = \frac{s+z}{s+p}$$

$$\omega_{\max} = \sqrt{|z|\ |p|}$$

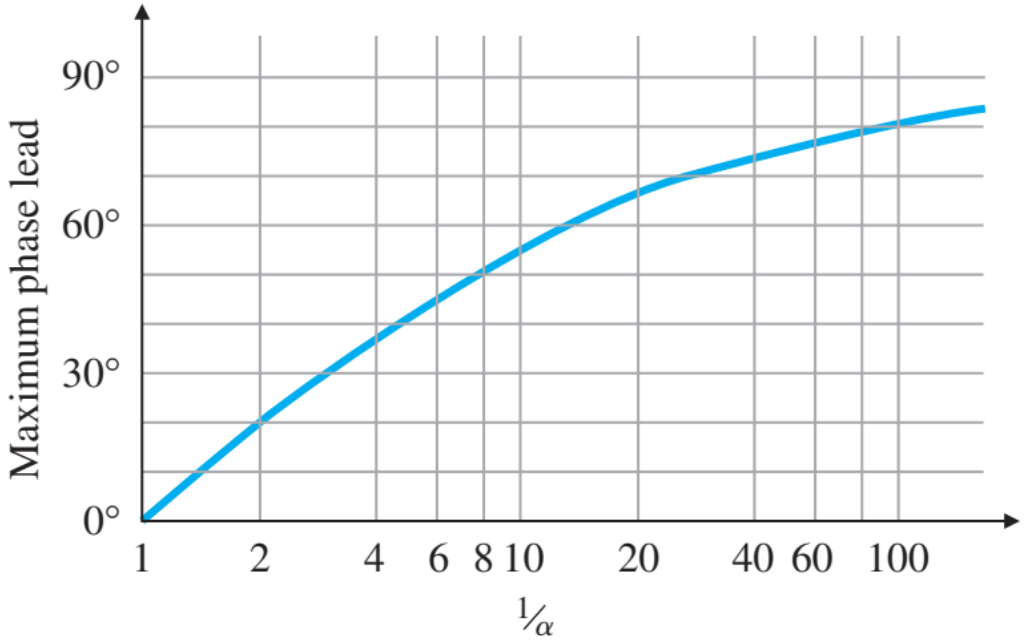
$$\log \omega_{\max} = \frac{1}{2}(\log |z| + \log |p|)$$



# 7. Compensation

**Lead compensator**:

- **Objective**: to select a value of 1/α that is a good compromise between an acceptable PM and an acceptable noise sensitivity at high frequencies
- **Primary design parameters**:
  1. The crossover frequency ωc, which determines bandwidth ωBW , rise time tr, and settling time ts
  2. The PM, which determines the damping coefficient ζ and the overshoot Mp
  3. The low-frequency gain, which determines the steady-state error characteristics.



# 7. Compensation

## Lead compensator: example

Compensator for a DC motor to obtain steady-state error of less than 0.1 for a unit-ramp input and overshoot Mp < 25%.

$$G(s) = \frac{1}{s(s+1)}$$

# 7. Compensation

## Lead compensator: example

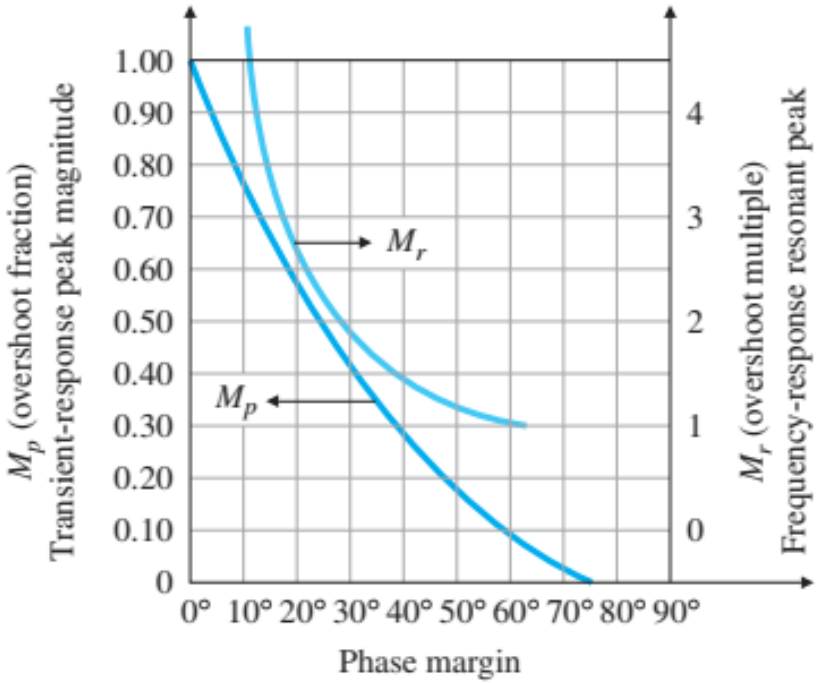
$$G(s) = \frac{1}{s(s+1)}$$

$$e_{ss} = \lim_{s\to 0} s\left[\frac{1}{1 + KD_c(s)G(s)}\right]R(s)$$

For a unit ramp:

$$e_{ss} = \lim_{s\to 0}\left\{\frac{1}{s + KD_c(s)[1/(s+1)]}\right\} = \frac{1}{KD_c(0)}$$

KDc(0) > 10 to obtain the desired error => K = 10

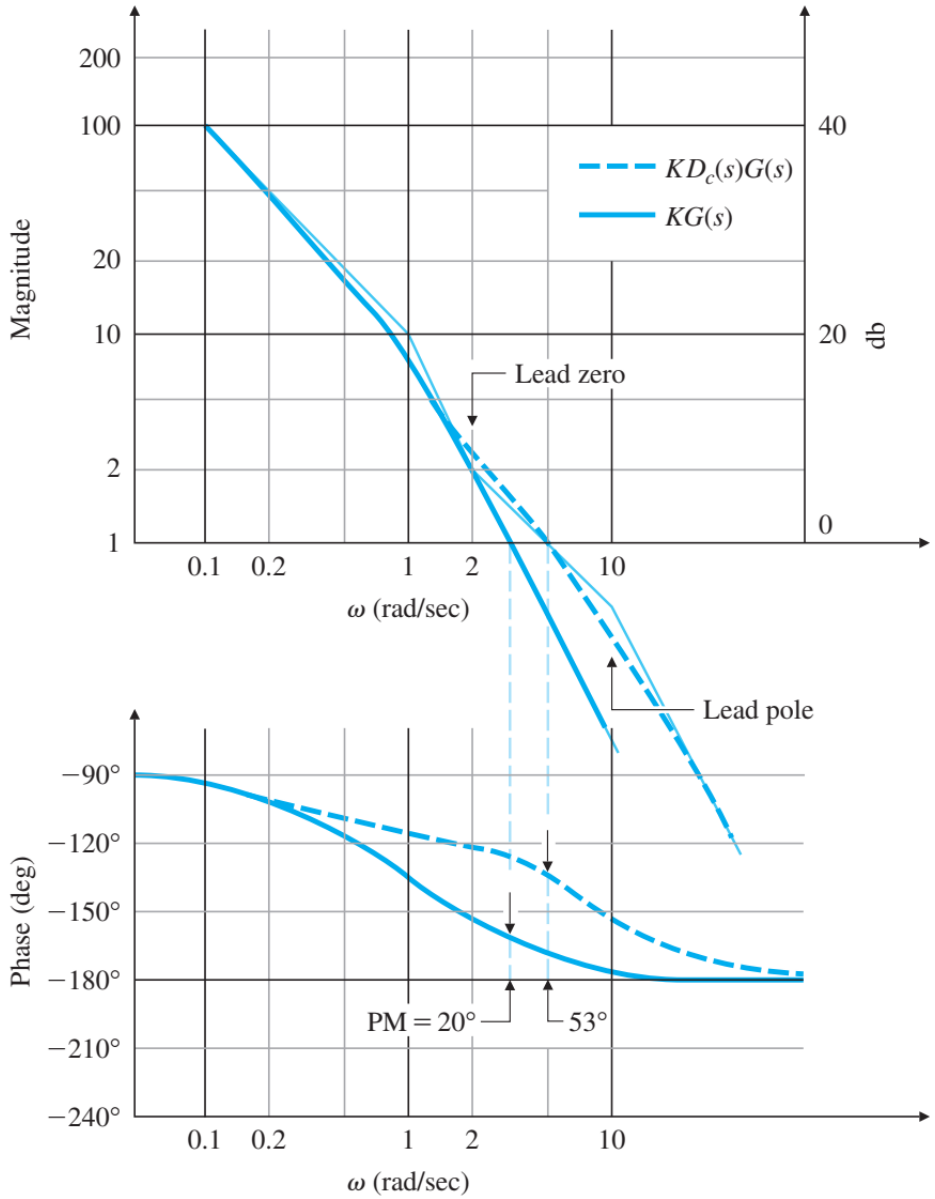



From the figure: with PM = 45° we obtain Mp = 20% aprox.

# 7. Compensation

**Lead compensator: example**

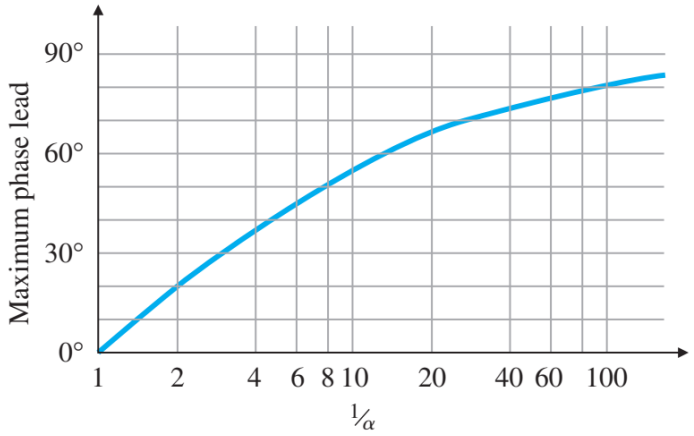
- PM without Dc(s): 20°
- We need to add 25° at $w_c$ = 3 rad/s (crossover frequency) without affecting the magnitued

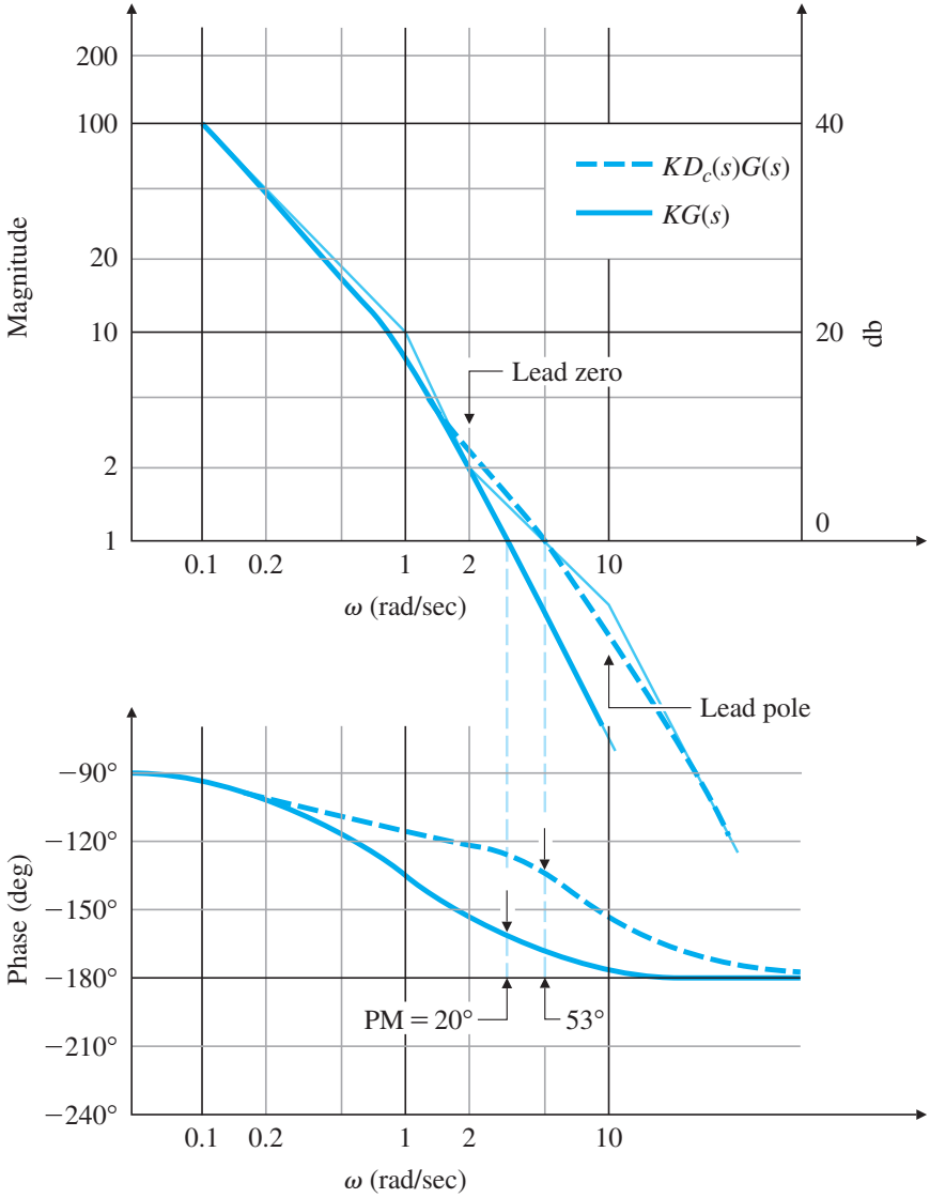


# 7. Compensation

**Lead compensator: example**

- We define a $\phi_{max}$ = 40° for the lead compensator => $1/\alpha$ = 5 (from a previous figure)
- With trial an error: $T_D$ = 1/2 (w = 2 rad/s); then $\alpha T_D$ = 1/10 (w = 10) to locate $\phi_{max}$ at $w_c$ = 3 rad/s



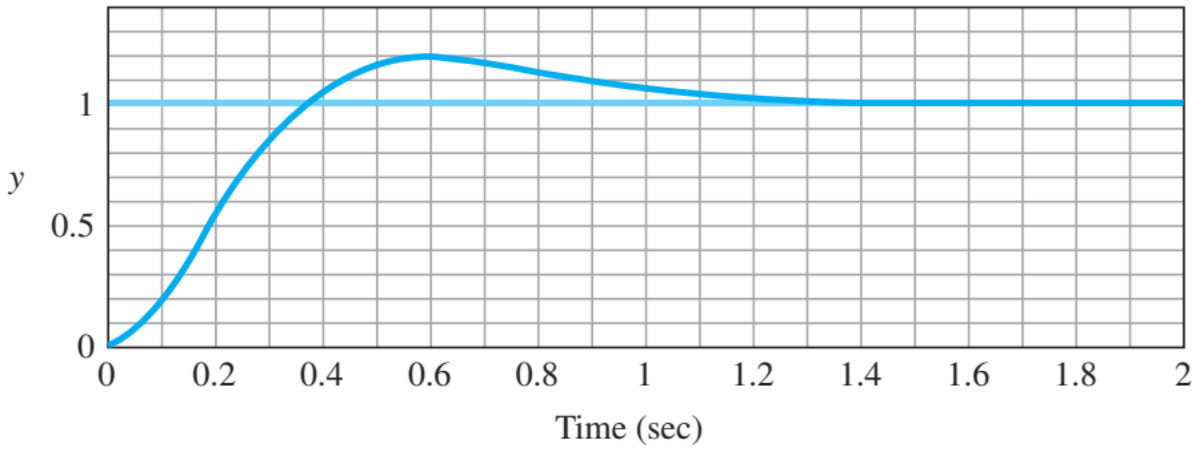



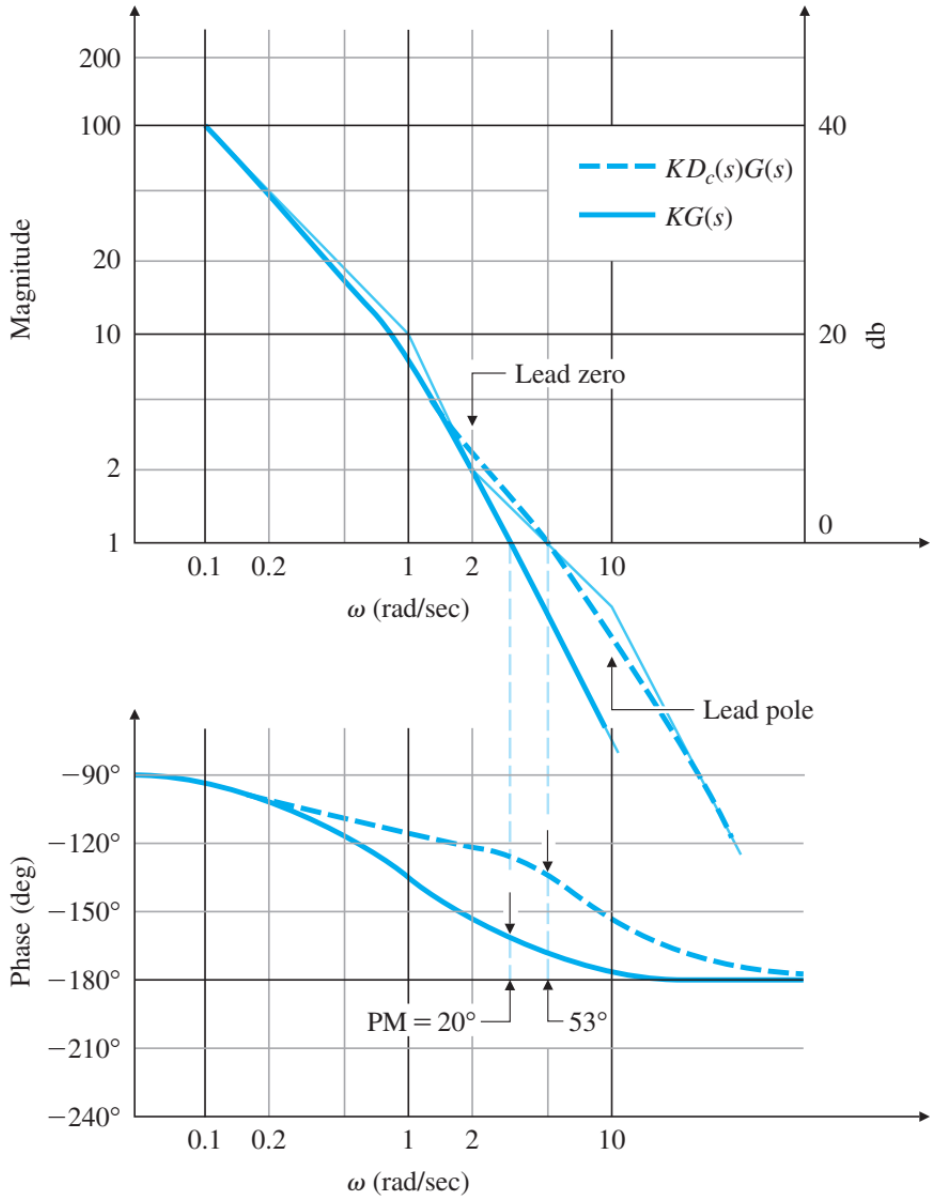
# 7. Compensation

## Lead compensator: example

$$KD_c(s) = 10\frac{s/2+1}{s/10+1}$$

Step response:



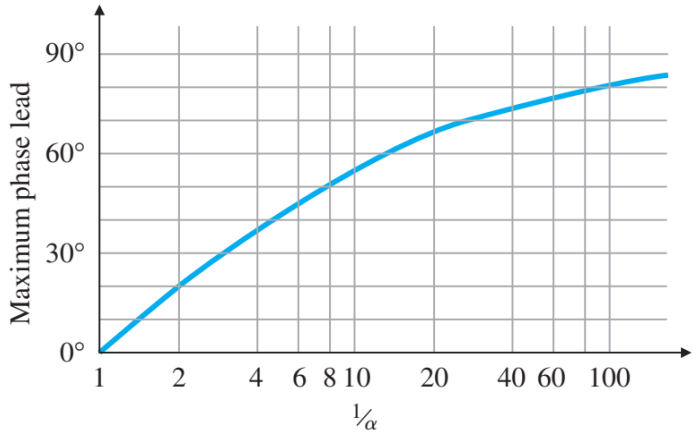



# 7. Compensation

## Lead compensator: design procedure

$$\sin\phi_{\max} = \frac{1-\alpha}{1+\alpha}$$

$$\alpha = \frac{1-\sin\phi_{\max}}{1+\sin\phi_{\max}}$$




1. Determine the gain $K$ to satisfy error or bandwidth requirements:
   (a) to meet error requirements, pick $K$ to satisfy error constants ($K_p$, $K_v$, or $K_a$) so $e_{ss}$ error specification is met, or alternatively,
   (b) to meet bandwidth requirements, pick $K$ so the open-loop crossover frequency is a factor of two below the desired closed-loop bandwidth.
2. Evaluate the PM of the uncompensated system using the value of $K$ obtained from Step 1.
3. Allow for extra margin (about 10°), and determine the needed phase lead $\phi_{\max}$.
4. Determine $\alpha$ from Eq. (6.40) or Fig. 6.53.
5. Pick $\omega_{\max}$ to be at the crossover frequency; thus the zero is at $1/T_D = \omega_{\max}\sqrt{\alpha}$ and the pole is at $1/\alpha T_D = \omega_{\max}/\sqrt{\alpha}$.
6. Draw the compensated frequency response and check the PM.
7. Iterate on the design. Adjust compensator parameters (poles, zeros, and gain) until all specifications are met. Add an additional lead compensator (that is, a double-lead compensation) if necessary.

# 7. Compensation

## Lead compensator: example Temperature Control System

Design a lead compensator such that Kp = 9 and the PM is at least 25°.

$$KG(s) = \frac{K}{(s/0.5+1)(s+1)(s/2+1)}$$

# 7. Compensation

$$KG(s) = \frac{K}{(s/0.5+1)(s+1)(s/2+1)}$$

## Lead compensator: example Temperature Control System

1. From the specification Kp (position error constant) for type 0 system

$$K_p = \lim_{s\to 0} KG(s) = K = 9$$

```
s = tf('s');
sysG = 9/((s/0.5 + 1)*(s + 1)*(s/2 + 1));
w=logspace(-1,1);
bode(sysG,w); grid;
[mag,phase] = bode(sysG,w);
```

2. Bode plot of the uncompensated system with K = 9

```
[GM,PM,Wcg,Wcp] = margin(mag,phase,w);
```

# 7. Compensation

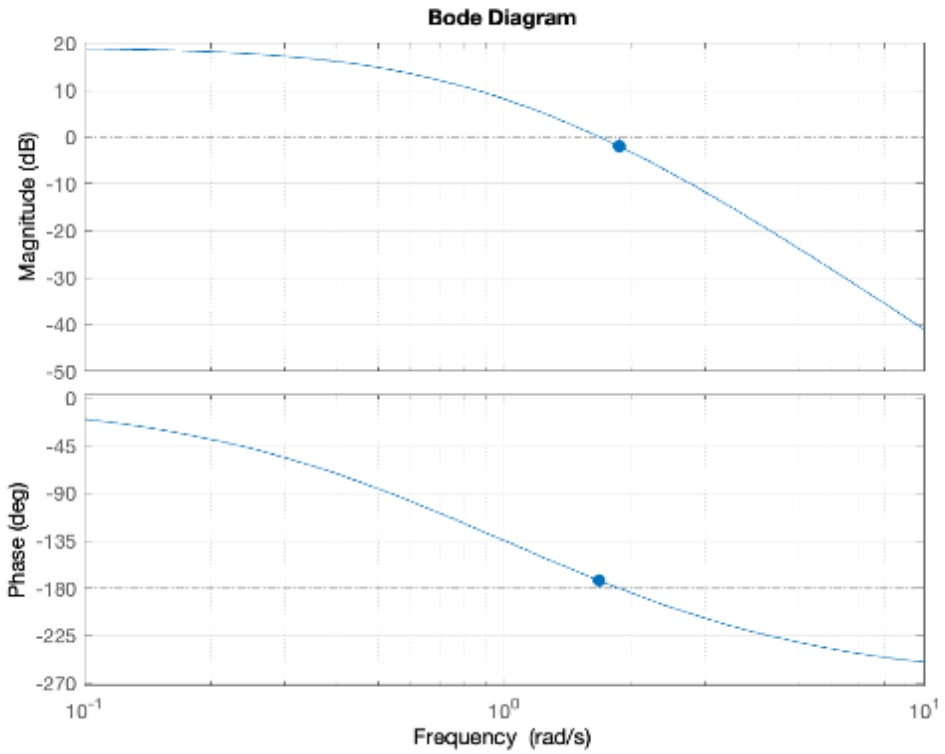
$$KG(s) = \frac{K}{(s/0.5+1)(s+1)(s/2+1)}$$

## Lead compensator: example Temperature Control System

1. From the specification Kp (position error constant) for type 0 system

$$K_p = \lim_{s\to 0} KG(s) = K = 9$$

2. Bode plot of the uncompensated system with K = 9




GM =1.25 db, **PM = 7.14°**, Wcg = 1.87 rad/s (w for ∠-180°), Wcp = 1.68 (crossover w) rad/s

# 7. Compensation

## Lead compensator: example Temperature Control System

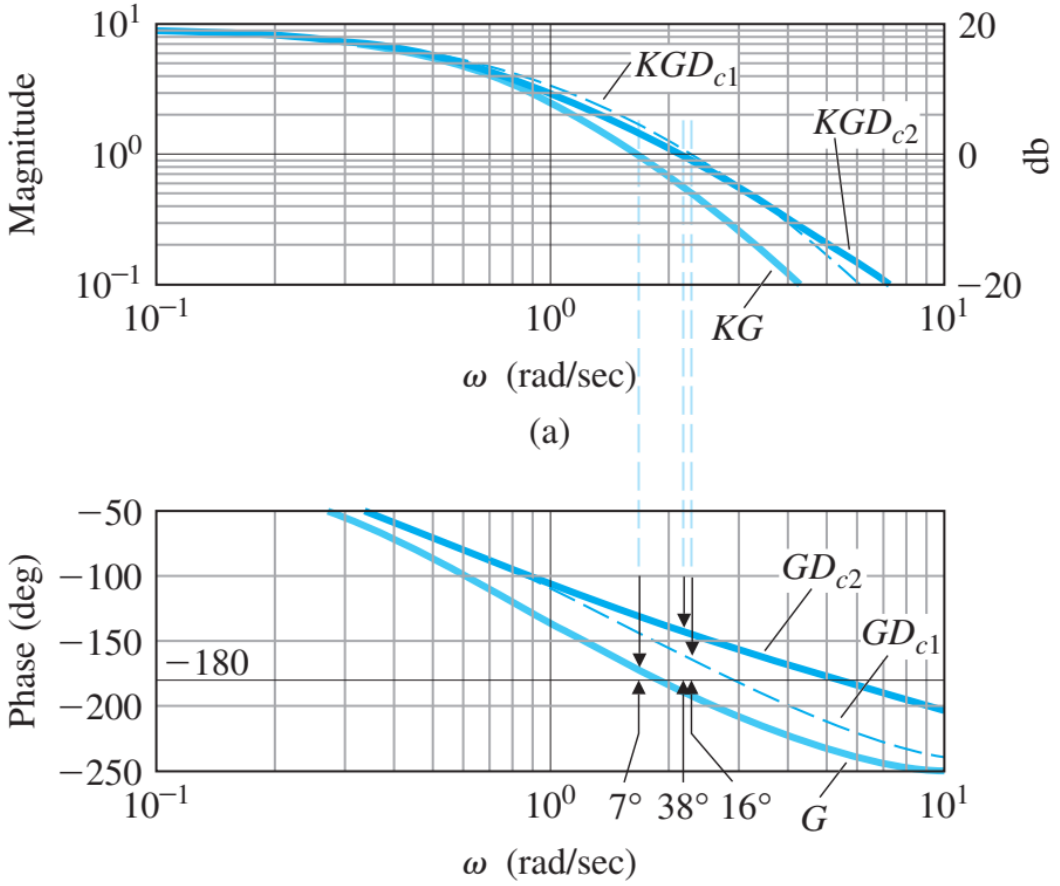
3. To obain a PM of 25° (specification) with 10° of extra PM, we need to add: 25° + 10° - 7° = 28°

4. From the figure we obtain $1/\alpha = 3$ to obtain PM = 30° => $\alpha = 1/3$

5. Trial 1: $T_D = 1$ (zero at 1 rad/s); then: $\alpha T_D = 1/3$ and $w_{max} = 1.73$ rad/s

$$\omega_{\max} = \frac{1}{T_D\sqrt{\alpha}}$$

$$D_{c1}(s) = \frac{s+1}{s/3+1} = \frac{1}{0.333}\left(\frac{s+1}{s+3}\right)$$



# 7. Compensation

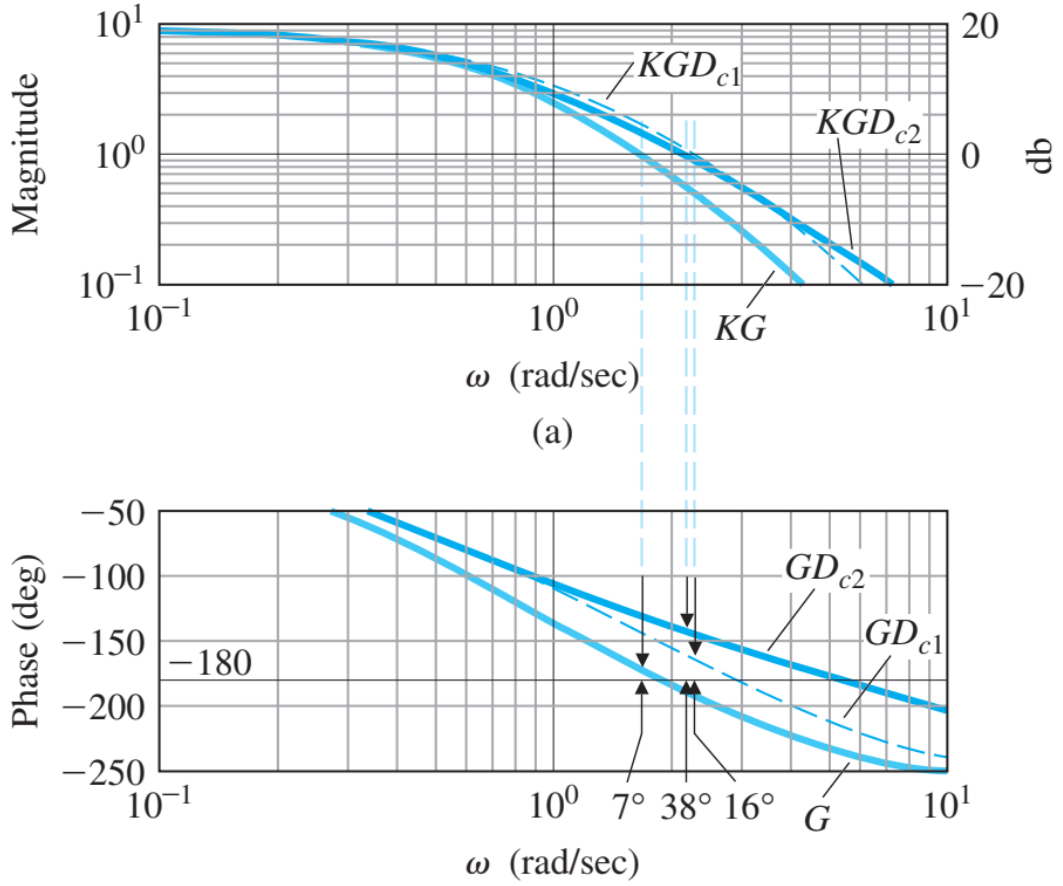
## Lead compensator: example Temperature Control System

6. Trial 2: increasing the increment in PM of the compensator to 55° => α = 1/10

And moving the zero to the right (at 1.5 rad/s) $T_D$ = 1/1.5; then: $\alpha T_D$ = 1/15 and $w_{max}$ = 2.10 rad/s c

$$D_{c2}(s) = \frac{s/1.5 + 1}{s/15 + 1} = \frac{1}{0.1}\left(\frac{s + 1.5}{s + 15}\right)$$

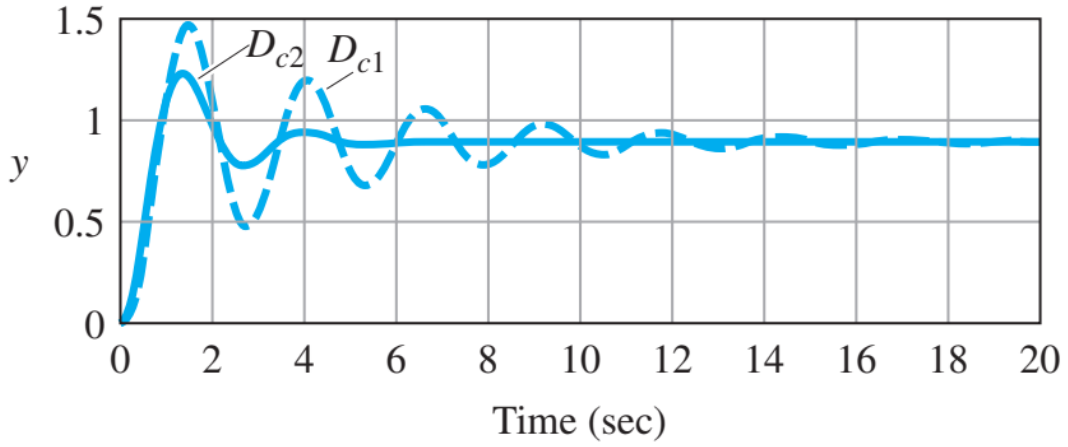
$$D_{c1}(s) = \frac{s+1}{s/3+1} = \frac{1}{0.333}\left(\frac{s+1}{s+3}\right)$$



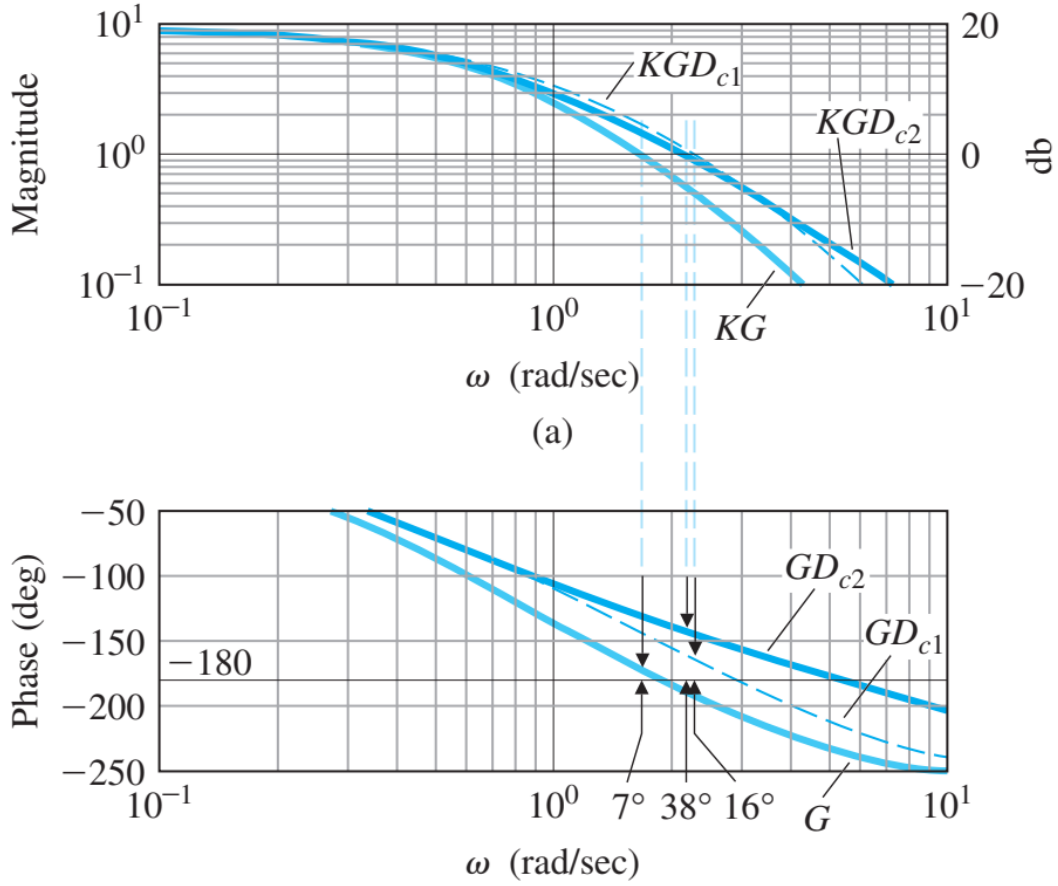
# 7. Compensation

## Lead compensator: example Temperature Control System

$$D_{c2}(s) = \frac{s/1.5 + 1}{s/15 + 1} = \frac{1}{0.1}\left(\frac{s + 1.5}{s + 15}\right)$$




$$D_{c1}(s) = \frac{s + 1}{s/3 + 1} = \frac{1}{0.333}\left(\frac{s + 1}{s + 3}\right)$$



# 7. Compensation

**Lead compensator: example Lead-Compensator Design for a Servomechanism System (Type 1)**

Design a lead compensator so that the PM = 45° and Kv = 10.

$$KG(s) = K\frac{10}{s(s/2.5+1)(s/6+1)}$$

# 7. Compensation

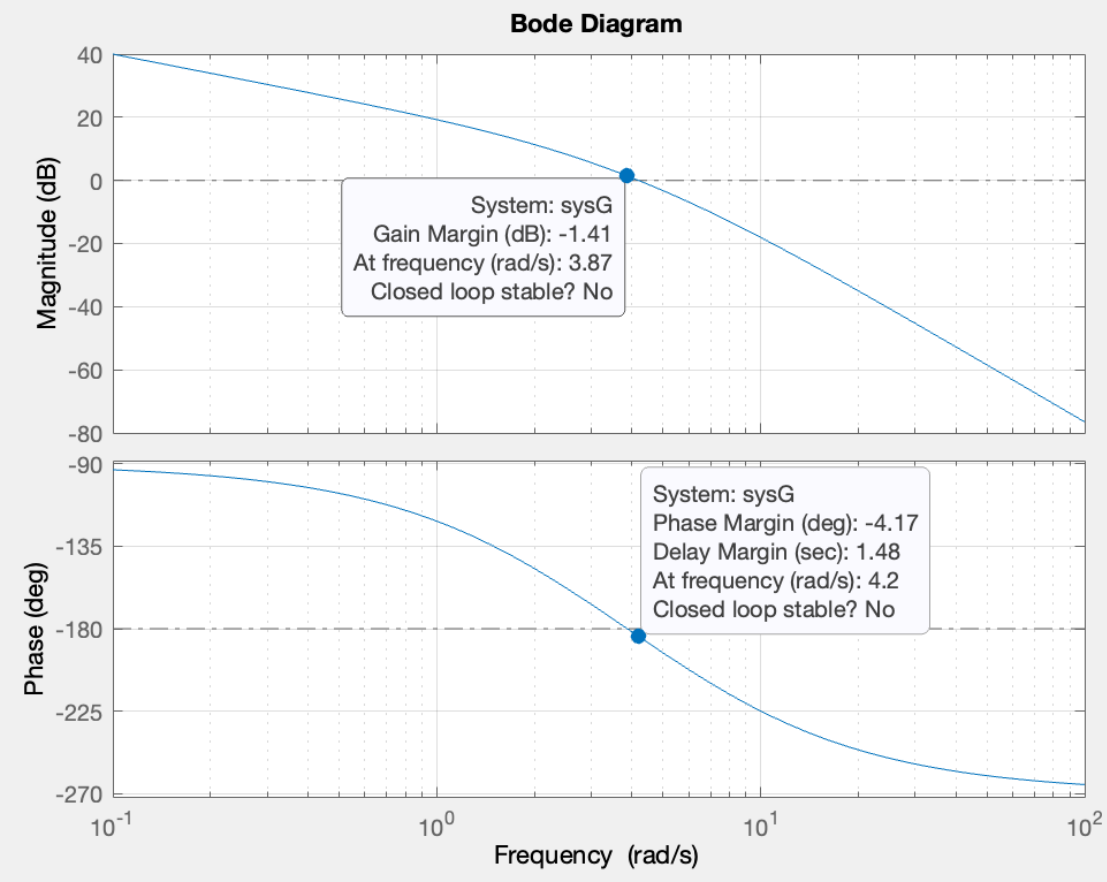
## Lead compensator: example (Type 1)

1. From the specification Kv (speed error constant) for type 1 system, then K = 1, so Kv = 10

$$K_v = \lim_{s \to 0} sGD_{cl}(s), \qquad n = 1$$

2. Bode plot of the uncompensated system with K = 1

GM = 0.85, **PM =** -4.1 °, Wcg = 3.87 (w for ∠-180°), $w_c$ = 4.19 (crossover w)



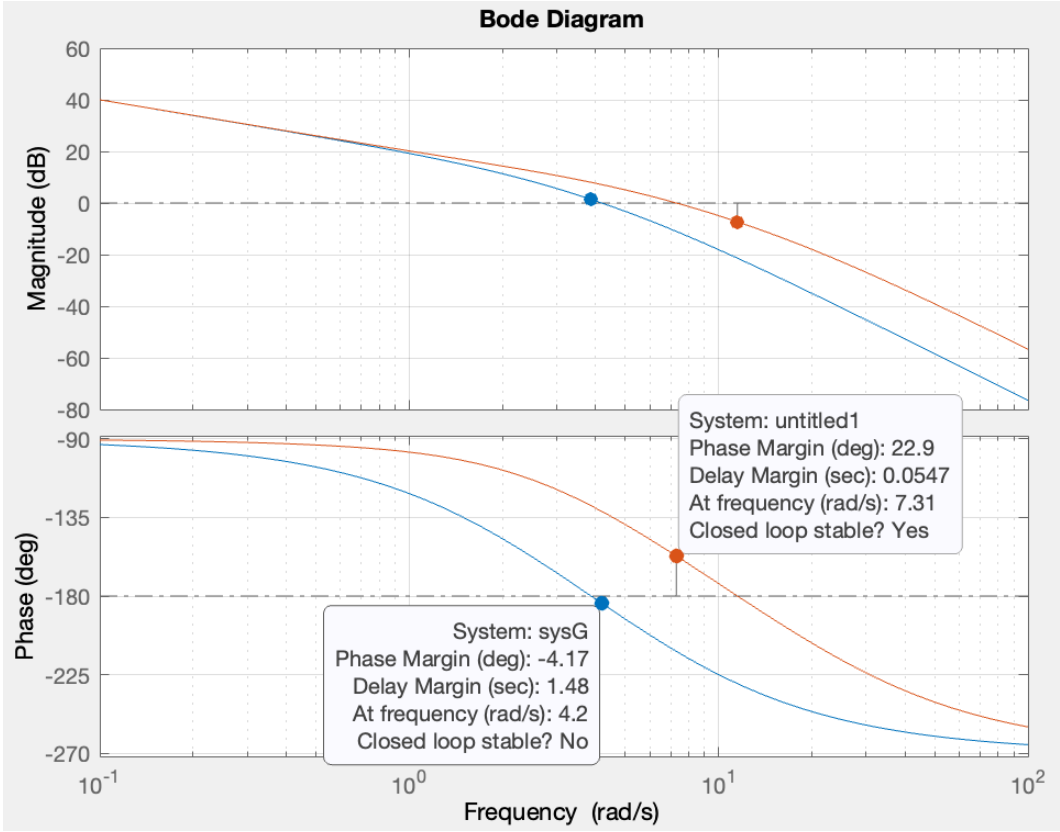
# 7. Compensation

## Lead compensator: example (Type 1)

3. To obain a PM of 45° (specification) with 5° of extra PM, we need to add:
45°+5°−(−4°) = 54°

4. From the figure we obtain $1/\alpha$ = 10 to obtain PM = 55° => $\alpha$ = 1/10

5. Crossover frequency will be higher with the compensator => we select the position of the zero at 2 rad/se: $T_D$ = 1/2 (zero at 2 rad/s); then: $\alpha T_D$ = 1/20 and $w_{max}$ = 6.32rad/s



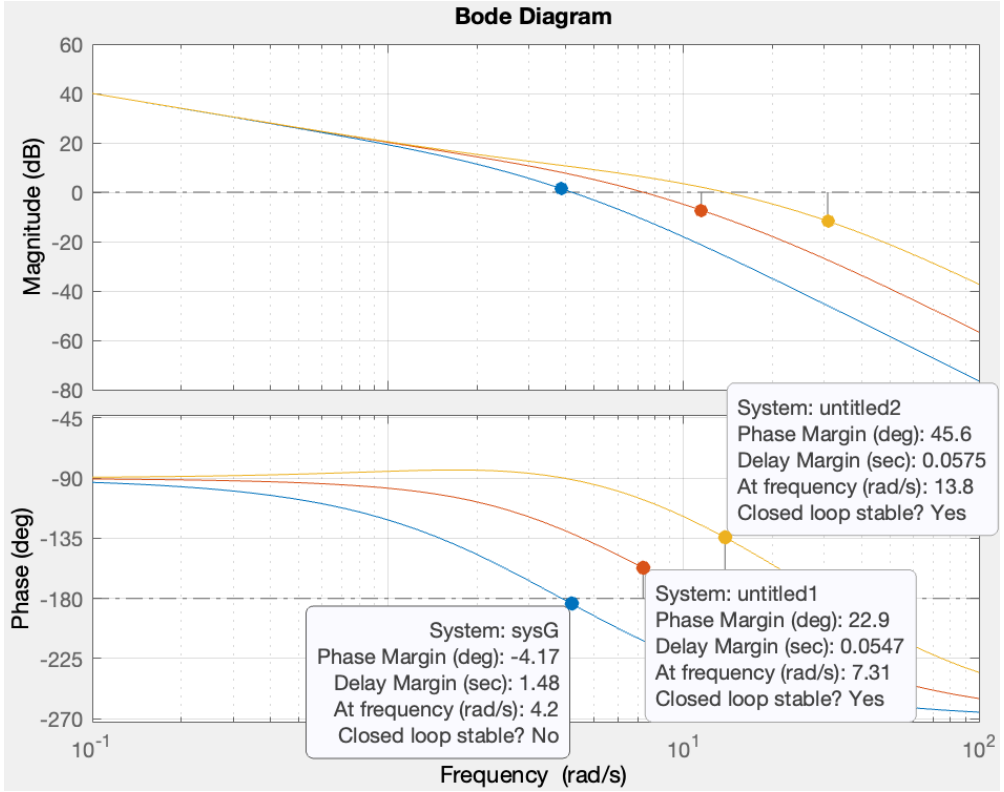
# 7. Compensation

## Lead compensator: example (Type 1)

6. PM = 22.9° => doesn't fulfill the specification. Evaluating other options shows that it is not possible to achieve the PM with one lead compensator.

7. Evaluate a double-lead compensator: PM = 45.6°

$$D_{c2}(s) = \frac{1}{(0.1)^2}\frac{(s+2)(s+4)}{(s+20)(s+40)} = \frac{(s/2+1)(s/4+1)}{(s/20+1)(s/40+1)}$$

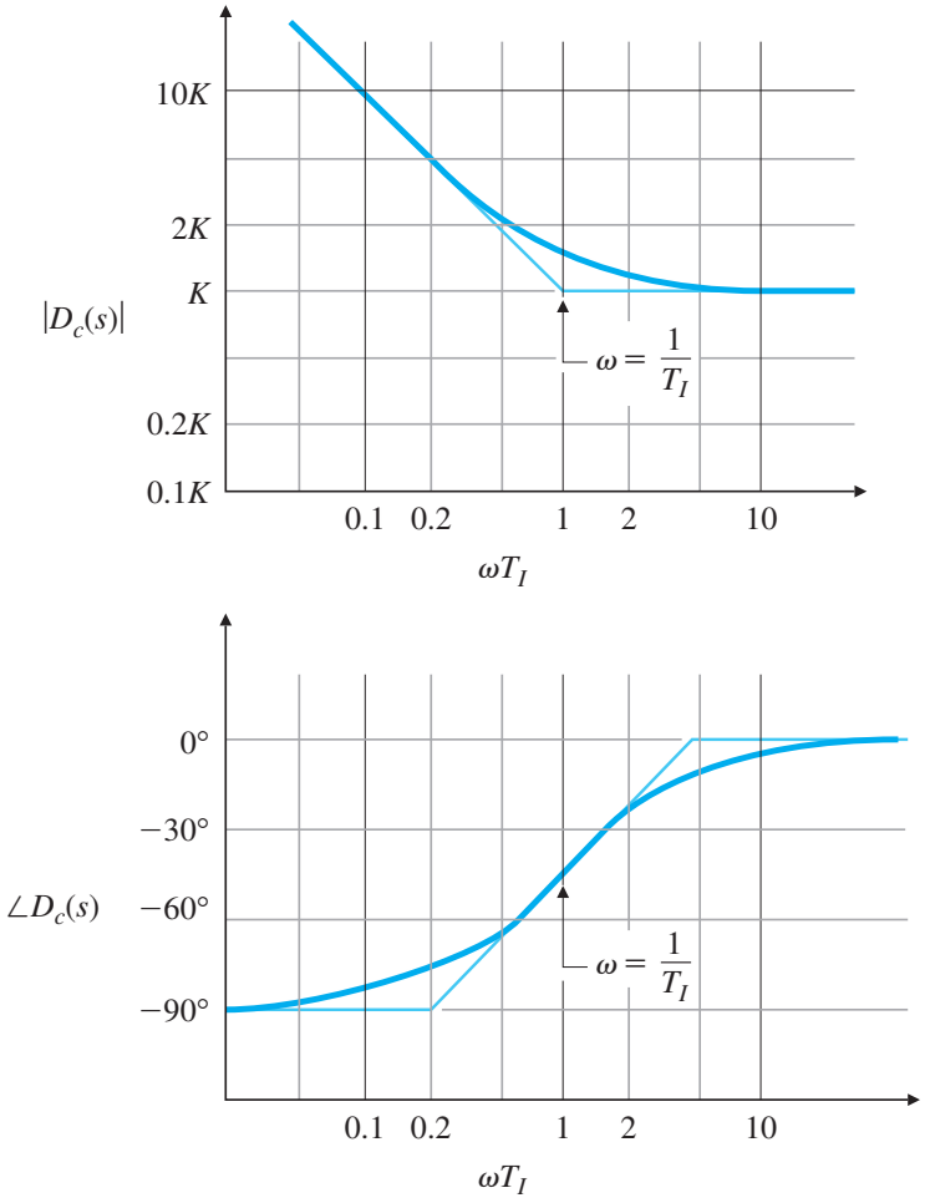


# 7. Compensation

**PI compensator:**

- Infinite gain at zero frequency
- Phase decrease at frequencies lower than the break point (not desired)
- The zero ($\omega = 1/T_I$) is usually located at a frequency << crossover frequency for not afecting the PM

$$D_c(s) = \frac{K}{s}\left(s + \frac{1}{T_I}\right)$$

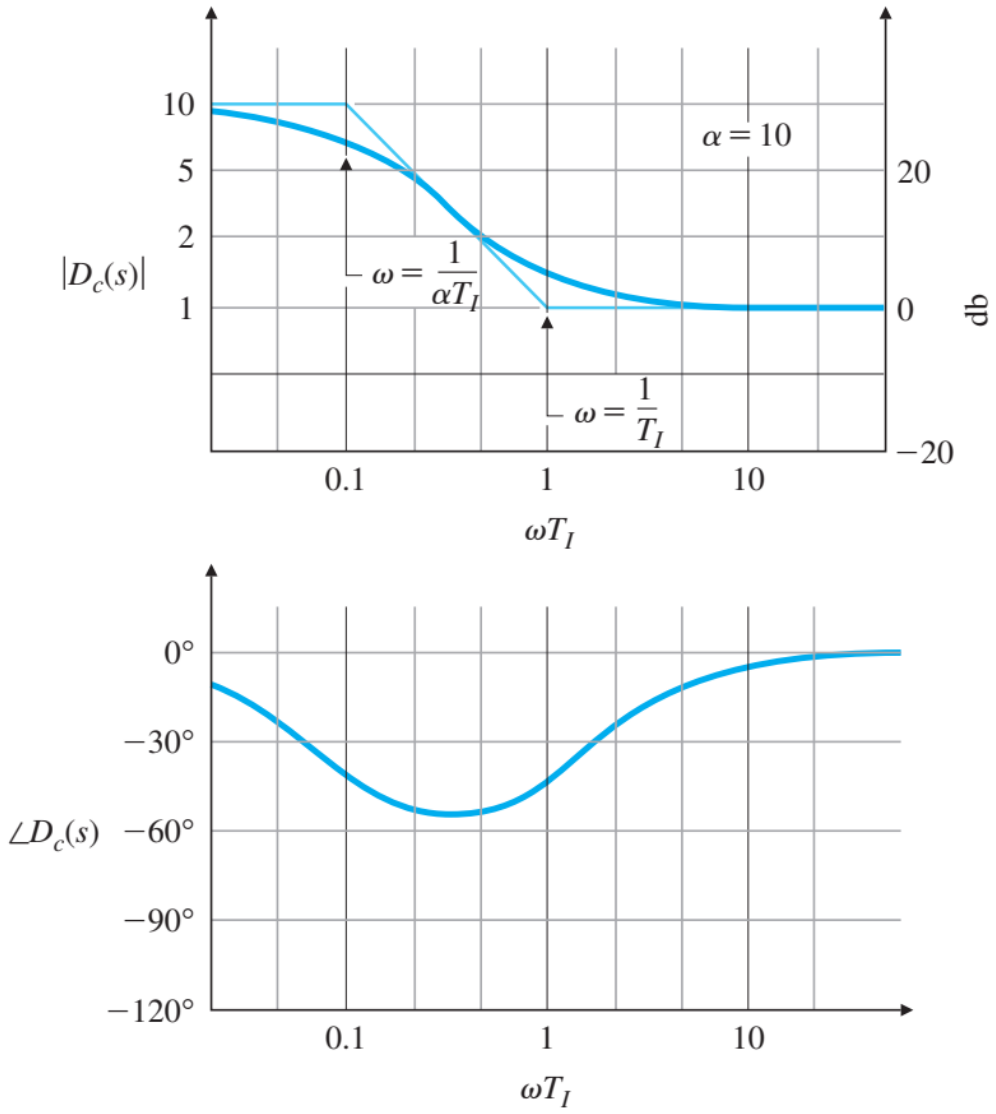


# 7. Compensation

**Lag compensator:**

- Additional gain of α in the low frequencies
- Phase decrease (lag) apparent in the frequency (**not desired**)
- w of the Pole (p) < w of the zero (z)
- If p and z are close to the origin (low frequencies)
  - Lag controller increases Kp, Kv, Ka by a factor α
  - Doesn't affect the transient response

$$D_c(s) = \alpha \frac{T_I s + 1}{\alpha T_I s + 1}, \qquad \alpha > 1$$



# 7. Compensation

## Lag compensator: design procedure

$$D_c(s) = \alpha \frac{T_I s + 1}{\alpha T_I s + 1}, \qquad \alpha > 1$$

1. Determine the open-loop gain $K$ that will meet the PM requirement without compensation.
2. Draw the Bode plot of the uncompensated system with crossover frequency from Step 1, and evaluate the low-frequency gain.
3. Determine $\alpha$ to meet the low-frequency gain error requirement.
4. Choose the corner frequency $\omega = {}^1/_{T_I}$ (the zero of the lag compensator) to be one octave to one decade below the new crossover frequency $\omega_c$.
5. The other corner frequency (the pole location of the lag compensator) is then $\omega = {}^1/_{\alpha T_I}$.
6. Iterate on the design. Adjust compensator parameters (poles, zeros, and gain) to meet all the specifications.

# 7. Compensation

## Lag compensator: example - Temperature Control System

Design a lag compensator so the PM is at least 40◦ and Kp = 9

$$KG(s) = \frac{K}{\left(\frac{1}{0.5}s + 1\right)(s+1)\left(\frac{1}{2}s + 1\right)}$$

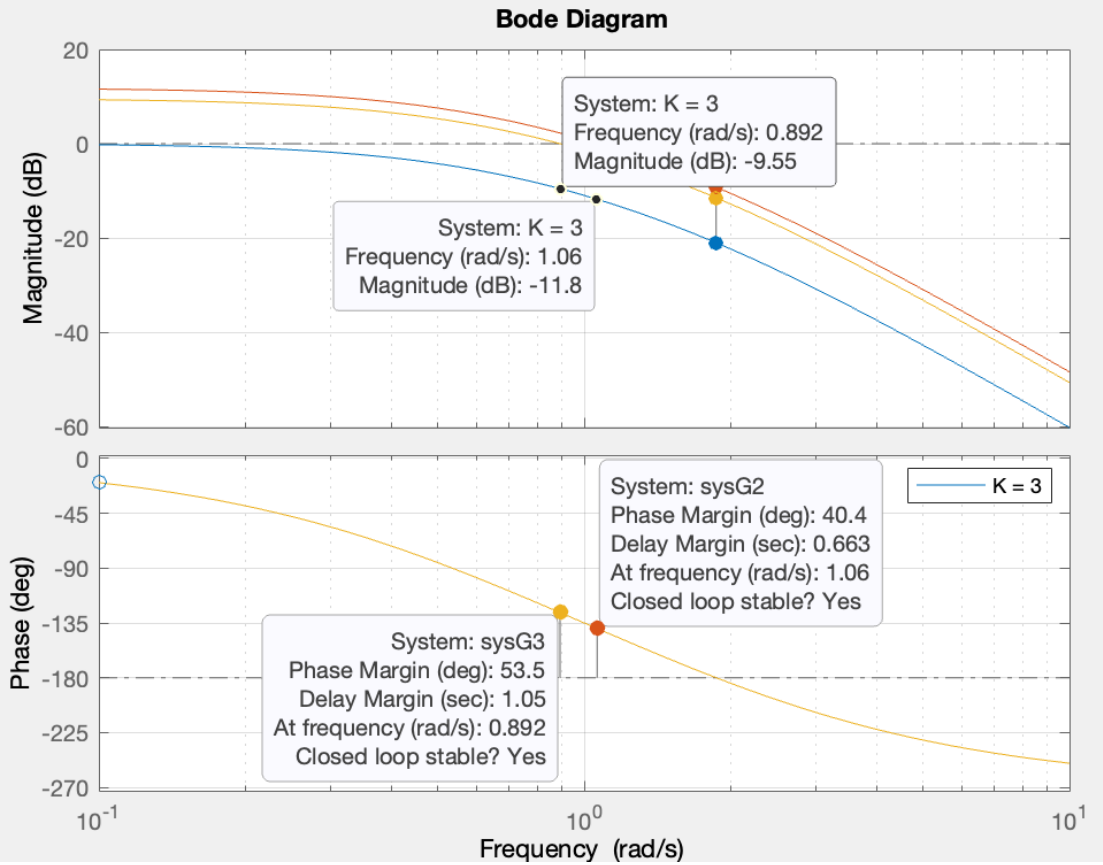
# 7. Compensation

## Lag compensator: example - Temperature Control System

1. From the bode plot of KG(s) for K = 9 it is possible to see that shifting the gain plot down 7.29 db the system has PM = 40°

$11.8 = 20 \log_{10}(K_1) \Rightarrow K_1 = 3.8905$
$9.55 = 20 \log_{10}(K_2) \Rightarrow K_2 = 3.0026$

For K = 3 the PM = 53° and wc = 0.892 rad/s



# 7. Compensation

## Lag compensator: example - Temperature Control System

2. The low frequency gain is almost 3 ($9.32 = 20 \log_{10}(|KG(s)|) \Rightarrow |KG(s)| = 2.92$)
The gain must be multiplied by 3 to obtain Kp = 9 => $\alpha = 3$

3. Defining the corner frequency for the zero as $w = 1/T_l = w_c/5$
Approximating $w_c = 0.892 = 1$ rad/s; then:
$w_c = 1/5$ rad/s => $w = 1/T_l = 0.2 \Rightarrow T_l = 5$

4. Defining the corner frequency of the pole as $\omega = 1/(\alpha T_l) = 1/(3*5) = 1/15$ rad/sec => the compensator is:

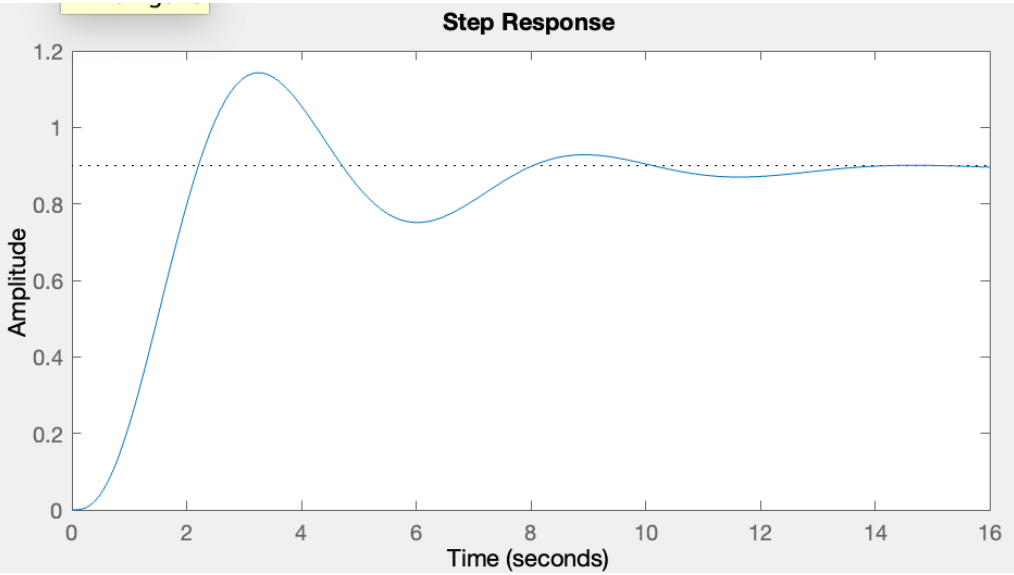
$$D_c(s) = 3\frac{5s+1}{15s+1}$$

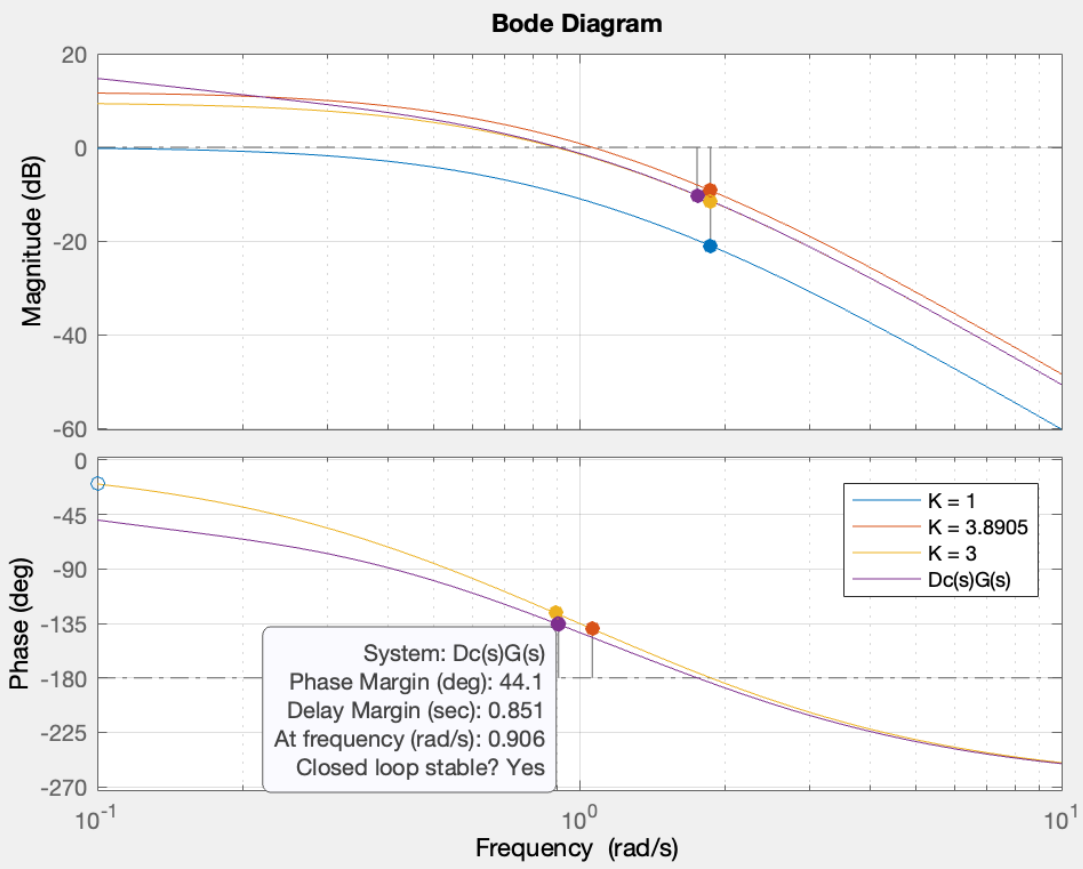
# 7. Compensation

## Lag compensator: example - Temperature Control System

$$D_c(s) = 3\frac{5s+1}{15s+1}$$

PM = 44°, ess = 0.1



# 7. Compensation

**Lag compensator: example - Temperature Control System**

- The same system was controlled with a
  - Lead compensator
  - Lag compensator
- Both compensators provide Kp = 9 (ess = 0.1)
- The lead compensator provide greater bandwidht (faster)

# 7. Compensation

## Lag compensator: example – DC motor

Design a lag compensator to obtain steady-state error of less than 0.1 for a **unit-ramp** input and overshoot Mp < 25%.

$$G(s) = \frac{1}{s(s+1)}$$

# 7. Compensation

## Lag compensator: example – DC motor (alternative design procedure

1. Define the value of $\alpha$:

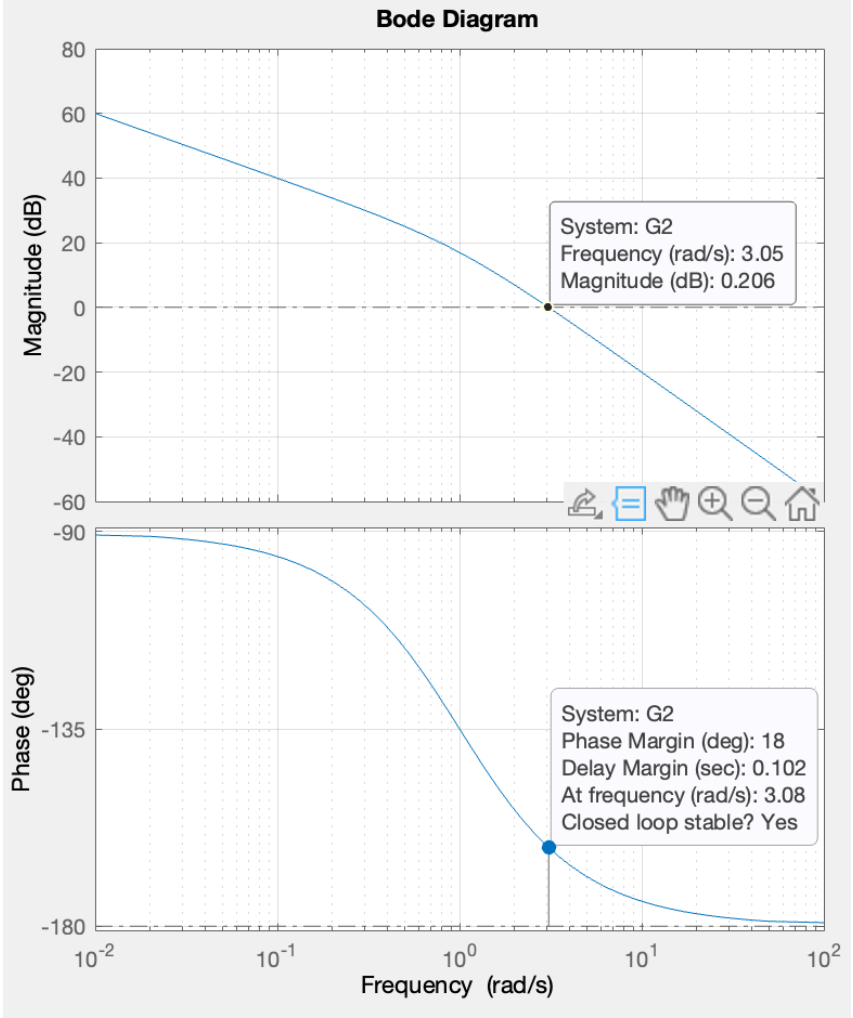
$$G(s) = \frac{1}{s(s+1)}$$

$$e_{ss} = \lim_{s\to 0} s\left[\frac{1}{1 + KD_c(s)G(s)}\right]R(s)$$

For a unit ramp:

$$e_{ss} = \lim_{s\to 0}\left\{\frac{1}{s + KD_c(s)[1/(s+1)]}\right\} = \frac{1}{KD_c(0)}$$

KDc(0) > 10 to obtain the desired error => $\alpha$ = 10




2. Evaluate $w_c$ and PM:
$w_c$ = 3 rad/sec and PM = 20◦

# 7. Compensation

## Lag compensator: example – DC motor (alternative design procedure

3. Design the break point of the pole and the zero to reduce $w_c$ and increase PM

Defining the corner frequency for the zero as $w = 1/T_l = w_c/10 = 3/10 = 0.3$; then, $T_l = 3.333$

4. Defining the corner frequency of the pole as $\omega = 1/(\alpha T_l) = 1/(10*3.333) = 1/33.33$ rad/sec => the compensator is:

$$D_c(s) = 10\,\frac{\frac{s}{3.333}+1}{\frac{s}{33.33}+1}$$

# 7. Compensation

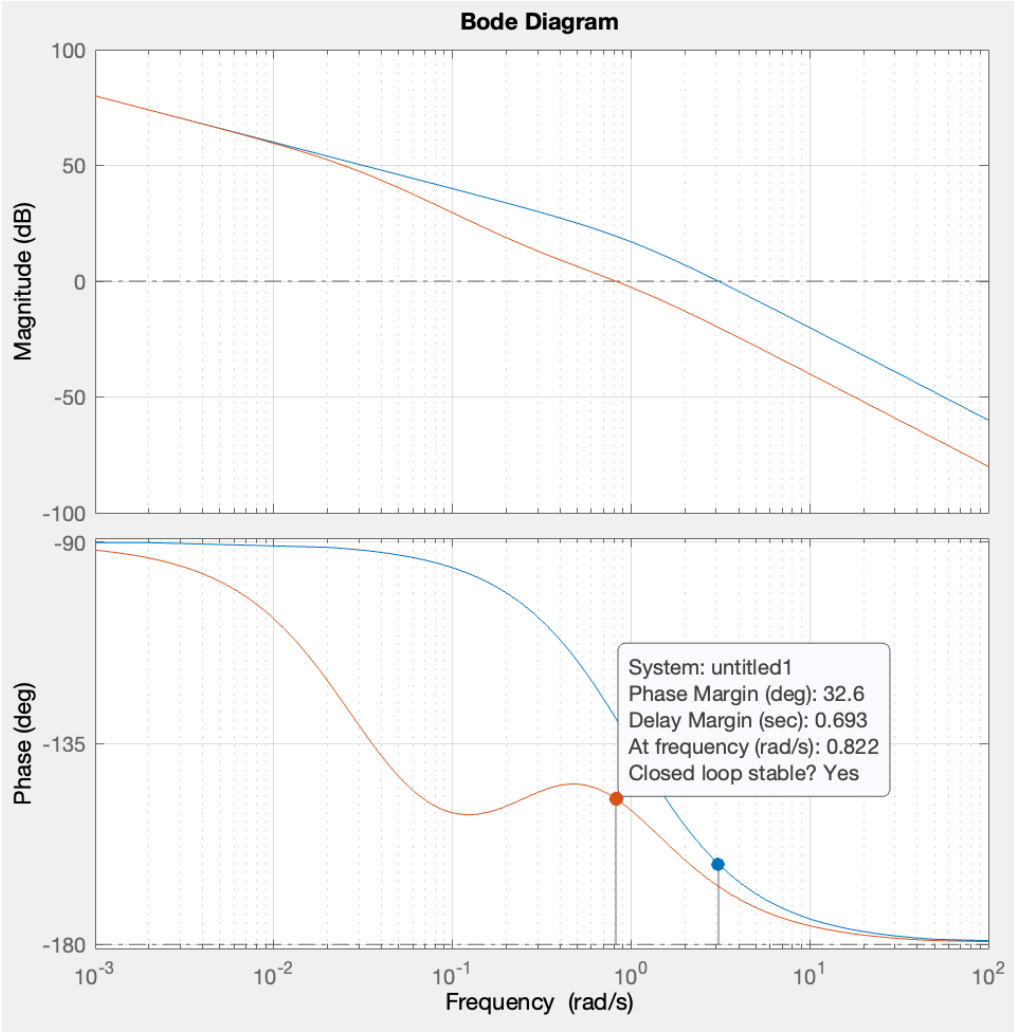
## Lag compensator: example – DC motor (alternative design procedure

Evaluating the response for

$$D_c(s) = 10\,\frac{\frac{s}{3.333}+1}{\frac{s}{33.33}+1}$$

$w_c$ = 0.822 rad/s
PM = 32.6°

**Does not fulfill the performance criteria**



# 7. Compensation

## Lag compensator: example – DC motor (alternative design procedure

5. Defining the corner frequency for the zero as $w = 1/T_l = w_c/10 = 3/10 = 0.3$ but approximating to 0.1

$w = 1/T_l = w_c/10 = 0.1$
$T_l = 10$

Then shifting to a lower frequency (reduce more wc to increase the PM)

4. Defining the corner frequency of the pole as $\omega = 1/(\alpha T_l) = 1/(10*10) = 1/100$ rad/sec => the compensator is:

$$D_c(s) = 10\frac{\frac{s}{10}+1}{\frac{s}{100}+1}$$

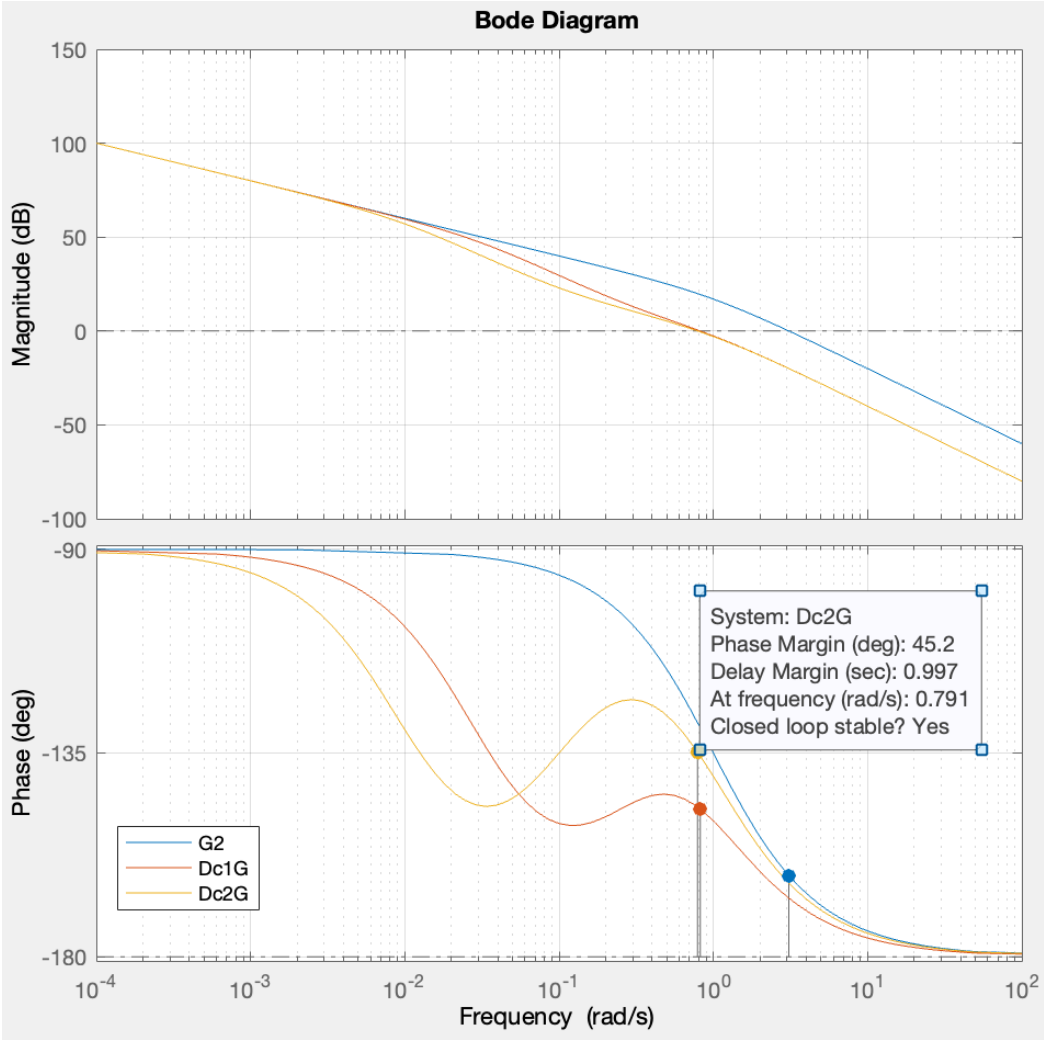
# 7. Compensation

## Lag compensator: example – DC motor (alternative design procedure

Evaluating the response for

$$D_c(s) = 10\,\frac{\frac{s}{10}+1}{\frac{s}{100}+1}$$

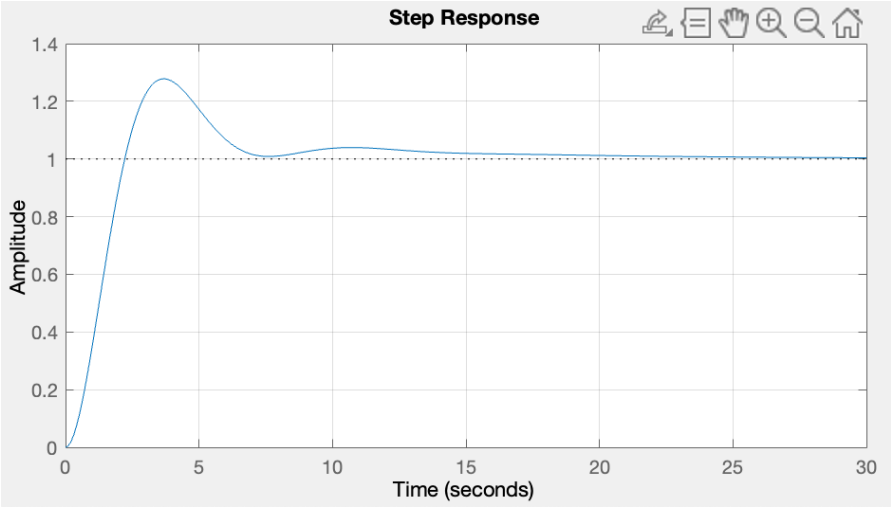
$w_c$ = 0.791 rad/s
PM = 45.2°

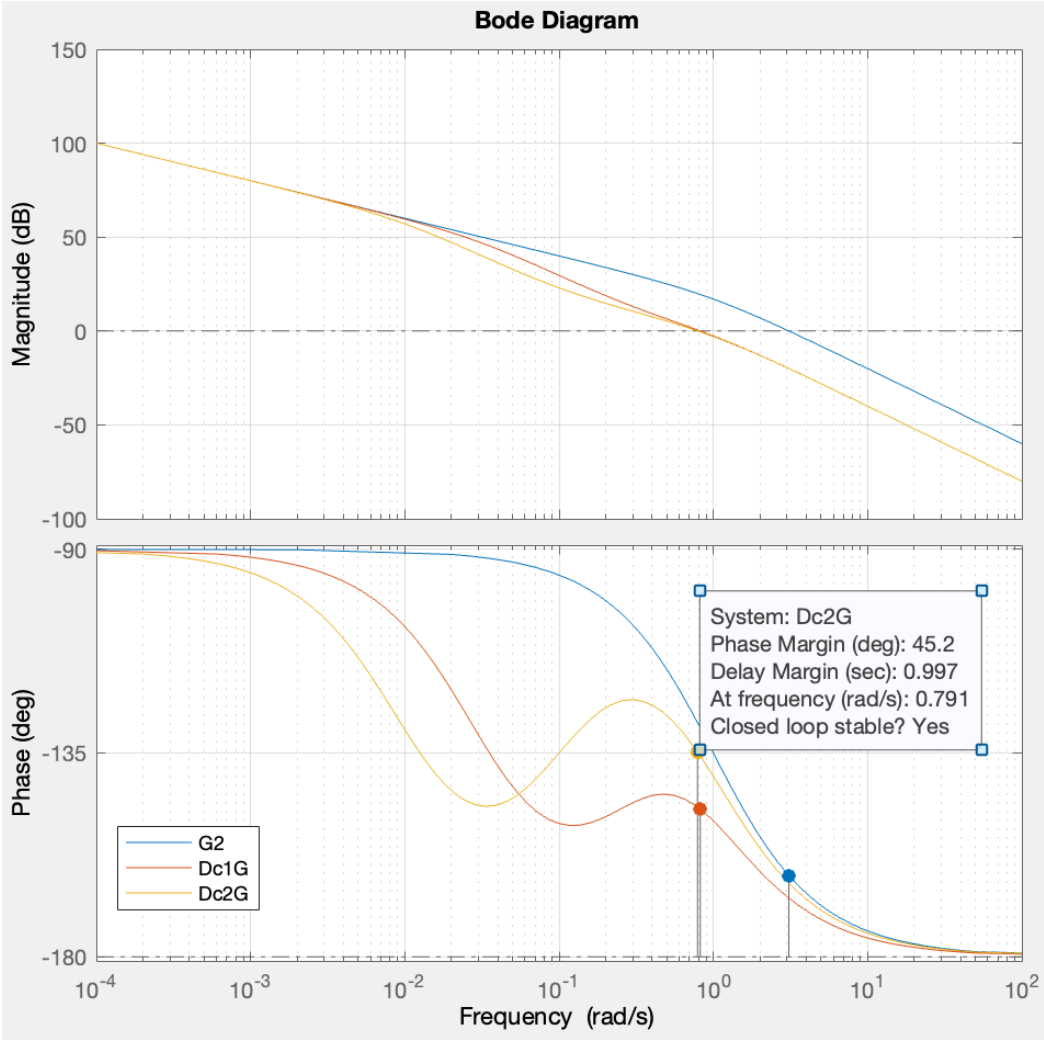


# 7. Compensation

## Lag compensator: example – DC motor (alternative design procedure

$$D_c(s) = 10\,\frac{\frac{s}{10}+1}{\frac{s}{100}+1}$$



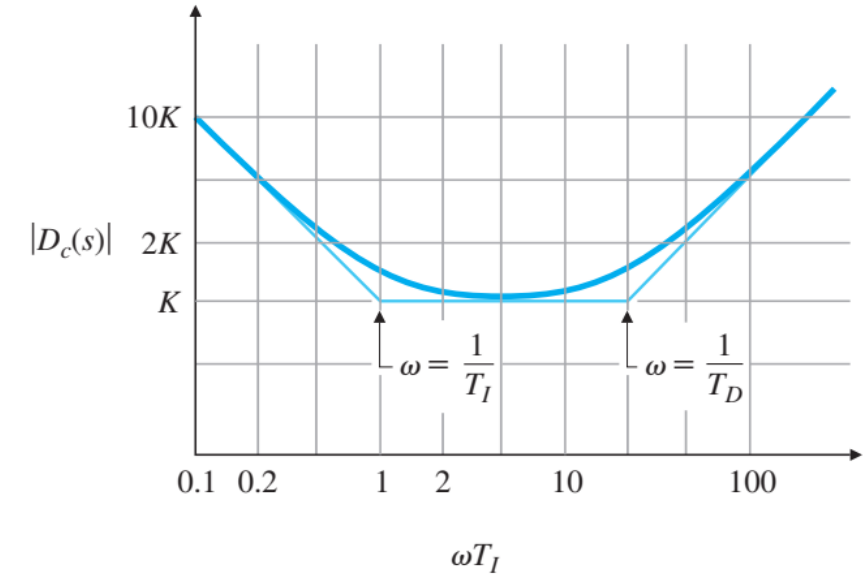



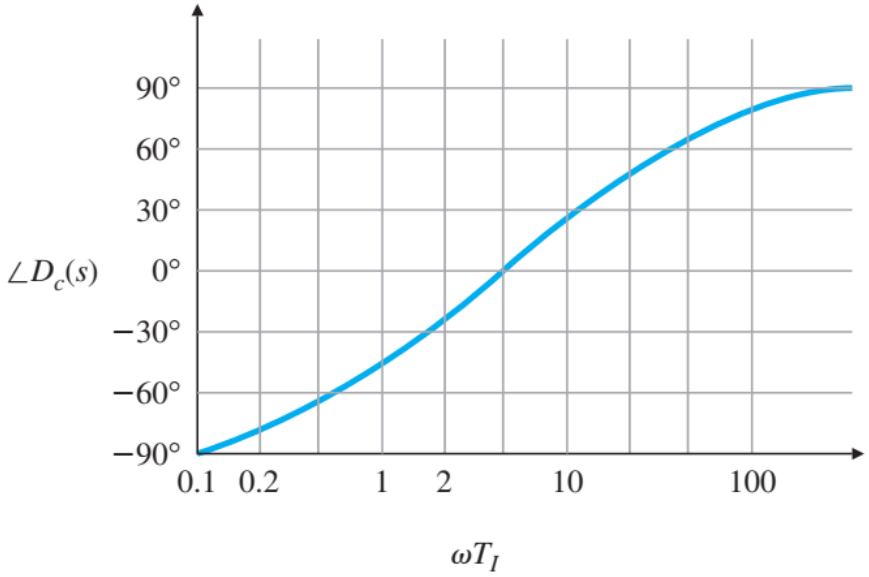
# 7. Compensation

**PID compensator (lead–lag compensator):**

- Combination of the lead and lag compensator
- It improves transient response
- It improves steady-state error

$$D_c(s) = \frac{K}{s}\left[(T_D s + 1)\left(s + \frac{1}{T_I}\right)\right]$$

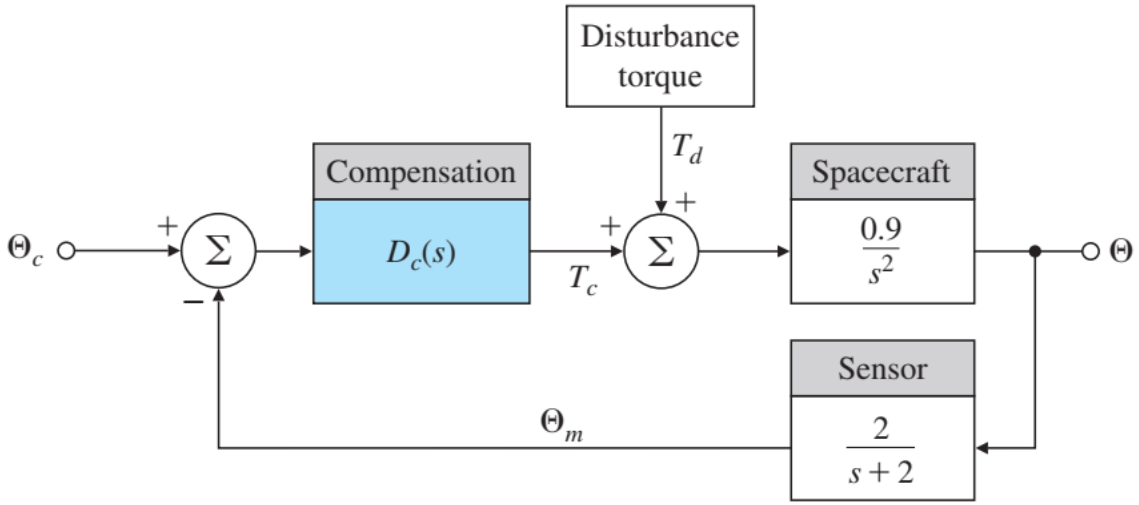


# 7. Compensation

**PID compensator (lead–lag compensator): example - Spacecraft Attitude Control**

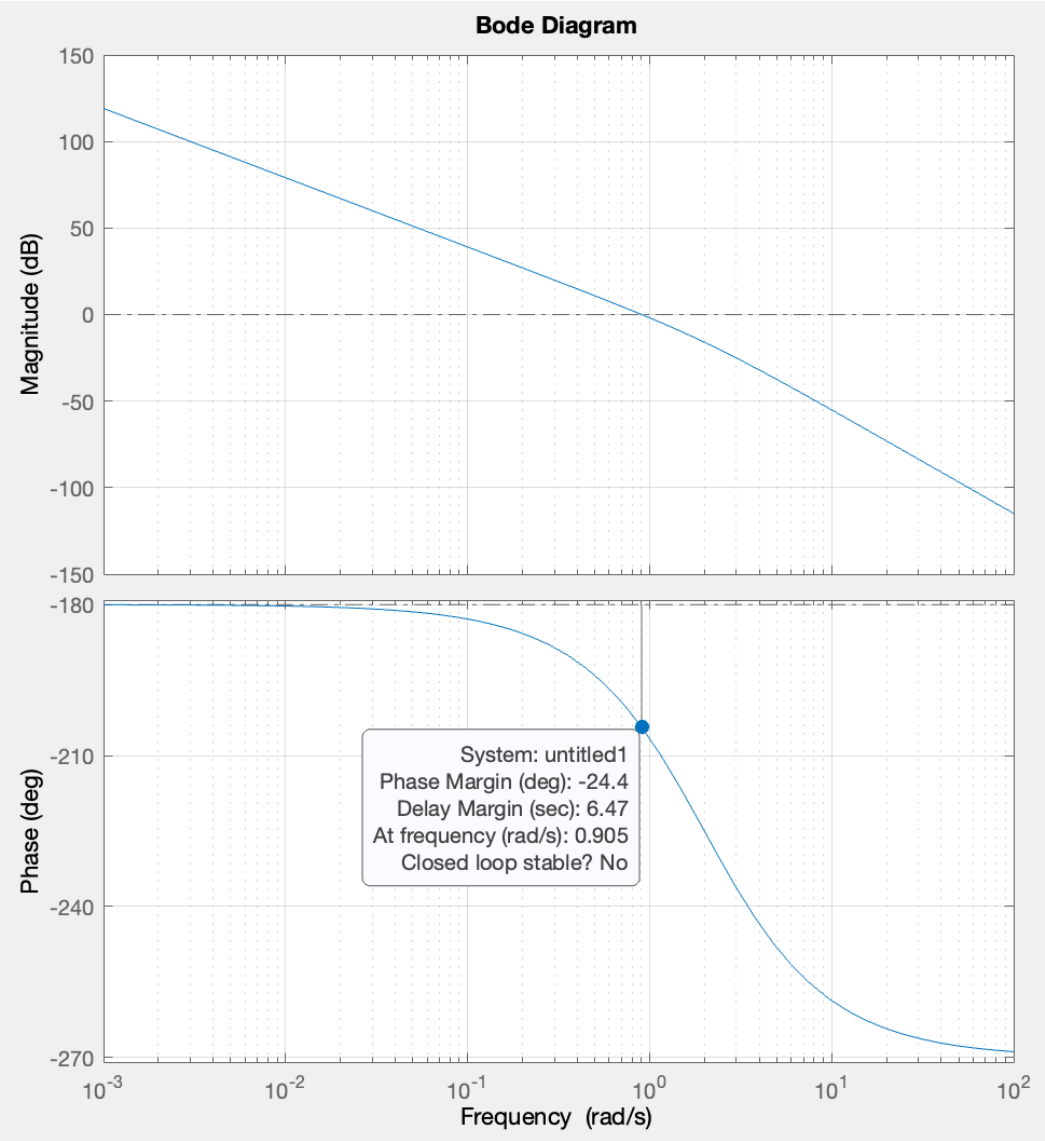
- Objectives:
  - 0 steady-state error for constant disturbance
  - PM = 65°
  - High bandwidth (as possible)



# 7. Compensation

**PID compensator (lead–lag compensator): example - Spacecraft Attitude Control**

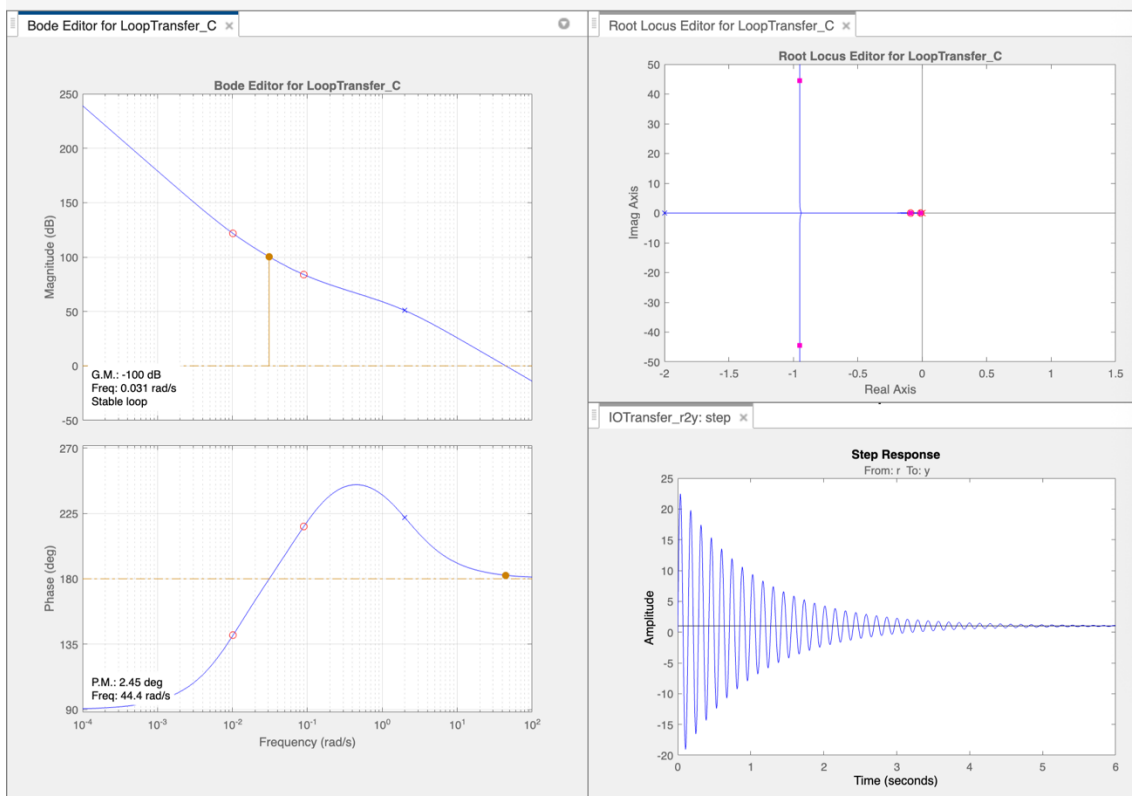
- 0 steady-state error => it can be obtained with an integrator (1/s) in Dc(s)
- Bode of G(s)H(s)
  - Negative PM
  - Dc(s) requires a zero to stabilize the system
  - Make the slope -20 db/dec at the crossover frequency



# 7. Compensation

**PID compensator (lead–lag compensator): example - Spacecraft Attitude Control**

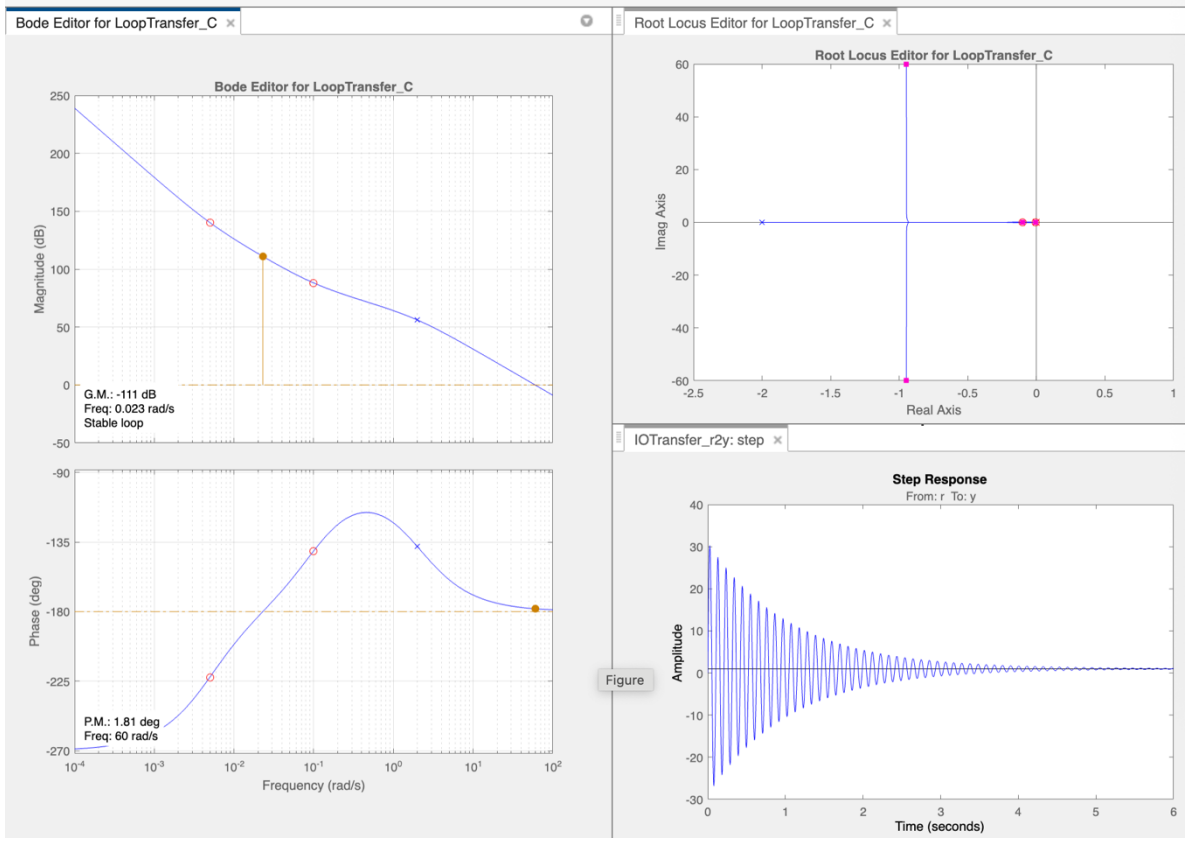
- $T_I$ at the left (lower f) of $T_D$
- PM = 65° (-115°) can be adjusted with $T_D$
  - $T_D > 1/0.1$ ($w_{TD} < 0.1$ rad/s) can provide PM = 65° (adjusting K)
  - $T_D < 1/0.1$ ($w_{TD} > 0.1$ rad/s) can provide PM < 65° (adjusting K)
  - $T_D = 1/2$ ($w_{TD} = 2$ rad/s) cancels the pole of H(s) but makes the system unstable
  - **$T_D = 1/0.1$ ($w_{TD} = 0.1$ rad/s) to obtain PM = 65° (adjusting K)**



# 7. Compensation

**PID compensator (lead–lag compensator): example - Spacecraft Attitude Control**

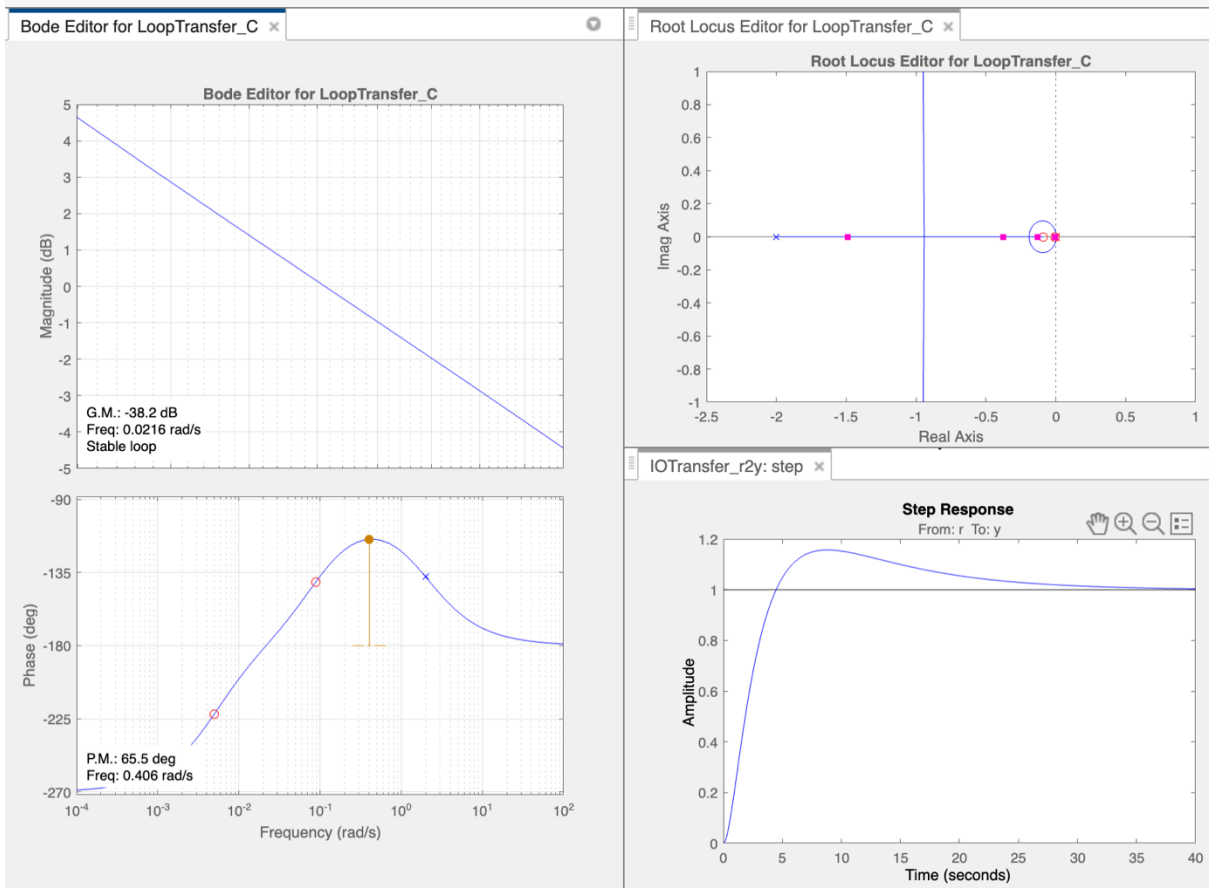
- $T_I$ at the left (lower f) of $T_D$
- $T_I$ is defined as:
  - $20T_D$: **$T_I$ = 20(1/0.1) = 200 = 1/0.005 ($w_{TI}$ = 0.005 rad/s)**
  - **Or**
  - $w_{TI} = w_{TD}/20$: **$w_{TI}$ = (0.1/20) = 0.005 ($w_{TI}$ = 0.005 rad/s)**



# 7. Compensation

**PID compensator (lead–lag compensator): example - Spacecraft Attitude Control**

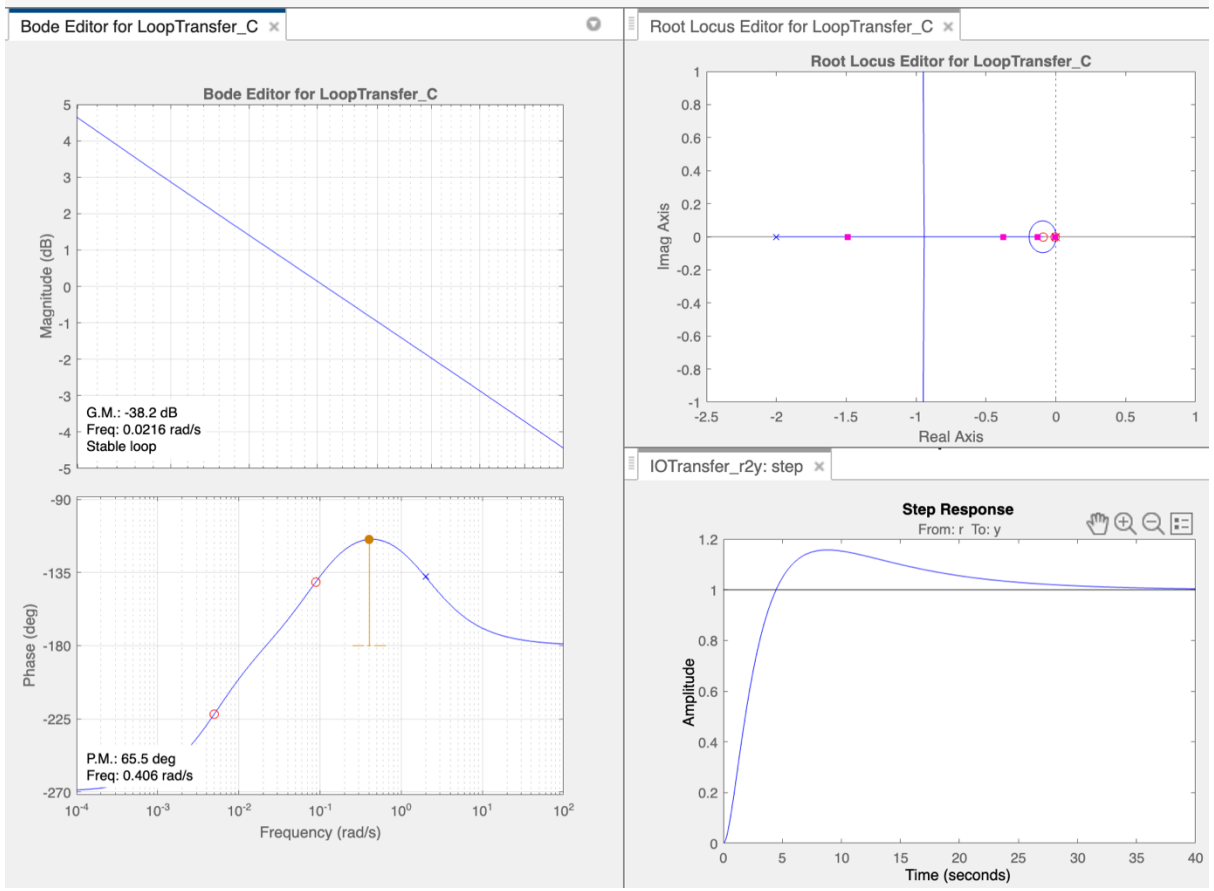
- K is adjusted to reduce the crossover frequency and obtain the desired PM
- K to reduce 74 db
  - -74 db = $20*\log_{10}(K)$
  - K = 10^(-74/20)
  - K = 1.9953e-04



# 7. Compensation

**PID compensator (lead–lag compensator): example - Spacecraft Attitude Control**

- The design is complete
  - **K = 1.9953e-04**
  - **TD = 1/0.1**
  - **TI = 1/0.005**



# 7. Compensation

## PID compensator (lead–lag compensator): example - Spacecraft Attitude Control

$$D_C(s) = \frac{1.9953e-04}{s}\left(\frac{1}{0.1}s+1\right)\left(\frac{1}{0.005}s+1\right)$$

$$\mathcal{T}(s) = \frac{\Theta}{\Theta_c} = \frac{D_c G}{1 + D_c GH}$$

$$\frac{\Theta}{T_d} = \frac{G}{1 + D_c GH}$$